

Exercises

Level - 1

(Problems Based on Fundamentals)

MEASUREMENT OF ANGLES

1. If the radius of the earth 4900 km, what is the length of its circumference?
2. The angles of a triangle are in the ratio 3 : 4 : 5. Find the smallest angle in degrees and the greatest angle in radians.
3. The angles of a triangle are in AP and the number of degrees in the least is to the number of radians in the greatest as 60 to π , find the angles in degrees.
4. The number of sides in two regular polygons are 5 : 4 and the difference between their angles is 9. Find the number of sides of the polygon.
5. The angles of a quadrilateral are in AP and the greatest is double the least. Express the least angles in radians.
6. Find the angle between the hour hand and the minute hand in circular measure at half past 4.
7. Find the length of an arc of a circle of radius 10 cm subtending an angle of 30° at the centre.
8. The minute hand of a watch is 35 cm long. How far does its tip move in 18 minutes ?
9. At what distance does a man, whose height is 2 m subtend an angle of 10° ?
10. Find the distance at which a globe $5\frac{1}{2}$ cm in diameter, will subtend an angle of 6° .
11. The radius of the earth being taken to 6400 km and the distance of the moon from the earth being 60 times the radius of the earth. Find the radius of the moon which subtends an angle of 16° at the earth.
12. The difference between the acute angles of a right angled triangle is $\frac{2\pi}{3}$ radians. Express the angle in degrees.
13. The angles of a quadrilateral are in AP and the greatest angle is 120° . Find the angles in radians.
14. At what distance does a man $5\frac{1}{2}$ ft in height, subtend an angle of $15''$?
15. Find the angle between the hour hand and minute-hand in circular measure at 4 o' clock.

TRIGONOMETRICAL RATIOS AND IDENTITIES

16. If $\sec \theta + \tan \theta = 3$, where θ lies in the first quadrant, then find the value of $\cos \theta$.
17. If $\operatorname{cosec} \theta - \cot \theta = \frac{1}{5}$, then find the value of $\sin \theta$.
18. If $a = c \cos \theta + d \sin \theta$ and $b = c \sin \theta - d \cos \theta$ such that $a^m + b^n = c^p + d^q$, where $m, n, p, q \in N$ then find the value of $m + n + p + q + 42$.
19. If $3 \sin \theta + 4 \cos \theta = 5$, then find the value of $3 \cos \theta - 4 \sin \theta$.
20. If $x = r \cos \theta \sin \phi$, $y = r \cos \theta \cos \phi$ and $z = r \sin \theta$ such that $x^m + y^n + z^p = r^2$, where $m, n, p \in N$, then find the value of $(m + n + p - 4)^{m+n+p+4}$.
21. If $x = \frac{2 \sin \alpha}{1 + \cos \alpha + 3 \sin \alpha}$, then find the value of $\frac{\sin \alpha - 3 \cos \alpha + 3}{2 - 2 \cos \alpha}$.
22. If $P = \sec^6 \theta - \tan^6 \theta - 3 \sec^2 \theta \tan^2 \theta$, $Q = \operatorname{cosec}^6 \theta - \cot^6 \theta - 3 \operatorname{cosec}^2 \theta \cot^2 \theta$ and $R = \sin^6 \theta + \cos^6 \theta + 3 \sin^2 \theta \cos^2 \theta$, then find the value of $(P + Q + R)^{(P+Q+R)}$.
23. If $3 \sin x + 4 \cos x = 5$, for all x in $\left(0, \frac{\pi}{2}\right)$, then find the value of $2 \sin x + \cos x + 4 \tan x$.
24. If $\sin A + \sin B + \sin C + 3 = 0$, then find the value of $\cos A + \cos B + \cos C + 10$.
25. If $(1 + \sin \theta)(1 + \cos \theta) = \frac{5}{4}$, then find the value of $(1 - \sin \theta)(1 - \cos \theta)$.
26. Find the minimum value of the expression $f(x) = \frac{9x^2 \sin^2 x + 4}{x \sin x}$, for all x in $(0, \pi)$.
27. If $\cos \theta + \sin \theta = \sqrt{2} \cos \theta$, then prove that $\cos \theta - \sin \theta = \sqrt{2} \sin \theta$.
28. If $\tan^2 \theta = 1 - e^2$ then prove that $\sec \theta + \tan^3 \theta \cdot \operatorname{cosec} \theta = (2 - e^2)^{3/2}$.
29. If $\sin \theta + \sin^2 \theta + \sin^3 \theta = 1$, then prove that, $\cos^6 \theta - 4 \cos^4 \theta + 8 \cos^2 \theta = 4$.
30. If $x = \frac{2 \sin \theta}{1 + \cos \theta + \sin \theta}$, then prove that $\frac{1 - \cos \theta + \sin \theta}{1 + \sin \theta} = x$.
31. Prove that $3(\sin x - \cos x)^4 + 6(\sin x + \cos x)^2 + 4(\sin^6 x + \cos^6 x) = 13$.
32. If $\sin x + \sin^2 x = 1$, then find the value of $\cos^8 x + 2 \cos^6 x + \cos^4 x$.

33. If $0 \leq \theta \leq 180^\circ$ and $81^{\sin^2 \theta} + 81^{\cos^2 \theta} = 30$, then find the value of θ .
34. Let $f_k(\theta) = \sin^k(\theta) + \cos^k(\theta)$. Then find the value of $\frac{1}{6} f_6(\theta) - \frac{1}{4} f_4(\theta)$.
35. If $x \sin^3 \alpha + y \cos^3 \alpha = \sin \alpha \cos \alpha$ and $x \sin \alpha = y \cos \alpha$ then prove that $x^2 + y^2 = 1$.
36. If $\tan \theta + \sin \theta = m$, $\tan \theta - \sin \theta = n$, then prove that $m^2 - n^2 = 4 \sqrt{mn}$.
37. If $\frac{\cos^4 x}{\cos^2 y} + \frac{\sin^4 x}{\sin^2 y} = 1$, then prove that $\frac{\cos^4 y}{\cos^2 x} + \frac{\sin^4 y}{\sin^2 x} = 1$.
38. If $f_n(\theta) = \sin^n \theta + \cos^n \theta$, prove that $2f_6(\theta) - 3f_4(\theta) + 1 = 0$.
39. If $\frac{\sin A}{\sin B} = p$, $\frac{\cos A}{\cos B} = q$, prove that $\tan A \cdot \tan B = \frac{p}{q} \left(\frac{q^2 - 1}{1 - p^2} \right)$.
40. If $\frac{\sin^4 \alpha}{a} + \frac{\cos^4 \alpha}{b} = \frac{1}{a+b}$, then prove that $\frac{\sin^8 \alpha}{a^3} + \frac{\cos^8 \alpha}{b^3} = \frac{1}{(a+b)^3}$.

MEASUREMENT OF DIFFERENT T-RATIOS

41. Find the value of
 - (i) $\sin 120^\circ$
 - (ii) $\sin 150^\circ$
 - (iii) $\sin 210^\circ$
 - (iv) $\sin 225^\circ$
 - (v) $\sin 300^\circ$
 - (vi) $\sin 330^\circ$
 - (vii) $\sin 405^\circ$
 - (viii) $\sin 650^\circ$
 - (ix) $\sin 1500^\circ$
 - (x) $\sin 2013^\circ$
42. Find the value of $\cos(1^\circ) \cdot \cos(2^\circ) \cdot \cos(3^\circ) \dots \cos(189^\circ)$.
43. Find the value of $\tan(1^\circ) \cdot \tan(2^\circ) \cdot \tan(3^\circ) \dots \tan(89^\circ)$.
44. Find the value of $\tan 35^\circ \cdot \tan 40^\circ \cdot \tan 45^\circ \cdot \tan 50^\circ \cdot \tan 55^\circ$.
45. Find the value of $\sin(10^\circ) + \sin(20^\circ) + \sin(30^\circ) + \sin(40^\circ) + \dots + \sin(360^\circ)$.
46. Find the value of $\cos(10^\circ) + \cos(20^\circ) + \cos(30^\circ) + \cos(40^\circ) + \dots + \cos(360^\circ)$.
47. Find the value of $\sin^2 5^\circ + \sin^2 10^\circ + \sin^2 15^\circ + \dots + \sin^2 90^\circ$.

48. Find the value of

$$\sin^2\left(\frac{\pi}{18}\right) + \sin^2\left(\frac{\pi}{9}\right) + \sin^2\left(\frac{4\pi}{9}\right) + \sin^2\left(\frac{7\pi}{18}\right)$$

49. Find the value of

$$\tan(20^\circ) \tan(25^\circ) \tan(45^\circ) \tan(65^\circ) \tan(70^\circ)$$

 50. Find the value of $\cos(\theta_1) + \cos(\theta_2) + \cos(\theta_3)$

$$\text{if } \sin(\theta_1) + \sin(\theta_2) + \sin(\theta_3) = 3$$

51. Find the value of

$$\sin^2 6^\circ + \sin^2 12^\circ + \dots + \sin^2 90^\circ$$

52. Find the value of

$$\sin^2 10^\circ + \sin^2 20^\circ + \dots + \sin^2 90^\circ$$

53. Find the value of

$$\sin^2 9^\circ + \sin^2 18^\circ + \dots + \sin^2 90^\circ$$

54. Find the value of

$$\tan 1^\circ \cdot \tan 2^\circ \cdot \tan 3^\circ \dots \tan 89^\circ$$

55. Find the value of

$$\cos 1^\circ \cdot \cos 2^\circ \cdot \cos 3^\circ \dots \cos 189^\circ$$

 56. Solve for θ ; $2 \sin^2 \theta + 3 \cos \theta = 0$

$$\text{where } 0 < \theta < 360^\circ.$$

 57. Solve for θ ; $\cos \theta + \sqrt{3} \sin \theta = 2$,

$$\text{where } 0 < \theta < 360^\circ.$$

 58. If $4n\alpha = \pi$, then prove that

$$\tan \alpha \tan 2\alpha \tan 3\alpha \dots \tan (2n-1)\alpha = 1.$$

59. Find the value of

$$\cos(18^\circ) + \cos(234^\circ) + \cos(162^\circ) + \cos(306^\circ).$$

60. Find the value of

$$\cos(20^\circ) + \cos(40^\circ) + \cos(60^\circ) + \dots + \cos(180^\circ)$$

61. Find the value of

$$\sin(20^\circ) + \sin(40^\circ) + \sin(60^\circ) + \dots + \sin(360^\circ)$$

62. Draw the graphs of

$$(i) f(x) = \sin x + 1$$

$$(ii) f(x) = \sin x - 1$$

$$(iii) f(x) = -\sin x$$

$$(iv) f(x) = 1 - \sin x$$

$$(v) f(x) = -1 - \sin x$$

$$(vi) f(x) = \sin 2x, \sin 3x$$

$$(vii) f(x) = \sin^2 x$$

$$(viii) f(x) = \cos^2 x$$

$$(ix) f(x) = \max\{\sin x, \cos x\}$$

$$(x) f(x) = \min\{\sin x, \cos x\}$$

$$(xi) f(x) = \min\left\{\sin x, \frac{1}{2}, \cos x\right\}$$

$$(xii) f(x) = \max\{\tan x, \cot x\}$$

$$(xiii) f(x) = \min\{\tan x, \cot x\}$$

63. Find the number of solutions of

$$(i) \sin x = \frac{1}{2}, \forall x \in [0, 2\pi]$$

$$(ii) \cos x = \frac{\sqrt{3}}{2}, \forall x \in [0, 3\pi]$$

$$(iii) 4 \sin^2 x - 1 = 0, \forall x \in [0, 3\pi]$$

$$(iv) \sin^2 x - 3 \sin x + 2 = 0, \forall x \in [0, 3\pi]$$

$$(v) \cos^2 x - \cos x - 2 = 0, \forall x \in [0, 3\pi]$$

COMPOUND ANGLES

64. Find the values of

$$(i) \sin(15^\circ),$$

$$(ii) \cos(15^\circ),$$

$$(iii) \tan(15^\circ)$$

 65. Find the value of $\tan(75^\circ) + \cot(75^\circ)$

 66. Prove that $\cos(9^\circ) + \sin(9^\circ) = \sqrt{2} \sin(54^\circ)$

 67. Prove that $\tan(70^\circ) = 2 \tan(50^\circ) + \tan(20^\circ)$

 68. Prove that $\frac{\cos 20^\circ - \sin 20^\circ}{\cos 20^\circ + \sin 20^\circ} = \tan 25^\circ$

 69. Prove that $\frac{\cos 7^\circ + \sin 7^\circ}{\cos 7^\circ - \sin 7^\circ} = \tan 52^\circ.$

 70. Prove that $\tan 20^\circ + \tan 25^\circ + \tan 20^\circ \tan 25^\circ = 1.$

 71. If $A + B = 45^\circ$, then find the value of $(1 + \tan A)(1 + \tan B)$

72. Find the value of

$$(1 + \tan 245^\circ)(1 + \tan 250^\circ)(1 + \tan 260^\circ)$$

$$(1 - \tan 200^\circ)(1 - \tan 205^\circ)(1 - \tan 215^\circ)$$

 73. Prove that $\tan 13A - \tan 9A - \tan 4A$

$$= \tan 4A \cdot \tan 9A \cdot \tan 13A$$

 74. Prove that $\tan 9A - \tan 7A - \tan 2A$

$$= \tan 2A \cdot \tan 7A \cdot \tan 9A$$

 75. If $\tan \alpha = \frac{m}{m+1}$, $\tan \beta = \frac{1}{2m+1}$,

$$\text{then prove that } \alpha + \beta = \frac{\pi}{4}.$$

 76. Prove that $\sin^2 B$

$$= \sin^2 A + \sin^2(A - B) - 2 \sin A \cos B \sin(A - B)$$

 77. Prove that $\cos(2x + 2y)$

$$= \cos 2x \cos 2y + \cos^2(x + y) - \cos^2(x - y)$$

 78. If $\sin \theta = \frac{x-y}{x+y}$,

$$\text{then prove that } \tan\left(\frac{\pi}{4} + \frac{\theta}{2}\right) = \pm \sqrt{\frac{x}{y}}$$

 79. If $\tan \alpha = \frac{Q \sin \beta}{P + Q \cos \beta}$,

$$\text{prove that } \tan(\beta - \alpha) = \frac{P \sin \alpha}{Q + P \cos \alpha}$$

 80. If $\cos(\alpha - \beta) + \cos(\beta - \gamma) + \cos(\gamma - \alpha) = -\frac{3}{2}$

$$\text{prove that } \cos \alpha + \cos \beta + \cos \gamma = 0$$

$$\text{and } \sin \alpha + \sin \beta + \sin \gamma = 0$$

 81. If $\tan(\alpha + \theta) = n \tan(\alpha - \theta)$, show that $\frac{\sin 2\theta}{\sin 2\alpha} = \frac{n-1}{n+1}$

 82. If $\sin \alpha + \sin \beta = a$ and $\cos \alpha + \cos \beta = b$, then show that

$$(i) \cos(\alpha + \beta) = \frac{b^2 - a^2}{b^2 + a^2}$$

$$(ii) \sin(\alpha + \beta) = \frac{2ab}{b^2 + a^2}$$

83. If α and β are the roots of $a \cos \theta + b \sin \theta = c$, then prove that

$$(i) \cos(\alpha + \beta) = \frac{a^2 - b^2}{a^2 + b^2}$$

$$(ii) \cos(\alpha - \beta) = \frac{2c - (a^2 + b^2)}{a^2 + b^2}$$

84. If α and β are the roots of $a \tan \theta + b \sec \theta = c$, then show that

$$\tan(\alpha + \beta) = \frac{2ac}{a^2 - c^2}$$

85. If $\tan(\pi \cos \theta) = \cot(\pi \sin \theta)$, prove that

$$\cos\left(\theta - \frac{\pi}{4}\right) = \pm \frac{1}{2\sqrt{2}}$$

86. If $\tan \theta = \frac{x \sin \phi}{1 - x \cos \phi}$ and $\tan \phi = \frac{y \sin \theta}{1 - y \cos \theta}$,

then prove that, $x \sin \phi = y \sin \theta$.

87. If $\tan \alpha + \tan \beta = a$, $\cot \alpha + \cot \beta = b$ and $\tan(\alpha + \beta) = c$ then find a relation in a , b and c .

88. If $\tan \beta = \frac{n \sin \alpha \cos \alpha}{1 - n \sin^2 \alpha}$,

then prove that $\tan(\alpha - \beta) = (1 - n) \tan \alpha$

89. If $x + y + z = 0$, then prove that

$$\begin{aligned} & \cot(x + y - z) \cdot \cot(y + z - x) \\ & + \cot(y + z - x) \cdot \cot(z + x - y) \\ & + \cot(z + x - y) \cdot \cot(x + y - z) = 1 \end{aligned}$$

90. If $2 \tan \alpha = 3 \tan \beta$, then show that

$$\tan(\alpha - \beta) = \frac{\sin 2\beta}{5 - \cos 2\beta}$$

TRANSFORMATION FORMULAE

91. Prove that:

$$(i) \frac{\sin 5A - \sin 3A}{\cos 5A + \cos 3A} = \tan A$$

$$(ii) \frac{\sin A + \sin 3A}{\cos A + \cos 3A} = \tan 2A$$

$$(iii) \frac{\sin A + \sin B}{\cos A + \cos B} = \tan\left(\frac{A+B}{2}\right)$$

92. Prove that:

$$(i) \sin 38^\circ + \sin 22^\circ = \sin 82^\circ$$

$$(ii) \sin 105^\circ + \cos 105^\circ = \cos 45^\circ$$

$$(iii) \cos 55^\circ + \cos 65^\circ + \cos 175^\circ = 0$$

$$(iv) \cos 20^\circ + \cos 100^\circ + \cos 140^\circ = 0$$

$$(v) \sin 50^\circ - \sin 70^\circ + \sin 10^\circ = 0$$

93. Prove that

$$\sin(47^\circ) + \cos(77^\circ) = \cos(17^\circ)$$

94. Prove that

$$\cos(80^\circ) + \cos(40^\circ) - \cos(20^\circ) = 0$$

95. Prove that

$$\begin{aligned} & \sin(10^\circ) + \sin(20^\circ) + \sin(40^\circ) + \sin(50^\circ) \\ & - \sin(70^\circ) - \sin(80^\circ) = 0 \end{aligned}$$

96. Prove that $(\cos \alpha + \cos \beta)^2 + (\sin \alpha + \sin \beta)^2$

$$= 4 \cos^2\left(\frac{\alpha - \beta}{2}\right)$$

97. Prove that $(\cos \alpha - \cos \beta)^2 + (\sin \alpha - \sin \beta)^2$

$$= 4 \sin^2\left(\frac{\alpha - \beta}{2}\right)$$

98. Prove that $(\sin \alpha - \sin \beta)^2 + (\cos \alpha - \cos \beta)^2$

$$= 4 \sin^2\left(\frac{\alpha - \beta}{2}\right)$$

99. Prove that $\cos 20^\circ \cdot \cos 40^\circ \cdot \cos 80^\circ = \frac{1}{8}$

100. Prove that $\sin 20^\circ \cdot \sin 40^\circ \cdot \sin 80^\circ = \frac{\sqrt{3}}{8}$

101. Prove that $\sin 10^\circ \cdot \sin 50^\circ \cdot \sin 60^\circ \cdot \sin 70^\circ = \frac{\sqrt{3}}{16}$

112. Prove that $\cos 10^\circ \cdot \cos 30^\circ \cdot \cos 50^\circ \cdot \cos 70^\circ = \frac{3}{16}$

113. Prove that:

$$(i) \frac{\sin A + \sin 3A + \sin 5A}{\cos A + \cos 3A + \cos 5A} = \tan 3A$$

$$(ii) \frac{\cos 4x + \cos 3x + \cos 12x}{\sin 4x + \sin 3x + \sin 2x} = \cot 3x$$

$$(iii) \frac{\sin A + \sin 3A + \sin 5A + \sin 7A}{\cos A + \cos 3A + \cos 5A + \cos 7A} = \tan 4A$$

$$(iv) \frac{\sin A + \sin 2A + \sin 4A + \sin 5A}{\cos A + \cos 2A + \cos 4A + \cos 5A} = \tan 3A$$

114. If $\sin A - \sin B = \frac{1}{2}$ and $\cos A - \cos B = \frac{1}{3}$, then find

$$\tan\left(\frac{A+B}{2}\right)$$

115. If $\cos A + \cos B = \frac{1}{2}$, $\sin A + \sin B = \frac{1}{4}$,

$$\text{then prove that } \tan\left(\frac{A+B}{2}\right) = \frac{1}{2}$$

116. If $\operatorname{cosec} A + \sec A = \operatorname{cosec} B + \sec B$, then

$$\text{prove that } \tan A \tan B = \cot\left(\frac{A+B}{2}\right)$$

117. If $\sin 2A = \lambda \sin 2B$, then prove that

$$\frac{\tan(A+B)}{\tan(A-B)} = \frac{\lambda+1}{\lambda-1}$$

118. Find the value of $\sqrt{3} \cot(20^\circ) - 4 \cos(20^\circ)$

119. If $\sin A + \sin B = a$ and $\cos A + \cos B = b$, then find $\cos(A+B)$
120. If $2 \cos A = x + \frac{1}{x}$, $2 \cos B = y + \frac{1}{y}$, then find $\cos(A-B)$ Ans. $\frac{1}{2} \left(xy + \frac{1}{xy} \right)$
121. Prove that $\sin(47^\circ) + \sin(61^\circ) - \sin(11^\circ) - \sin(25^\circ) = \cos(7^\circ)$.
122. If $\tan \alpha = \frac{m}{m+1}$, and $\tan \beta = \frac{1}{2m+1}$, then find $\tan(\alpha + \beta)$
123. Find the number of integral values of k for which $7 \cos x + 5 \sin x = 2k + 1$ has a solution.
124. Prove that $\cos \alpha + \cos \beta + \cos \gamma + \cos(\alpha + \beta + \gamma)$
 $= 4 \cos\left(\frac{\alpha + \beta}{2}\right) \cos\left(\frac{\beta + \gamma}{2}\right) \cos\left(\frac{\gamma + \alpha}{2}\right)$

MULTIPLE ANGLES

125. Prove that $\left(\frac{1 - \cos 2\theta}{\sin 2\theta} \right) = \tan \theta$
126. Prove that $\left(\frac{1 + \cos 2\theta}{\sin 2\theta} \right) = \cot \theta$
127. Prove that $\cot \theta - \tan \theta = 2 \cot(2\theta)$
128. Prove that $\tan \theta + 2 \tan(2\theta) + 4 \tan(4\theta) + 8 \cot 8\theta = \cot \theta$
129. If $\tan \theta = \frac{b}{a}$,
 prove that $a \cos(2\theta) + b \sin(2\theta) = a$
130. Prove that $\sqrt{3} \operatorname{cosec}(20^\circ) - \sec(20^\circ) = 4$
131. Prove that $\tan(9^\circ) - \tan(27^\circ) - \tan(63^\circ) + \tan(81^\circ) = 4$
132. Prove that $\left(\frac{\sec 8A - 1}{\sec 4A - 1} \right) = \frac{\tan 8A}{\tan 2A}$
133. Prove that
 (i) $\cos^2(\theta) + \cos^2\left(\frac{2\pi}{3} - \theta\right) + \cos^2\left(\frac{2\pi}{3} + \theta\right) = \frac{3}{2}$
 (ii) $\cos^2 \theta + \cos^2\left(\frac{\pi}{3} - \theta\right) + \cos^2\left(\frac{\pi}{3} + \theta\right) = \frac{3}{2}$
134. Prove that $\sin^2 \theta + \sin^2(120^\circ + \theta) + \sin^2(240^\circ + \theta) = \frac{3}{2}$
135. Prove that $4 \sin(\theta) \sin(60^\circ + \theta) \sin(60^\circ - \theta) = \sin 3\theta$
136. Prove that $\sin(20^\circ) \cdot \sin(40^\circ) \cdot \sin(80^\circ) = \frac{\sqrt{3}}{8}$

137. Prove that $4 \cos(\theta) \cdot \cos(60^\circ - \theta) \cdot \cos 60^\circ + \theta = \cos(3\theta)$
138. Prove that $\cos(10^\circ) \cdot \cos(50^\circ) \cdot \cos(70^\circ) = \frac{\sqrt{3}}{8}$
139. Prove that $\tan(\theta) + \tan(60^\circ + \theta) - \tan(60^\circ - \theta) = 3 \tan(3\theta)$
140. Prove that $\cos(\theta) \cos(2\theta) \cos(2^2\theta) \cos(2^3\theta) \dots \cos(2^{n-1}\theta) = \frac{\sin(2^n\theta)}{2^n \sin \theta}$
141. Prove that $\cos\left(\frac{2\pi}{7}\right) \cos\left(\frac{4\pi}{7}\right) \cos\left(\frac{8\pi}{7}\right) = \frac{1}{8}$
142. Prove that $\cos\left(\frac{2\pi}{7}\right) + \cos\left(\frac{4\pi}{7}\right) + \cos\left(\frac{6\pi}{7}\right) = -\frac{1}{2}$
143. If $M = \sqrt{a^2 \cos^2 \theta + b^2 \sin^2 \theta} + \sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}$
 such that maximum $(M^2) = m_1$ and minimum $(M^2) = m_2$,
 then find the value of $m_1 - m_2$.
144. Prove that $\tan 4\theta = \frac{4 \tan \theta - 4 \tan^3 \theta}{1 - 6 \tan^2 \theta + \tan^4 \theta}$
145. Prove that $\left(\frac{\sin x}{\cos 3x} + \frac{\sin 3x}{\cos 9x} + \frac{\sin 9x}{\cos 27x} \right) = \frac{1}{2} (\tan 27x - \tan x)$
146. Prove that $\tan\left(\frac{\theta}{2}\right)(1 + \sec \theta)(1 + \sec 2\theta)(1 + \sec 2^2\theta) \dots (1 + \sec(2^n\theta)) = \tan(2^n\theta)$
 Maximum or minimum values of $f(x) = \theta \cos x + b \sin x$
147. Find the maximum and minimum values of
 (i) $f(x) = 3 \sin x + 4 \cos x + 10$
 (ii) $f(x) = 3 \sin(100^\circ)x + 4 \cos(100^\circ)x + 10$
 (iii) $f(x) = 3 \sin x + 4$
 (iv) $f(x) = 2 \cos x + 5$
 (v) $f(x) = \sin x + \cos x$
 (vi) $f(x) = \sin x - \cos x$
 (vii) $f(x) = \sin(\sin x)$
 (viii) $f(x) = \cos(\cos x)$
 (ix) $f(x) = \sin(\sin x) + \cos(\sin x)$
 (x) $f(x) = \cos(\sin x) + \sin(\cos x)$
148. Find the range of $f(x) = \sin x + \cos x + 3$
149. Find the greatest and the least values of $2 \sin^2 \theta + 3 \cos^2 \theta$
150. Prove that $-4 < 5 \cos \theta + 3 \cos\left(\theta + \frac{\pi}{3}\right) + 3 < 10$

151. Find the maximum and minimum values of

$$\cos^2 \theta - 6 \sin \theta \cos \theta + 3 \sin^2 \theta + 2$$

152. Find the least value of

$$\operatorname{cosec}^2 x + 25 \sec^2 x$$

153. Find the ratio of the greatest value of

$$2 - \cos x + \sin^2 x$$
 to its least value.

154. If
- $y = 4 \sin^2 \theta - \cos 2\theta$
- , then
- y
- lies in the interval.....

155. If
- m
- is the minimum value of
- $f(x) = 3 \sin x + 5$
- and
- n
- is the maximum value of
- $g(x) = 3 - 2 \sin x$
- then find the value of
- $(m + n + 2)$
- .

156. Find the maximum and the minimum values of

$$f(x) = \sin^2 x + \cos^4 x.$$

157. Find the maximum and the minimum values of

$$f(x) = \cos^2 x + \sin^4 x.$$

158. Find the maximum and minimum values of

$$f(x) = \sin^4 x + \cos^4 x.$$

159. Find the maximum and minimum values of

$$f(x) = \sin^6 x + \cos^6 x.$$

160. If
- $A = \cos^2 \theta + \sin^4 \theta$
- and

$$B = \cos^4 \theta + \sin^2 \theta$$
 such that

$$m_1 = \text{Maximum of } A \text{ and } m_2 = \text{Minimum of } B$$

$$\text{then find the value of } m_1^2 + m_2^2 + m_1 m_2$$

161. Find the maximum and minimum values of

$$f(x) = (\sin x + \cot x + \operatorname{cosec} 2x)^3$$

$$\text{where } x \in \left(0, \frac{\pi}{2}\right)$$

162. Find the maximum and minimum values of

$$f(x) = \frac{5}{\sin^2 \theta - 6 \sin \theta \cos \theta + 3 \cos^2 \theta}$$

163. Find the minimum value of

$$f(x) = \frac{a^2}{\cos^2 x} + \frac{b^2}{\sin^2 x}, x \in \left(0, \frac{\pi}{2}\right)$$

164. Find the minimum value of

$$f(x) = \frac{x^2 \sin^2 x + 4}{x \sin x},$$

$$\text{where } x \in \left(0, \frac{\pi}{2}\right)$$

165. Find the maximum and minimum values of

$$f(x) = \log y + \log x, \text{ where } x > 1, y > 1$$

166. Find the minimum values of

$$f(x) = 2 \log_{10} x - \log_x (0.01), x > 1$$

167. Find the minimum value of

$$f(x, y, z) = \frac{(x^2 + 1)(y^2 + 1)(z^2 + 1)}{xyz}, x, y, z > 0$$

168. Find the minimum value of

$$f(x, y, z) = \frac{(x^3 + 2)(y^3 + 2)(z^3 + 2)}{xyz}, x, y, z > 0$$

169. Find the minimum value of

$$f(a, b, c, d) = \frac{(a^2 + 1)(b^2 + 1)(c^2 + 1)(d^2 + 1)}{abcd}$$

$$\text{where } a, b, c, d > 0$$

SUB-MULTIPLE ANGLES

170. Prove that

$$(i) \tan \frac{\theta}{2} + \cot \frac{\theta}{2} = 2 \operatorname{cosec} \theta$$

$$(ii) \operatorname{cosec} \theta - \cot \theta = \tan \frac{\theta}{2}$$

$$(iii) \frac{1 + \sin \theta - \cos \theta}{1 + \sin \theta + \cos \theta} = \tan \frac{\theta}{2}$$

$$(iv) \sec \theta + \tan \theta = \tan \left(\frac{\pi}{4} + \frac{\theta}{2} \right)$$

$$(v) (\cos A + \cos B)^2 + (\sin A + \sin B)^2 = 4 \cos^2 \left(\frac{A+B}{2} \right)$$

$$171. \text{ Prove that } \sin^2 (24^\circ) - \sin^2 (6^\circ) = \frac{(\sqrt{5}-1)}{8}$$

$$172. \text{ Prove that } \sin^2 (48^\circ) - \cos^2 (12^\circ) = \frac{(\sqrt{5}+1)}{8}$$

$$173. \text{ Prove that } \sin (12^\circ) \cdot \sin (48^\circ) \cdot \sin (54^\circ) = \frac{1}{8}$$

$$174. \text{ Prove that } \sin (6^\circ) \cdot \sin (42^\circ) \cdot \sin (66^\circ) \cdot \sin (78^\circ) = \frac{1}{16}$$

175. Prove that

$$4(\sin (24^\circ) + \cos (6^\circ)) = (1 + \sqrt{5})$$

176. Prove that

$$\tan (6^\circ) \cdot \tan (42^\circ) \cdot \tan (66^\circ) \cdot \tan (78^\circ) = 1$$

177. Prove that

$$\left(1 + \cos \frac{\pi}{8}\right) \left(1 + \cos \frac{3\pi}{8}\right) \left(1 + \cos \frac{5\pi}{8}\right) \left(1 + \cos \frac{7\pi}{8}\right) = \frac{1}{8}$$

178. Prove that

$$\sin^4 \left(\frac{\pi}{8} \right) + \sin^4 \left(\frac{3\pi}{8} \right) + \sin^4 \left(\frac{5\pi}{8} \right) + \sin^4 \left(\frac{7\pi}{8} \right) = \frac{3}{2}$$

179. Prove that

$$\cos^4 \left(\frac{\pi}{8} \right) + \cos^4 \left(\frac{3\pi}{8} \right) + \cos^4 \left(\frac{5\pi}{8} \right) + \cos^4 \left(\frac{7\pi}{8} \right) = \frac{3}{2}$$

$$180. \text{ Prove that } \tan 20^\circ \tan 80^\circ = \sqrt{3} \tan 50^\circ$$

$$181. \text{ Prove that } \tan (10^\circ) \tan (70^\circ) = \frac{1}{\sqrt{3}} \times \tan (40^\circ)$$

182. Prove that

$$\sin 55^\circ - \sin 19^\circ + \sin 53^\circ - \sin 17^\circ = \cos 1^\circ$$

$$183. \text{ Prove that } \cos \frac{2\pi}{7} \cdot \cos \frac{4\pi}{7} \cdot \cos \frac{6\pi}{7} = \frac{1}{8}$$

184. Prove that: $\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} = -\frac{1}{2}$
185. Find the value of $\tan \left(7\frac{1}{2}^\circ\right) + \cot \left(7\frac{1}{2}^\circ\right)$.
186. If $\cos \left(\frac{x}{2}\right) - \sqrt{3} \sin \left(\frac{x}{2}\right)$ takes its minimum value then find its x .
187. If α and β be two different roots $a \cos \theta + b \sin \theta = c$, then prove that $\sin (\alpha + \beta) = \frac{2ab}{a^2 + b^2}$.

CONDITIONAL IDENTITIES

188. If $A + B + C = \pi$, then prove that,
 $\sin 2A + \sin 2B + \sin 2C = 4 \sin A \sin B \sin C$
189. If $A + B + C = \pi$, then prove that,
 $\cos 2A + \cos 2B + \cos 2C = -1 - 4 \cos A \cos B \cos C$
190. If $A + B + C = \pi$, then prove that
 $\sin^2 A + \sin^2 B - \sin^2 C = 2 \sin A \sin B \cos C$
191. If $A + B + C = \pi$, then prove that,
 (i) $\sin^2 A + \sin^2 B + \sin^2 C = 2 + \cos A \cos B \cos C$
 (ii) $\cos^2 A + \cos^2 B + \cos^2 C = 2 + \sin A \sin B \sin C$
192. If $A + B + C = \pi$, then prove that

$$\frac{\sin 2A + \sin 2B + \sin 2C}{\cos A + \cos B + \cos C - 1}$$

$$= 8 \cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2}$$
193. In a $\triangle ABC$ prove that
 (i) $\tan A + \tan B + \tan C = \tan A \cdot \tan B \cdot \tan C$
 (ii) $\cot A \cdot \cot B + \cot B \cdot \cot C + \cot C \cdot \cot A = 1$.
194. If A, B, C and D be the angles of a quadrilateral, then prove that

$$\frac{\tan A + \tan B + \tan C + \tan D}{\cot A + \cot B + \cot C + \cot D}$$

$$= \tan A \cdot \tan B \cdot \tan C \cdot \tan D$$
194. In a $\triangle ABC$ prove that
 $(\cot A + \cot B)(\cot B + \cot C)$
 $(\cot C + \cot A) = \operatorname{cosec} A \operatorname{cosec} B \operatorname{cosec} C$.
195. If $xy + yz + zx = 1$, then prove that

$$\frac{x}{1-x^2} + \frac{y}{1-y^2} + \frac{z}{1-z^2} = \frac{4xyz}{(1-x^2)(1-y^2)(1-z^2)}$$
196. If $xy + yz + zx = 1$, then prove that,

$$\frac{x}{1+x^2} + \frac{y}{1+y^2} + \frac{z}{1+z^2} = \frac{2}{\sqrt{(1+x^2)(1+y^2)(1+z^2)}}$$
197. Prove that
 $\tan (\alpha - \beta) + \tan (\beta - \gamma) + \tan (\gamma - \alpha)$
 $= \tan (\alpha - \beta) \tan (\beta - \gamma) \tan (\gamma - \alpha)$
198. In a $\triangle ABC$, if $\cot A + \cot B + \cot C = \sqrt{3}$, then prove that the triangle is an equilateral.

199. If $x + y + z = xyz$, then prove that

$$\frac{3x - x^3}{1 - 3x^2} + \frac{3y - y^3}{1 - 3y^2} + \frac{3z - z^3}{1 - 3z^2}$$

$$= \frac{3x - x^3}{1 - 3x^2} \cdot \frac{3y - y^3}{1 - 3y^2} \cdot \frac{3z - z^3}{1 - 3z^2}$$

200. Prove that $1 + \cos 56^\circ + \cos 58^\circ - \cos 66^\circ$
 $= 4 \cos 28^\circ \cos 29^\circ \sin 33^\circ$.

TRIGONOMETRICAL SERIES

201. Prove that
 $\sin \alpha + \sin (\alpha + \beta) + \sin (\alpha + 2\beta)$
 $+ \sin (\alpha + 3\beta) + \dots + \sin (\alpha + (n-1)\beta)$
 $= \frac{\sin \left(\frac{n\beta}{2}\right)}{\sin \left(\frac{\beta}{2}\right)} \times \sin \left[\alpha + (n-1)\frac{\beta}{2}\right]$
202. Prove that
 $\sin \alpha + \sin 2\alpha + \sin 3\alpha + \dots + \sin n\alpha$
 $= \frac{\sin \left(\frac{n\alpha}{2}\right)}{\sin \left(\frac{\alpha}{2}\right)} \times \sin (n+1)\left(\frac{\alpha}{2}\right)$
203. Prove that
 $\sin \theta + \sin 3\theta + \dots + \sin (2n-1)\theta = \frac{\sin^2 n\theta}{\sin \theta}$
204. Prove that
 $\cos \alpha + \cos (\alpha + \beta) + \cos (\alpha + 2\beta)$
 $+ \cos (\alpha + 3\beta) + \dots + \cos (\alpha + (n-1)\beta)$
 $= \frac{\sin \left(\frac{n\beta}{2}\right)}{\sin \left(\frac{\beta}{2}\right)} \times \cos \left(\alpha + (n-1)\frac{\beta}{2}\right)$
205. Prove that $\cos \alpha + \cos 2\alpha + \cos 3\alpha + \dots$
 $\dots + \cos n\alpha = \frac{\sin \left(\frac{n\alpha}{2}\right)}{\sin \left(\frac{\alpha}{2}\right)} \times \cos \left(\frac{(n+1)\alpha}{2}\right)$
206. Find the sum of n -terms of the series

$$\frac{\sin x}{\sin 2x \cdot \sin 3x} + \frac{\sin x}{\sin 3x \cdot \sin 4x}$$

$$+ \frac{\sin x}{\sin 4x \cdot \sin 5x} + \dots \text{to } n \text{ terms}$$
207. Prove that $\frac{1}{\cos \theta + \cos 2\theta} + \frac{1}{\cos \theta + \cos 5\theta}$
 $+ \frac{1}{\cos \theta + \cos 7\theta} + \dots \text{to } n \text{ terms}$
 $= \operatorname{cosec} \theta [\tan (n+1) \theta - \tan \theta]$

Hints & Solutions**Level - 1**

1. Given $r = 4900$ km,
Circumference $= 2\pi r$
- $$= 2 \times \frac{22}{7} \times 4900$$
- $$= 44 \times 700$$
- $$= 30800 \text{ km}$$

2. Let the three angles be $3x$, $4x$ and $5x$, respectively
Thus, $3x + 4x + 5x = 180^\circ$
 $\Rightarrow 12x = 180^\circ$
 $\Rightarrow x = 15^\circ$
 Therefore, the smallest angle
 $= 3x = 3 \times 15^\circ = 45^\circ$
 and the greatest angle
 $= 5x = 5 \times 15^\circ = 75^\circ$
- $$= \left(75 \times \frac{\pi}{180} \right) \text{ radius}$$
- $$= \left(\frac{5\pi}{12} \right) \text{ radius}$$

3. Let the three angles be $a + d$, a , $a - d$
Thus, $a + d + a + a - d = 180^\circ$
 $\Rightarrow 3a = 180^\circ$
 $\Rightarrow a = \frac{180^\circ}{3} = 60^\circ$
 It is given that,

$$(a - d)^\circ : (a + d) \times \frac{\pi}{180} = \frac{60}{\pi}$$

$$\Rightarrow \frac{(a - d)}{(a + d)} \times \frac{180}{\pi} = \frac{60}{\pi}$$

$$\Rightarrow \frac{(a-d)}{(a+d)} = \frac{1}{3}$$

$$\Rightarrow a+d = 3a-3d$$

$$\Rightarrow 4d = 2a$$

$$\Rightarrow d = \frac{a}{2} = 30^\circ$$

Hence, the three angles are $90^\circ, 60^\circ, 30^\circ$.

4. Let the number of sides of the given polygons be $5x$ and $4x$ respectively.

It is given that,

$$\left(\frac{2 \times 5x - 4}{5x} - \frac{2 \times 4x - 4}{4x} \right) \times 90 = 9$$

$$\Rightarrow \left(\frac{10x - 4}{5x} - \frac{2x - 1}{x} \right) = \frac{1}{10}$$

$$\Rightarrow \left(\frac{10x - 4 - 10x + 5}{5x} \right) = \frac{1}{10}$$

$$\Rightarrow \left(\frac{1}{x} \right) = \frac{1}{2}$$

$$\Rightarrow x = 2$$

Hence, the number of sides of the polygons would be 10 and 8 respectively.

5. Let the angles of the quadrilateral be

$$a - 3d, a - d, a + d, a + 3d$$

It is given that, $a + 3d = 2(a - 3d)$

$$\Rightarrow a + 3d = 2a - 6d$$

$$\Rightarrow a = 9d$$

$$\text{Also, } a + 3d + a - d + a + d + a + 3d = 360$$

$$\Rightarrow 4a = 360$$

$$\Rightarrow a = 90$$

$$\text{and } d = 10$$

$$\text{Hence, the smallest angle} = 90^\circ - 30^\circ = 60^\circ$$

$$= \left(\frac{\pi}{3} \right)^c$$

6. Clearly, at half past 4, the hour hand will be at $4\frac{1}{2}$ and minute hand will be at 6.

In 1 hour angle made by the hour hand 30° .

In $4\frac{1}{2}$ hours angle made by the hour hand

$$= \frac{9}{2} \times 30^\circ = 135^\circ$$

In 1 minute angle made by the minute hand $= 6^\circ$

In 30° minutes, angle made by the minute hand $= 6 \times 30^\circ = 180^\circ$

Thus, the angle between the hour hand and the minute hand $= 180^\circ - 135^\circ$

$$= 45^\circ$$

7. Angle subtended at the centre

$$= 30^\circ = \left(30 \times \frac{\pi}{180} \right) = \frac{\pi}{6}$$

$$\text{Hence, } l = 10 \times \frac{\pi}{6} = \frac{5\pi}{3}$$

8. The angle traced by a minute hand in 60 minutes

$$= 360^\circ = 2\pi \text{ radians}$$

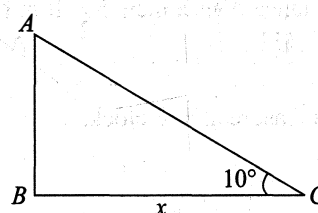
Thus the angle traced by minute hand in 18 minutes

$$= 2\pi \times \frac{18}{60} = \frac{3\pi}{5} \text{ radians}$$

Hence, the distance moved by the tip in 18 minutes

$$= l = 35 \times \frac{3\pi}{5} = 21 \times \frac{22}{7} = 66 \text{ cm}$$

9. Let AB be the height of the man and the required distance be x , where $BC = x$



$$\text{Therefore, } \frac{2}{x} \times \frac{180}{\pi} = \frac{10}{60}$$

$$\Rightarrow x = \frac{2}{10} \times \frac{180}{\pi} \times 60$$

$$\Rightarrow x = \frac{12 \times 180}{\pi}$$

$$\Rightarrow x = \frac{12 \times 180}{\frac{22}{7}} = \frac{12 \times 180 \times 7}{22}$$

$$\Rightarrow x = \frac{42 \times 180}{11} = 687.3$$

10. Let the required distance be x cm.

According to the question,

$$6^\circ = \frac{11}{2 \times x} \times \frac{180}{\pi}$$

$$\Rightarrow \frac{6}{60} = \frac{11}{2 \times x} \times \frac{180}{\pi}$$

$$\Rightarrow x = \frac{11}{2} \times \frac{180}{\pi} \times \frac{60}{6}$$

$$\Rightarrow x = \frac{11}{2} \times \frac{180 \times 7}{22} \times 10$$

$$\Rightarrow x = 45 \times 7 \times 10 = 3150$$

Hence, the required distance be 3150 cms.

11. Let the radius of the moon be x km

$$\text{It is given that, } \frac{16}{60} = \frac{2x}{60 \times 6400} \times \frac{180}{\pi}$$

$$\Rightarrow x = \frac{16 \times 6400 \times \pi}{180 \times 2}$$

$$\Rightarrow x = \frac{4 \times 640 \times \pi}{9}$$

$$\Rightarrow x = \frac{4 \times 640 \times 22}{9 \times 7}$$

$$\Rightarrow x = 894$$

Hence, the radius of the moon be 894 km.

12. The difference between the the acute angles of a right angled triangle is $\frac{2\pi}{3}$ radians. Express the angles in degrees. Ans. $81^\circ, 9^\circ$

13. The angles of a quadrilateral are in AP and the greatest angle is 120° . Find the angles in radians.

$$\text{Ans. } \frac{\pi}{3}, \frac{4\pi}{9}, \frac{5\pi}{9}, \frac{2\pi}{3}$$

14. At what distance does a man $5\frac{1}{2}$ ft in height, subtend an angle of 15° ? Ans. 14.32 miles

15. Find the angle between the hour hand and minute-hand in circular measure at 4 O'clock.

$$\text{Ans. } \frac{4\pi}{3}$$

16. Given $\sec \theta + \tan \theta = 3$... (i)

$$\Rightarrow (\sec \theta - \tan \theta) = \frac{1}{(\sec \theta + \tan \theta)} = \frac{1}{3} \quad \dots (ii)$$

Adding (i) and (ii) we get,

$$2 \sec \theta = 3 + \frac{1}{3} = \frac{10}{3}$$

$$\Rightarrow \sec \theta = \frac{5}{3}$$

$$\Rightarrow \cos \theta = \frac{3}{5}$$

17. Given, $\operatorname{cosec} \theta - \cot \theta = \frac{1}{5}$... (i)

$$\Rightarrow \operatorname{cosec} \theta - \cot \theta = \frac{1}{\operatorname{cosec} \theta + \cot \theta} = 5 \quad \dots (ii)$$

Adding (i) and (ii) we get,

$$2 \operatorname{cosec} \theta = 5 + \frac{1}{5} = \frac{26}{5}$$

$$\Rightarrow \operatorname{cosec} \theta = \frac{13}{5}$$

$$\Rightarrow \sin \theta = \frac{5}{13}$$

18. Given, $a = c \cos \theta + d \sin \theta$... (i)

$$\text{and } b = c \sin \theta - d \cos \theta \quad \dots (ii)$$

Squaring and adding (i) and (ii) we get,

$$a^2 + b^2 = (c \cos \theta + d \sin \theta)^2 + (c \sin \theta - d \cos \theta)^2$$

$$\Rightarrow a^2 + b^2 = (c^2 \cos^2 \theta + d^2 \sin^2 \theta) + (c^2 \sin^2 \theta + d^2 \cos^2 \theta)$$

$$\Rightarrow a^2 + b^2 = c^2 + d^2$$

$$\Rightarrow m = 2, n = 2, p = 2, q = 2$$

Hence, the value of $m + n + p + q + 42 = 50$

19. Let $x = 3 \cos \theta - 4 \sin \theta$... (i)

$$\text{and } 5 = 3 \sin \theta - 4 \cos \theta \quad \dots (ii)$$

Squaring and adding (i) and (ii) we get,

$$x^2 + 5^2 = (3 \cos \theta + 4 \sin \theta)^2 + (3 \sin \theta - 4 \cos \theta)^2$$

$$\Rightarrow x^2 + 5^2$$

$$= (9 \cos^2 \theta + 16 \sin^2 \theta + 24 \sin \theta \cos \theta)$$

$$+ (9 \sin^2 \theta + 16 \cos^2 \theta - 24 \sin \theta \cos \theta)$$

$$= (9 \cos^2 \theta + 16 \sin^2 \theta) + (9 \sin^2 \theta + 16 \cos^2 \theta)$$

$$= 9(\cos^2 \theta + \sin^2 \theta) + 16(\cos^2 \theta + \sin^2 \theta)$$

$$\Rightarrow x^2 + 25 = 25$$

$$\Rightarrow x^2 = 0$$

$$\Rightarrow x = 0$$

$$\Rightarrow 3 \cos \theta - 4 \sin \theta = 0$$

20. We have $x^2 + y^2 + z^2$

$$= (r \cos \theta \cos \phi)^2 + (r \cos \theta \sin \phi)^2 + (r \sin \theta)^2$$

$$\Rightarrow x^2 + y^2 + z^2$$

$$= (r^2 \cos^2 \theta \cos^2 \phi) + (r^2 \cos^2 \theta \sin^2 \phi) + (r^2 \sin^2 \theta)$$

$$\Rightarrow x^2 + y^2 + z^2$$

$$= r^2 \cos^2 \theta (\cos^2 \phi + \sin^2 \phi) + (r^2 \sin^2 \theta)$$

$$= r^2 \cos^2 \theta + r^2 \sin^2 \theta$$

$$= r^2 (\cos^2 \theta + \sin^2 \theta) = r^2$$

$$\Rightarrow x^2 + y^2 + z^2 = r^2$$

$$\Rightarrow m = 2, n = 2, p = 2$$

Thus, the value of $(m + n + p - 4)^{(m+n+p+4)}$

$$= 2^{10} = 1024$$

21. Given, $x = \frac{2 \sin \alpha}{1 + \cos \alpha + 3 \sin \alpha}$

$$\text{We have } \frac{\sin \alpha - 3 \cos \alpha + 3}{2 - 2 \cos \alpha}$$

$$= \frac{\sin \alpha + 3(1 - \cos \alpha)}{2(1 - \cos \alpha)}$$

$$= \frac{\sin \alpha}{2(1 - \cos \alpha)} + \frac{3}{2}$$

$$= \frac{\sin \alpha (1 + \cos \alpha)}{2(1 - \cos^2 \alpha)} + \frac{3}{2}$$

$$= \frac{\sin \alpha (1 + \cos \alpha)}{2 \sin^2 \alpha} + \frac{3}{2}$$

$$= \frac{(1 + \cos \alpha)}{2 \sin \alpha} + \frac{3}{2}$$

$$= \frac{(1 + \cos \alpha + 3 \sin \alpha)}{2 \sin \alpha}$$

$$= \frac{1}{x}$$

22. We have

$$P = \sec^6 \theta - \tan^6 \theta - 3 \sec^2 \theta \tan^2 \theta$$

$$= (\sec^2 \theta - \tan^2 \theta)^3 = 1$$

$$Q = \operatorname{cosec}^6 \theta - \cot^6 \theta - 3 \operatorname{cosec}^2 \theta \cot^2 \theta$$

$$= (\operatorname{cosec}^2 \theta - \cot^2 \theta)^3 = 1$$

and $R = \sin^6 \theta + \cos^6 \theta + 3 \sin^2 \theta \cos^2 \theta$

$$= (\sin^2 \theta + \cos^2 \theta)^3 = 1$$

Hence, the value of $(P + Q + R)^{(P+Q+R)}$

$$= 3^3 = 27$$

23. We have $3 \sin x + 4 \cos x = 5$

Let $y = 3 \cos x - 4 \sin x$

Now, $y^2 + 5^2 = (3 \cos x - 4 \sin x)^2$

$$+ (3 \sin x + 4 \cos x)^2$$

$$\Rightarrow y^2 + 25 = 9 \cos^2 x + 16 \sin^2 x - 24 \sin x \cos x$$

$$+ 9 \sin^2 x + 16 \cos^2 x + 24 \sin x \cos x$$

$$\Rightarrow y^2 + 25 = 25(\cos^2 x + \sin^2 x) = 25$$

$$\Rightarrow y^2 = 0$$

$$\Rightarrow y = 0$$

$$\Rightarrow 3 \cos x - 4 \sin x = 0$$

$$\Rightarrow 3 \cos x = 4 \sin x$$

$$\Rightarrow \tan x = 3/4$$

Hence, the value of $2 \sin x + \cos x + 4 \tan x$

$$= 2\left(\frac{3}{5}\right) + \left(\frac{4}{5}\right) + 4\left(\frac{3}{4}\right) = 2 + 3 = 5$$

24. Given, $\sin A + \sin B + \sin C = -3$

$$\Rightarrow \sin A = -1, \sin B = -1, \sin C = -1$$

$$\Rightarrow A = -\frac{\pi}{2}, B = -\frac{\pi}{2}, C = -\frac{\pi}{2}$$

Hence, the value of $\cos A + \cos B + \cos C + 10$

$$= 0 + 0 + 0 + 10 = 10.$$

25. We have $(1 + \sin \theta)(1 + \cos \theta) = \frac{5}{4}$

$$\Rightarrow 1 + \sin \theta + \cos \theta + \sin \theta \cos \theta = \frac{5}{4}$$

$$\Rightarrow 1 + t + \left(\frac{t^2 - 1}{2}\right) = \frac{5}{4} \quad (\sin \theta + \cos \theta = t, \text{ say})$$

$$\Rightarrow t + \left(\frac{t^2 - 1}{2}\right) = \frac{1}{4}$$

$$\Rightarrow t^2 + 2t - 1 = \frac{1}{2}$$

$$\Rightarrow 2t^2 + 4t - 3 = 0$$

$$\Rightarrow t = \frac{-4 \pm \sqrt{16 + 24}}{4}$$

$$= \frac{-4 \pm 2\sqrt{10}}{4} = -1 \pm \frac{1}{2}\sqrt{10}$$

$$\Rightarrow t = -1 + \frac{1}{2}\sqrt{10}$$

$$\Rightarrow \sin \theta + \cos \theta = -1 + \frac{1}{2}\sqrt{10}$$

Now, $(1 - \sin \theta)(1 - \cos \theta)$

$$= 1 - \sin \theta - \cos \theta + \sin \theta \cos \theta$$

$$= 1 - (\sin \theta + \cos \theta) + \sin \theta \cos \theta$$

$$= 1 - \left(-1 + \frac{\sqrt{10}}{2}\right) + \frac{1}{2}\left(\frac{10}{4} - \sqrt{10}\right)$$

$$= \left(2 + \frac{5}{4}\right) - \sqrt{10}$$

$$= \left(\frac{13}{4} - \sqrt{10}\right)$$

26. Given, $f(x) = \frac{9x^2 \sin^2 x + 4}{x \sin x}$

$$= 9x \sin x + \frac{4}{x \sin x}$$

Applying AM $\geq$ GM we get,

$$\left(\frac{9x \sin x + \frac{4}{x \sin x}}{2}\right) \geq \sqrt{9x \sin x \times \frac{4}{x \sin x}}$$

$$\Rightarrow \left(\frac{9x \sin x + \frac{4}{x \sin x}}{2}\right) \geq 6$$

$$\Rightarrow \left(9x \sin x + \frac{4}{x \sin x}\right) \geq 12$$

Hence, the minimum value of $f(x)$ is 12.

27. We have, $\cos \theta + \sin \theta = \sqrt{2} \cos \theta$

$$\Rightarrow \sin \theta = (\sqrt{2} - 1) \cos \theta$$

$$\Rightarrow \cos \theta = \frac{\sin \theta}{(\sqrt{2} - 1)}$$

$$\Rightarrow \cos \theta = (\sqrt{2} + 1) \sin \theta$$

$$\Rightarrow \cos \theta - \sin \theta = \sqrt{2} \sin \theta$$

28. We have, $\tan^2 \theta = 1 - e^2$

$$\Rightarrow 1 + \tan^2 \theta = 1 + 1 - e^2 = 2 - e^2$$

$$\Rightarrow \sec^2 \theta = 2 - e^2$$

$$\Rightarrow \sec \theta = \sqrt{2 - e^2}$$

Now, $\sec \theta + \tan^3 \theta \cdot \operatorname{cosec} \theta$

$$= \sec \theta + \frac{\sin^3 \theta}{\cos^3 \theta} \cdot \frac{1}{\sin \theta}$$

$$= \sec \theta + \frac{\sin^2 \theta}{\cos^3 \theta}$$

$$= \sec \theta + \frac{\sin^2 \theta}{\cos^2 \theta} \cdot \frac{1}{\cos \theta}$$

$$= \sec \theta + \tan^2 \theta \cdot \sec \theta$$

$$= \sec \theta (1 + \tan^2 \theta)$$

$$= \sec^3 \theta$$

$$= (2 - e^2)^{3/2}$$

29. Given, $\sin \theta + \sin^2 \theta + \sin^3 \theta = 1$

$$\begin{aligned} \Rightarrow (\sin \theta + \sin^3 \theta) &= 1 - \sin^2 \theta = \cos^2 \theta \\ \Rightarrow (\sin \theta + \sin^3 \theta)^2 &= (\cos^2 \theta)^2 \\ \Rightarrow (\sin \theta + \sin^3 \theta)^2 &= \cos^4 \theta \\ \Rightarrow (1 - \cos^2 \theta)(2 - \cos^2 \theta)^2 &= \cos^4 \theta \\ \Rightarrow (1 - \cos^2 \theta)(4 - 4\cos^2 \theta + \cos^4 \theta) &= \cos^4 \theta \\ \Rightarrow 4 - 4\cos^2 \theta + \cos^4 \theta - 4\cos^2 \theta + 4\cos^4 \theta & \\ - \cos^6 \theta &= \cos^4 \theta \\ \Rightarrow \cos^6 \theta - 4\cos^4 \theta + 8\cos^2 \theta &= 4 \end{aligned}$$

30. Given, $x = \frac{2 \sin \theta}{1 + \cos \theta + \sin \theta}$

$$\begin{aligned} &= \frac{2 \sin \theta}{(1 + \sin \theta) + \cos \theta} \\ &= \frac{2 \sin \theta((1 + \sin \theta) - \cos \theta)}{((1 + \sin \theta) + \cos \theta)((1 + \sin \theta) - \cos \theta)} \\ &= \frac{2 \sin \theta((1 + \sin \theta) - \cos \theta)}{((1 + \sin \theta)^2 - \cos^2 \theta)} \\ &= \frac{2 \sin \theta((1 + \sin \theta) - \cos \theta)}{(1 + \sin^2 \theta + 2 \sin \theta - \cos^2 \theta)} \\ &= \frac{2 \sin \theta((1 + \sin \theta) - \cos \theta)}{(\sin^2 \theta + 2 \sin \theta + (1 - \cos^2 \theta))} \\ &= \frac{2 \sin \theta((1 + \sin \theta) - \cos \theta)}{(2 \sin \theta + 2 \sin^2 \theta)} \\ &= \frac{2 \sin \theta((1 + \sin \theta) - \cos \theta)}{2 \sin \theta(1 + \sin \theta)} \\ &= \frac{((1 + \sin \theta) - \cos \theta)}{(1 + \sin \theta)} \\ &= \frac{(1 - \cos \theta + \sin \theta)}{(1 + \sin \theta)} \end{aligned}$$

31. We have $3(\sin x - \cos x)^4 + 6(\sin x + \cos x)^2 + 4(\sin^6 x + \cos^6 x)$

$$\begin{aligned} &= 3(\sin^4 x - 4\sin^3 x \cos x + 6\sin^2 x \cos^2 x \\ &\quad - 4\sin x \cos^3 x + \cos^4 x) \\ &\quad + 6(\sin^2 x + \cos^2 x + 2\sin x \cos x) \\ &\quad + 4\{(\sin^2 x)^3 + (\cos^2 x)^3\} \\ &= 3(\sin^4 x + \cos^4 x - 4\sin x \cos x \\ &\quad (\sin^2 x + \cos^2 x) + 6\sin^2 x \cos^2 x) \\ &\quad + 6(1 + 2\sin x \cos x) \\ &\quad + 4(\sin^2 x + \cos^2 x)^2 \\ &\quad - 12\sin^2 x \cos^2 x \\ &= 3 - 6\sin^2 x \cos^2 x - 12\sin x \cos x \\ &\quad + 18\sin^2 x \cos^2 x + 6 + 12\sin x \cos x \\ &\quad + 4 - 12\sin^2 x \cos^2 x \\ &= 3 + 6 + 4 \\ &= 13 \end{aligned}$$

32. We have $\sin x + \sin^2 x = 1$

$$\begin{aligned} \Rightarrow \sin x &= 1 - \sin^2 x = \cos^2 x \\ \text{Now, } \cos^8 x + 2\cos^6 x + \cos^4 x & \\ &= (\cos^4 x)^2 + 2 \cdot \cos^4 x \cdot \cos^2 x + (\cos^2 x)^2 \\ &= (\cos^4 x + \cos^2 x)^2 \\ &= (\sin^2 x + \sin x)^2 \\ &= (1)^2 = 1 \end{aligned}$$

33. We have $81^{\sin^2 \theta} + 81^{\cos^2 \theta} = 30$

$$\begin{aligned} \Rightarrow 81^{\sin^2 \theta} + 81^{1 - \sin^2 \theta} &= 30 \\ \Rightarrow 81^{\sin^2 \theta} + \frac{81}{81^{\sin^2 \theta}} &= 30 \\ \Rightarrow a + \frac{81}{a} = 30, a = 81^{\sin^2 \theta} & \\ \Rightarrow a^2 - 30a + 81 &= 0 \\ \Rightarrow (a - 27)(a - 3) &= 0 \\ \Rightarrow a = 3, 27 & \end{aligned}$$

When $a = 3$

$$\begin{aligned} \Rightarrow 81^{\sin^2 \theta} &= 3 \\ \Rightarrow 3^{4\sin^2 \theta} &= 3 \\ \Rightarrow 4\sin^2 \theta &= 1 \\ \Rightarrow \sin^2 \theta &= \left(\frac{1}{2}\right)^2 \\ \Rightarrow \sin^2 \theta &= \sin^2 \left(\frac{\pi}{6}\right) \\ \Rightarrow \theta &= \left(n\pi \pm \frac{\pi}{6}\right) \\ \Rightarrow \theta &= \frac{\pi}{6}, \frac{5\pi}{6} \end{aligned}$$

When $a = 27$

$$\begin{aligned} \Rightarrow 81^{\sin^2 \theta} &= 27 \\ \Rightarrow 3^{4\sin^2 \theta} &= 3^3 \\ \Rightarrow 4\sin^2 \theta &= 3 \\ \Rightarrow \sin^2 \theta &= \left(\frac{\sqrt{3}}{2}\right)^2 \\ \Rightarrow \sin^2 \theta &= \sin^2 \left(\frac{\pi}{3}\right) \\ \Rightarrow \theta &= \left(n\pi \pm \frac{\pi}{3}\right) \\ \Rightarrow \theta &= \frac{\pi}{3}, \frac{2\pi}{3} \end{aligned}$$

Hence, the values of θ are $\frac{\pi}{6}, \frac{\pi}{3}, \frac{2\pi}{3}, \frac{5\pi}{6}$.

34. We have $f_6(\theta) = \sin^6 \theta + \cos^6 \theta$

$$= 1 - 3\sin^2 \theta \cos^2 \theta$$

Also $f_4(\theta) = \sin^4 \theta + \cos^4 \theta$

$$= 1 - 2\sin^2 \theta \cos^2 \theta$$

$$\begin{aligned}
 \text{Now, } \frac{1}{6}f_6(\theta) - \frac{1}{4}f_4(\theta) &= \frac{1}{6}(1 - 3\sin^2\theta \cos^2\theta) - \frac{1}{4}(1 - 2\sin^2\theta \cos^2\theta) \\
 &= \frac{1}{6} - \frac{1}{2}\sin^2\theta \cos^2\theta - \frac{1}{4} + \frac{1}{2}\sin^2\theta \cos^2\theta \\
 &= \frac{1}{6} - \frac{1}{4} \\
 &= -\frac{1}{12}
 \end{aligned}$$

35. Given, $x \sin \alpha = y \cos \alpha$

$$\Rightarrow \frac{x}{\cos \alpha} = \frac{y}{\sin \alpha} = k$$

$$\text{Also, } x \sin^3 \alpha + y \cos^3 \alpha = \sin \alpha \cos \alpha$$

$$\Rightarrow k \sin^3 \alpha \cos \alpha + k \sin \alpha \cos^3 \alpha = \sin \alpha \cos \alpha$$

$$\Rightarrow k \sin \alpha \cos \alpha (\sin^2 \alpha + \cos^2 \alpha) = \sin \alpha \cos \alpha$$

$$\Rightarrow k \sin \alpha \cos \alpha = \sin \alpha \cos \alpha$$

$$\Rightarrow k = 1$$

$$\text{Thus, } x = k \cos \alpha = \cos \alpha, y = k \sin \alpha = \sin \alpha$$

$$\text{Now, } x^2 + y^2 = \cos^2 \alpha + \sin^2 \alpha = 1$$

36. Given, $\tan \theta + \sin \theta = m, \tan \theta - \sin \theta = n,$

$$\Rightarrow mn = \tan^2 \theta - \sin^2 \theta$$

$$= \sin^2 \theta \left(\frac{1 - \cos^2 \theta}{\cos^2 \theta} \right) = \frac{\sin^4 \theta}{\cos^2 \theta}$$

$$\text{Now, } m^2 - n^2$$

$$= (\tan \theta + \sin \theta)^2 - (\tan \theta - \sin \theta)^2$$

$$= 4 \tan \theta \sin \theta$$

$$= 4 \frac{\sin^2 \theta}{\cos \theta}$$

$$= 4\sqrt{mn}$$

37. Given, $\frac{\cos^4 x}{\cos^2 y} + \frac{\sin^4 x}{\sin^2 y} = 1$

$$\Rightarrow \frac{\cos^4 x}{\cos^2 y} + \frac{\sin^4 x}{\sin^2 y} = \cos^2 x + \sin^2 x$$

$$\Rightarrow \frac{\cos^4 x}{\cos^2 y} - \cos^2 x = \sin^2 x - \frac{\sin^4 x}{\sin^2 y}$$

$$\Rightarrow \cos^2 x \left(\frac{\cos^2 x}{\cos^2 y} - 1 \right) = \sin^2 x \left(1 - \frac{\sin^2 x}{\sin^2 y} \right)$$

$$\Rightarrow \frac{\cos^2 x}{\cos^2 y} (\cos^2 x - \cos^2 y)$$

$$= \frac{\sin^2 x}{\sin^2 y} (\sin^2 y - \sin^2 x)$$

$$= \frac{\sin^2 x}{\sin^2 y} (\cos^2 x - \cos^2 y)$$

$$\Rightarrow (\cos^2 x - \cos^2 y) \left(\frac{\cos^2 x}{\cos^2 y} - \frac{\sin^2 x}{\sin^2 y} \right) = 0$$

$$\Rightarrow (\cos^2 x - \cos^2 y) = 0, \left(\frac{\cos^2 x}{\cos^2 y} - \frac{\sin^2 x}{\sin^2 y} \right) = 0$$

$$\Rightarrow \cos^2 x = \cos^2 y, \sin^2 x = \sin^2 y$$

$$\text{Now, } \frac{\cos^4 y}{\cos^2 x} + \frac{\sin^4 y}{\sin^2 x}$$

$$= \frac{\cos^4 y}{\cos^2 y} + \frac{\sin^4 y}{\sin^2 y}$$

$$= \cos^2 y + \sin^2 y$$

$$= 1$$

38. We have $2f_6(\theta) - 3f_4(\theta) + 1$

$$= 2(\sin^6 \theta + \cos^6 \theta) - 3(\sin^4 \theta + \cos^4 \theta) + 1$$

$$= 2(1 - 3\sin^2 \theta \cos^2 \theta) - (1 - 2\sin^2 \theta \cos^2 \theta) + 1$$

$$= 2 - 3 + 1$$

$$= 0$$

39. We have $\frac{\sin A}{\sin B} = p, \frac{\cos A}{\cos B} = q$

$$\Rightarrow \frac{\sin A}{\cos A} \cdot \frac{\sin B}{\cos B} = \frac{p}{q}$$

$$\Rightarrow \frac{\tan A}{\tan B} = \frac{p}{q}$$

$$\Rightarrow \frac{\tan A}{p} = \frac{\tan B}{q} = \lambda$$

$$\text{Also, } \frac{\sin A}{\sin B} = p, \frac{\cos A}{\cos B} = q$$

$$\Rightarrow \frac{\sin A \cos A}{\sin B \cos B} = pq$$

$$\Rightarrow \frac{2 \sin A \cos A}{2 \sin B \cos B} = pq$$

$$\Rightarrow \frac{\sin 2A}{\sin 2B} = pq$$

$$\Rightarrow \frac{2 \tan A}{1 + \tan^2 A} \cdot \frac{2 \tan B}{1 + \tan^2 B} = pq$$

$$\Rightarrow \frac{2p\lambda}{1 + p^2\lambda^2} \cdot \frac{2q\lambda}{1 + q^2\lambda^2} = pq$$

$$\Rightarrow \frac{p}{(1 + p^2\lambda^2)} \times \frac{(1 + q^2\lambda^2)}{q} = pq$$

$$\Rightarrow \frac{(1 + q^2\lambda^2)}{(1 + p^2\lambda^2)} = q^2$$

$$\Rightarrow (1 + q^2\lambda^2) = q^2(1 + p^2\lambda^2)$$

$$\Rightarrow \lambda^2(1-p^2)q^2 = q^2 - 1$$

$$\Rightarrow \lambda^2 = \frac{(q^2 - 1)}{(1 - p^2)q^2}$$

$$\Rightarrow \lambda = \pm \frac{1}{q} \sqrt{\frac{(q^2 - 1)}{(1 - p^2)}}$$

$$\text{Therefore, } \tan A = \pm \frac{p}{q} \sqrt{\frac{(q^2 - 1)}{(1 - p^2)}}$$

$$\text{and } \tan B = \pm \sqrt{\frac{(q^2 - 1)}{(1 - p^2)}}$$

$$40. \text{ We have } \frac{\sin^4 \alpha}{a} + \frac{\cos^4 \alpha}{b} = \frac{1}{a+b}$$

$$\Rightarrow \left(\frac{a+b}{a}\right) \sin^4 \alpha + \left(\frac{a+b}{b}\right) \cos^4 \alpha = 1$$

$$\Rightarrow \left(1 + \frac{b}{a}\right) \sin^4 \alpha + \left(1 + \frac{a}{b}\right) \cos^4 \alpha = 1$$

$$\Rightarrow \left(\frac{b}{a} \sin^4 \alpha + \frac{a}{b} \cos^4 \alpha\right) + (\sin^4 \alpha + \cos^4 \alpha) = 1$$

$$\Rightarrow \left(\frac{b}{a} \sin^4 \alpha + \frac{a}{b} \cos^4 \alpha\right) + (1 - 2 \sin^2 \alpha \cdot \cos^2 \alpha) = 1$$

$$\Rightarrow \left(\frac{b}{a} \sin^4 \alpha + \frac{a}{b} \cos^4 \alpha - 2 \sin^2 \alpha \cdot \cos^2 \alpha\right) = 0$$

$$\Rightarrow \left(\sqrt{\frac{b}{a}} \sin^2 \alpha\right)^2 + \left(\sqrt{\frac{a}{b}} \cos^2 \alpha\right)^2$$

$$- 2 \sqrt{\frac{b}{a}} \sin^2 \alpha \cdot \sqrt{\frac{a}{b}} \cos^2 \alpha = 0$$

$$\Rightarrow \left(\sqrt{\frac{b}{a}} \sin^2 \alpha - \sqrt{\frac{a}{b}} \cos^2 \alpha\right)^2 = 0$$

$$\Rightarrow \left(\sqrt{\frac{b}{a}} \sin^2 \alpha - \sqrt{\frac{a}{b}} \cos^2 \alpha\right) = 0$$

$$\Rightarrow \sqrt{\frac{b}{a}} \sin^2 \alpha = \sqrt{\frac{a}{b}} \cos^2 \alpha$$

$$\Rightarrow \frac{\sin^2 \alpha}{a} = \frac{\cos^2 \alpha}{b} = \frac{1}{a+b}$$

$$\Rightarrow \sin^2 \alpha = \frac{a}{a+b}, \cos^2 \alpha = \frac{b}{a+b}$$

$$\text{Now, } \frac{\sin^8 \alpha}{a^3} + \frac{\cos^8 \alpha}{b^3}$$

$$\begin{aligned} &= \frac{(\sin^2 \alpha)^4}{a^3} + \frac{(\cos^2 \alpha)^4}{b^3} \\ &= \frac{\left(\frac{a}{a+b}\right)^4}{a^3} + \frac{\left(\frac{b}{a+b}\right)^4}{b^3} \\ &= \frac{a^4}{a^3(a+b)^4} + \frac{b^4}{b^3(a+b)^4} \\ &= \frac{a}{(a+b)^4} + \frac{b}{(a+b)^4} \\ &= \frac{a+b}{(a+b)^4} \\ &= \frac{1}{(a+b)^3} \end{aligned}$$

41.

$$(i) \sin(120^\circ) = \sin(90 \times 1 + 30^\circ)$$

$$= \cos(30^\circ) = \frac{\sqrt{3}}{2}$$

$$(ii) \sin(150^\circ) = \sin(90 \times 2 - 30^\circ)$$

$$= \sin(30^\circ) = \frac{1}{2}$$

$$(iii) \sin(210^\circ) = \sin(90 \times 2 + 30^\circ)$$

$$= -\sin(30^\circ) = -\frac{1}{2}$$

$$(iv) \sin(225^\circ) = \sin(90 \times 2 + 45^\circ)$$

$$= -\sin(45^\circ) = -\frac{1}{\sqrt{2}}$$

$$(v) \sin(300^\circ) = \sin(90 \times 3 + 30^\circ)$$

$$= -\cos(30^\circ) = -\frac{\sqrt{3}}{2}$$

$$(vi) \sin(330^\circ) = \sin(90 \times 3 + 60^\circ)$$

$$= -\cos(60^\circ) = -\frac{1}{2}$$

$$(vii) \sin(405^\circ) = \sin(90 \times 4 + 45^\circ)$$

$$= \sin(45^\circ) = \frac{1}{\sqrt{2}}$$

$$(viii) \sin(660^\circ) = \sin(90 \times 7 + 30^\circ)$$

$$= \sin(30^\circ) = \frac{1}{2}$$

$$(ix) \sin(1500^\circ) = \sin(90 \times 16 + 60^\circ)$$

$$= \sin(60^\circ) = \frac{\sqrt{3}}{2}$$

$$\begin{aligned} \text{(x)} \quad \sin(2013^\circ) &= \sin(90 \times 22 + 33^\circ) \\ &= -\sin(33^\circ) \end{aligned}$$

42. We have

$$\begin{aligned} &\cos(1^\circ) \cos(2^\circ) \cos(3^\circ) \dots \cos(189^\circ) \\ &= \cos(1^\circ) \cos(2^\circ) \cos(3^\circ) \dots \cos(89^\circ) \\ &\cos(90^\circ) \cos(91^\circ) \dots \cos(189^\circ) \\ &= \cos(1^\circ) \cos(2^\circ) \cos(3^\circ) \dots \cos(89^\circ) \\ &\quad \times 0 \times \cos(91^\circ) \dots \cos(189^\circ) \\ &= 0 \end{aligned}$$

43. We have

$$\begin{aligned} &\tan(1^\circ) \tan(2^\circ) \tan(3^\circ) \dots \tan(89^\circ) \\ &= \tan(1^\circ) \tan(2^\circ) \tan(3^\circ) \dots \tan(44^\circ) \\ &\tan(45^\circ) \tan(46^\circ) \dots \tan(87^\circ) \tan(88^\circ) \tan(89^\circ) \\ &= \{\tan(1^\circ) \times \tan(89^\circ)\} \cdot \{\tan(2^\circ) \times \tan(88^\circ)\} \\ &\quad \dots \{\tan(44^\circ) \times \tan(46^\circ)\}, \tan(45^\circ) \\ &= 1 \end{aligned}$$

44. We have

$$\begin{aligned} &\tan 35^\circ \cdot \tan 40^\circ \cdot \tan 45^\circ \cdot \tan 50^\circ \cdot \tan 55^\circ \\ &= \{\tan 35^\circ \times \tan 55^\circ\} \{\tan 40^\circ \times \tan 50^\circ\} \cdot \tan 45^\circ \\ &= \{\tan 35^\circ \times \cot 35^\circ\} \cdot \{\tan 40^\circ \times \cot 40^\circ\} \times \tan 45^\circ \\ &= 1 \end{aligned}$$

45. We have $\sin(10^\circ) + \sin(20^\circ) + \sin(30^\circ)$

$$\begin{aligned} &+ \sin(40^\circ) + \dots + \sin(360^\circ) \\ &= \sin(10^\circ) + \sin(20^\circ) + \sin(30^\circ) \\ &\quad + \sin(40^\circ) + \dots + \sin(150^\circ) \\ &\quad + \sin(340^\circ) + \sin(350^\circ) + \sin(360^\circ) \\ &= \sin(10^\circ) + \sin(20^\circ) + \sin(30^\circ) \\ &\quad + \sin(40^\circ) + \dots + \sin(80^\circ) \\ &\quad + \sin(90^\circ) + \sin(100^\circ) \\ &\quad + \sin(360^\circ - 40^\circ) + \sin(360^\circ - 30^\circ) \\ &\quad + \sin(360^\circ - 20^\circ) + \sin(360^\circ - 10^\circ) \\ &\quad + \sin(360^\circ) \\ &= \sin(10^\circ) + \sin(20^\circ) + \sin(30^\circ) \\ &\quad + \sin(40^\circ) + \dots + \sin(80^\circ) \\ &\quad + \sin(90^\circ) + \sin(100^\circ) \\ &\quad - \sin(40^\circ) - \sin(30^\circ) \\ &\quad - \sin(20^\circ) - \sin(10^\circ) + \sin(180^\circ) \\ &= 0 \end{aligned}$$

46. We have $\cos(10^\circ) + \cos(20^\circ) + \cos(30^\circ)$

$$\begin{aligned} &+ \cos(40^\circ) + \dots + \cos(360^\circ) \\ &= \cos 20^\circ + \cos 30^\circ + \cos 40^\circ + \dots + \\ &\cos 140^\circ + \cos 150^\circ + \cos 160^\circ + \cos 170^\circ \\ &\quad + \cos 180^\circ + (\cos 190^\circ + \cos 200^\circ + \\ &\cos 210^\circ + \cos 220^\circ + \dots + \cos 360^\circ) \\ &= \cos 10^\circ + \cos 20^\circ + \cos 30^\circ + \cos 40^\circ \\ &\quad + \dots - \cos 40^\circ - \cos 50^\circ - \cos 60^\circ \\ &\quad - \cos 70^\circ + \cos 180^\circ + (\cos 190^\circ \end{aligned}$$

$$\begin{aligned} &+ \cos 200^\circ + \cos 210^\circ + \cos 220^\circ \\ &+ \dots + \cos 360^\circ) \\ &= \cos 180^\circ + \cos 360^\circ \\ &= -1 + 1 \\ &= 0 \end{aligned}$$

47. We have

$$\begin{aligned} &\sin^2 5^\circ + \sin^2 10^\circ + \sin^2 15^\circ + \dots + \sin^2 90^\circ \\ &= \sin^2 5^\circ + \sin^2 10^\circ + \sin^2 15^\circ + \dots + \sin^2 40^\circ + \sin^2 45^\circ \\ &\quad + \sin^2 50^\circ + \sin^2 80^\circ + \sin^2 85^\circ + \sin^2 90^\circ \\ &= (\sin^2 5^\circ + \sin^2 85^\circ) \\ &\quad + (\sin^2 10^\circ + \sin^2 80^\circ) \\ &\quad + (\sin^2 15^\circ + \sin^2 75^\circ) \\ &\quad + \dots + (\sin^2 40^\circ + \sin^2 50^\circ) \\ &\quad + (\sin^2 45^\circ + \sin^2 90^\circ) \\ &= (\sin^2 5^\circ + \cos^2 5^\circ) \\ &\quad + (\sin^2 10^\circ + \cos^2 10^\circ) \\ &\quad + (\sin^2 15^\circ + \cos^2 15^\circ) + \dots \\ &\quad \dots + (\sin^2 40^\circ + \cos^2 40^\circ) \\ &\quad + (\sin^2 45^\circ + \sin^2 90^\circ) \\ &= (1 + 1 + \dots 8 \text{ times}) + \left(\frac{1}{2} + 1\right) \\ &= \left(8 + 1 + \frac{1}{2}\right) \\ &= 9\frac{1}{2} \end{aligned}$$

48. We have $\sin^2\left(\frac{\pi}{18}\right) + \sin^2\left(\frac{\pi}{9}\right)$

$$\begin{aligned} &+ \sin^2\left(\frac{4\pi}{9}\right) + \sin^2\left(\frac{7\pi}{18}\right) \\ &= \sin^2\left(\frac{\pi}{18}\right) + \sin^2\left(\frac{2\pi}{18}\right) \\ &\quad + \sin^2\left(\frac{8\pi}{18}\right) + \sin^2\left(\frac{7\pi}{18}\right) \\ &= \sin^2\left(\frac{\pi}{18}\right) + \sin^2\left(\frac{\pi}{2} - \frac{7\pi}{18}\right) \\ &\quad + \sin^2\left(\frac{\pi}{2} - \frac{\pi}{18}\right) + \sin^2\left(\frac{7\pi}{18}\right) \\ &= \sin^2\left(\frac{\pi}{18}\right) + \cos^2\left(\frac{7\pi}{18}\right) \\ &\quad + \cos^2\left(\frac{\pi}{18}\right) + \sin^2\left(\frac{7\pi}{18}\right) \\ &= \left\{\sin^2\left(\frac{\pi}{18}\right) + \cos^2\left(\frac{\pi}{18}\right)\right\} \\ &\quad + \left\{\cos^2\left(\frac{7\pi}{18}\right) + \sin^2\left(\frac{7\pi}{18}\right)\right\} \\ &= 1 + 1 \\ &= 2 \end{aligned}$$

49. We have

$$\begin{aligned} & \tan(20^\circ) \tan(25^\circ) \tan(45^\circ) \tan(65^\circ) \tan(70^\circ) \\ &= \tan(20^\circ) \tan(25^\circ) \tan(45^\circ) \tan(90^\circ - 25^\circ) \tan(90^\circ - 20^\circ) \\ &= \tan(20^\circ) \tan(25^\circ) \tan(45^\circ) \cot(25^\circ) \cot(20^\circ) \\ &= \tan(45^\circ) \\ &= 1 \end{aligned}$$

50. Given, $\sin(\theta_1) + \sin(\theta_2) + \sin(\theta_3) = 3$

It is possible only when each term of the above equation will provide the maximum value

Thus, $\sin(\theta_1) = 1$, $\sin(\theta_2) = 1$, $\sin(\theta_3) = 1$

$$\Rightarrow \theta_1 = \frac{\pi}{2}, \theta_2 = \frac{\pi}{2}, \theta_3 = \frac{\pi}{2}$$

Hence, the value of

$$\begin{aligned} & \cos(\theta_1) + \cos(\theta_2) + \cos(\theta_3) \\ &= \cos\left(\frac{\pi}{2}\right) + \cos\left(\frac{\pi}{2}\right) + \cos\left(\frac{\pi}{2}\right) \\ &= 0 \end{aligned}$$

51. We have $\sin^2 6^\circ + \sin^2 12^\circ + \dots + \sin^2 90^\circ$

$$\begin{aligned} &= (\sin^2 6^\circ + \sin^2 84^\circ) + (\sin^2 12^\circ + \sin^2 78^\circ) + \dots \\ &\quad + (\sin^2 42^\circ + \sin^2 48^\circ) + \sin^2 90^\circ \\ &= 7 \times 1 + 1 \\ &= 8 \end{aligned}$$

52. We have $\sin^2 10^\circ + \sin^2 20^\circ + \dots + \sin^2 90^\circ$

$$\begin{aligned} &= (\sin^2 10^\circ + \sin^2 80^\circ) + (\sin^2 20^\circ + \sin^2 70^\circ) + \dots \\ &\quad + (\sin^2 40^\circ + \sin^2 50^\circ) + \sin^2 90^\circ \\ &= 4 \times 1 + 1 \\ &= 5 \end{aligned}$$

53. We have $\sin^2 9^\circ + \sin^2 18^\circ + \dots + \sin^2 90^\circ$

$$\begin{aligned} &= (\sin^2 9^\circ + \sin^2 81^\circ) + (\sin^2 18^\circ + \sin^2 72^\circ) + \dots \\ &\quad + (\sin^2 36^\circ + \sin^2 54^\circ) + \sin^2 45^\circ + \sin^2 90^\circ \\ &= 4 \times 1 + \frac{1}{2} + 1 \\ &= 5\frac{1}{2} \end{aligned}$$

54. We have $\tan 1^\circ \cdot \tan 2^\circ \cdot \tan 3^\circ \dots \tan 89^\circ$

$$\begin{aligned} &= (\tan 1^\circ \cdot \tan 89^\circ) (\tan 2^\circ \cdot \tan 88^\circ) (\tan 3^\circ \cdot \tan 87^\circ) \dots (\tan 44^\circ \cdot \tan 46^\circ) \tan 45^\circ \\ &= 1.1.1 \dots 1 \\ &= 1 \end{aligned}$$

55. We have $\cos 1^\circ \cdot \cos 2^\circ \cdot \cos 3^\circ \dots \cos 189^\circ$

$$\begin{aligned} &= \cos 1^\circ \cdot \cos 2^\circ \cdot \cos 3^\circ \dots \cos 89^\circ \cos 90^\circ \dots \cos 180^\circ \\ &= 0 \end{aligned}$$

56. Given equation is $2 \sin^2 \theta + 2 \cos \theta = 0$

$$\Rightarrow 2 - 2 \cos^2 \theta + 2 \cos \theta = 0$$

$$\Rightarrow 2 \cos^2 \theta - 3 \cos \theta - 2 = 0$$

$$\Rightarrow 2 \cos^2 \theta - 4 \cos \theta + \cos \theta - 2 = 0$$

$$\Rightarrow 2 \cos \theta (\cos \theta - 2) + 1(\cos \theta - 2) = 0$$

$$\Rightarrow (2 \cos \theta + 1)(\cos \theta - 2) = 0$$

$$\Rightarrow \cos \theta = -\frac{1}{2}, 2$$

$$\Rightarrow \theta = 120^\circ, 240^\circ$$

57. Given equation is $\cos \theta + \sqrt{3} \sin \theta = 2$

$$\Rightarrow \frac{1}{2} \cos \theta + \frac{\sqrt{3}}{2} \sin \theta = 1$$

$$\Rightarrow \cos\left(\theta - \frac{\pi}{3}\right) = 1$$

$$\Rightarrow \left(\theta - \frac{\pi}{3}\right) = 0$$

$$\Rightarrow \theta = \frac{\pi}{3}$$

58. Given, $4n\alpha = \pi$

$$2n\alpha = \frac{\pi}{2}$$

Now, $\tan \alpha \tan \alpha \cdot \tan \alpha \cdot \tan 3\alpha \tan 3\alpha \tan (2n-1)\alpha$

$$= \tan \alpha \tan \alpha (2n-1)\alpha (\tan 2\alpha \tan 2(2n-2)\alpha) \cdot (\tan 3\alpha \cdot \tan (2n-3)\alpha) \dots$$

$$= (\tan \alpha \cdot \tan (2n\alpha - \alpha)) (\tan 2\alpha \cdot \tan (2n\alpha - 2\alpha)) (\tan 3\alpha \cdot \tan (2n\alpha - 3\alpha)) \dots$$

$$= \left(\tan \alpha \cdot \tan\left(\frac{\pi}{2} - \alpha\right)\right) \cdot \left(\tan 2\alpha \cdot \tan\left(\frac{\pi}{2} - 2\alpha\right)\right)$$

$$\cdot \left(\tan 3\alpha \cdot \tan\left(\frac{\pi}{2} - 3\alpha\right)\right) \dots$$

$$= (\tan \alpha \times \cot \alpha) \cdot (\tan 2\alpha \cdot \cot 2\alpha) \cdot (\tan 3\alpha \cdot \cot 3\alpha) = 1$$

59. We have

$$\cos(18^\circ) + \cos(234^\circ) + \cos(162^\circ) + \cos(306^\circ)$$

$$=$$

$$= \cos(18^\circ) - \cos(54^\circ) - \cos(18^\circ) + \cos(54^\circ)$$

$$= 0$$

60. We have

$$\cos(20^\circ) + \cos(40^\circ) + \cos(60^\circ) + \dots + \cos(180^\circ)$$

$$= \cos(20^\circ) + \cos(160^\circ) + \cos(40^\circ) + \cos(140^\circ)$$

$$+ \cos(60^\circ) + \cos(120^\circ) + \cos(80^\circ)$$

$$+ \cos(100^\circ) + \cos(180^\circ)$$

$$= \cos(180^\circ) = -1$$

61. We have

$$\sin(20^\circ) + \sin(40^\circ) + \sin(60^\circ) + \dots + \sin(360^\circ)$$

$$= \sin(20^\circ) + \sin(340^\circ) + \sin(40^\circ)$$

$$+ \sin(320^\circ) + \dots + \sin(180^\circ) + \sin(360^\circ)$$

$$= 0$$

63.

$$(i) \text{ Given equation is } \sin x = \frac{1}{2}$$

As we know the period of $\sin x$ is 2π
So, there is two solutions in $[0, 2\pi]$

(ii) Given equation is $\cos x = \frac{\sqrt{3}}{2}$

As we know the period of $\cos x$ is 2π

For each 2π , there is 2 solutions

So, it has 3 solutions

(iii) Given equation is $4 \sin^2 x - 1 = 0$

$$\Rightarrow \sin^2 x = \frac{1}{4}$$

The period of $\sin^2 x$ is π .

For each π , there is two solutions.

So, it has 6 solutions.

(iv) Given equation is $\sin^2 x - 3 \sin x + 2 = 0$

$$\Rightarrow (\sin x - 1)(\sin x - 2) = 0$$

$$\Rightarrow \sin x = 1, 2$$

$$\Rightarrow \sin x = 1$$

For each 2π , there is 2 solutions.

So, it has 3 solutions.

(v) Given equation is

$$\cos^2 x - \cos x - 2 = 0$$

$$\Rightarrow \cos^2 x - 2 \cos x + \cos x - 2 = 0$$

$$\Rightarrow \cos x(\cos x - 2) + 1(\cos x - 2) = 0$$

$$\Rightarrow (\cos x + 1)(\cos x - 2) = 0$$

$$\Rightarrow \cos x = -1, 2$$

$$\Rightarrow \cos x = -1$$

For each 2π , there are 2 solutions.

So, it has 3 solutions.

64. We have,

$$\begin{aligned} \text{(i)} \quad \sin(15^\circ) &= \sin(45^\circ - 30^\circ) \\ &= \sin(45^\circ) \cos(30^\circ) - \cos(45^\circ) \sin(30^\circ) \\ &= \frac{1}{2} \cdot \frac{\sqrt{3}}{2} - \frac{1}{2} \cdot \frac{1}{2} \\ &= \frac{\sqrt{3} - 1}{2\sqrt{2}} \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad \cos(15^\circ) &= \cos(45^\circ - 30^\circ) \\ &= \cos(45^\circ) \cos(30^\circ) + \sin(45^\circ) \sin(30^\circ) \\ &= \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \cdot \frac{1}{2} \\ &= \frac{\sqrt{3} + 1}{2\sqrt{2}} \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad \tan(15^\circ) &= \tan(45^\circ - 30^\circ) \\ &= \frac{\tan 45^\circ - \tan 30^\circ}{1 + \tan 45^\circ \cdot \tan 30^\circ} \\ &= \frac{1 - \frac{1}{\sqrt{3}}}{1 + \frac{1}{\sqrt{3}}} \end{aligned}$$

$$\begin{aligned} &= \frac{\sqrt{3} - 1}{\sqrt{3} + 1} \\ &= \frac{(\sqrt{3} - 1)^2}{3 - 1} \\ &= \frac{3 + 1 - 2\sqrt{3}}{2} \\ &= \frac{4 - 2\sqrt{3}}{2} \\ &= 2 - \sqrt{3} \end{aligned}$$

Note:

$$\begin{aligned} \text{(i)} \quad \cot(15^\circ) &= \frac{1}{\tan(15^\circ)} = \frac{1}{2 - \sqrt{3}} = 2 + \sqrt{3} \\ \text{(ii)} \quad \tan(105^\circ) &= -\cot(15^\circ) = -(2 + \sqrt{3}) \\ \text{(iii)} \quad \cot(105^\circ) &= -\tan(15^\circ) = -(2 - \sqrt{3}) \\ &= \sqrt{3} - 2 \end{aligned}$$

65. We have $\tan(75^\circ) + \cot(75^\circ)$

$$= \cot(15^\circ) + \tan(15^\circ)$$

$$= (2 + \sqrt{3}) + (2 - \sqrt{3})$$

$$= 4$$

66. We have $\cos(9^\circ) + \sin(9^\circ)$

$$= \sqrt{2} \left(\frac{1}{\sqrt{2}} \cos(9^\circ) + \frac{1}{\sqrt{2}} \sin(9^\circ) \right)$$

$$= \sqrt{2} (\sin(45^\circ) \cos(9^\circ) + \cos(45^\circ) \sin(9^\circ))$$

$$= \sqrt{2} (\sin(45^\circ + 9^\circ))$$

$$= \sqrt{2} \sin(54^\circ)$$

67. We have $\tan(70^\circ) = \tan(50^\circ + 20^\circ)$

$$\Rightarrow \tan(70^\circ) = \frac{\tan 50^\circ + \tan 20^\circ}{1 - \tan 50^\circ \tan 20^\circ}$$

$$\Rightarrow \tan(70^\circ) - \tan(70^\circ) \cdot \tan(50^\circ) \cdot \tan(20^\circ) = \tan 50^\circ + \tan 20^\circ$$

$$\Rightarrow \tan(70^\circ) - \tan(70^\circ) \tan(50^\circ) \cdot \cot(70^\circ) = \tan 50^\circ + \tan 20^\circ$$

$$\Rightarrow \tan(70^\circ) - \tan(50^\circ) = \tan 50^\circ + \tan 20^\circ$$

$$\Rightarrow \tan(70^\circ) = 2 \tan 50^\circ + \tan 20^\circ$$

68. We have $\frac{\cos 20^\circ - \sin 20^\circ}{\cos 20^\circ + \sin 20^\circ}$

$$= \frac{1 - \tan 20^\circ}{1 + \tan 20^\circ}$$

$$= \frac{\tan 45^\circ - \tan 20^\circ}{1 + \tan 45^\circ \tan 20^\circ}$$

$$= \tan(45^\circ - 20^\circ)$$

$$= \tan(25^\circ)$$

69. We have $\frac{\cos 7^\circ + \sin 7^\circ}{\cos 7^\circ - \sin 7^\circ} = \frac{1 + \tan(7^\circ)}{1 - \tan(7^\circ)}$

$$= \tan(45^\circ + 7^\circ)$$

$$= \tan(52^\circ)$$

70. We have $\tan(45^\circ)$

$$\Rightarrow \tan(25^\circ + 20^\circ) = 1$$

$$\Rightarrow \frac{\tan 25^\circ + \tan 20^\circ}{1 - \tan 25^\circ \tan 20^\circ} = 1$$

$$\Rightarrow \tan 25^\circ + \tan 20^\circ = 1 - \tan 25^\circ \tan 20^\circ$$

$$\Rightarrow \tan 25^\circ + \tan 20^\circ + \tan 25^\circ \tan 20^\circ = 1$$

71. We have $A + B = 45^\circ$

$$\Rightarrow \tan(A + B) = \tan(45^\circ)$$

$$\Rightarrow \tan(A + B) = 1$$

$$\Rightarrow \frac{\tan A + \tan B}{1 - \tan A \cdot \tan B} = 1$$

$$\Rightarrow \tan A + \tan B = 1 - \tan A \cdot \tan B$$

$$\Rightarrow \tan A + \tan B + \tan A \cdot \tan B = 1$$

$$\Rightarrow 1 + \tan A + \tan B + \tan A \cdot \tan B = 1 + 1 = 2$$

$$\Rightarrow (1 + \tan A) + \tan B (1 + \tan A) = 2$$

$$\Rightarrow (1 + \tan A) (1 + \tan B) = 2$$

72. We have $(1 + \tan 245^\circ) (1 + \tan 250^\circ)$

$$(1 + \tan 260^\circ) (1 - \tan 200^\circ)$$

$$(1 - \tan 205^\circ) (1 - \tan 215^\circ)$$

$$= \{(1 + \tan 245^\circ) (1 + \tan(-200^\circ))\}$$

$$\{(1 + \tan 250^\circ) (1 + \tan(-205^\circ))\}$$

$$\{(1 + \tan 260^\circ) (1 + \tan(-215^\circ))\}$$

$$= 2 \times 2 \times 2$$

$$= 8$$

73. Now, $\tan 13A = \tan(9A + 4A)$

$$= \frac{\tan 9A + \tan 4A}{1 - \tan 9A \tan 4A}$$

$$\tan 13A - \tan 4A \tan 9A \tan 13A$$

$$= \tan 9A + \tan 4A$$

$$\tan 13A - \tan 9A - \tan 4A$$

$$= \tan 4A \cdot \tan 9A \cdot \tan 13A$$

74. Do your self.

75. We have $\tan(\alpha + \beta)$

$$= \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \cdot \tan \beta}$$

$$= \frac{\frac{m}{m+1} + \frac{1}{2m+1}}{1 - \frac{m}{m+1} \cdot \frac{1}{2m+1}}$$

$$= \frac{\frac{m}{m+1} + \frac{1}{2m+1}}{1 - \frac{m}{m+1} \cdot \frac{1}{2m+1}}$$

$$= \frac{\frac{2m^2 + m + m + 1}{(m+1)(2m+1)}}{\frac{2m^2 + 3m + 1 - m}{(m+1)(2m+1)}}$$

$$= \frac{2m^2 + 2m + 1}{2m^2 + 3m + 1 - m}$$

$$= \frac{2m^2 + 2m + 1}{2m^2 + 2m + 1} = 1$$

$$\tan(\alpha + \beta) = 1$$

$$(\alpha + \beta) = \frac{\pi}{4}$$

76. We have

$$\sin^2 A + \sin^2(A - B) - 2 \sin A \cos B \sin(A - B)$$

$$= \sin^2 A + \sin^2(A - B) (\sin(A - B) - 2 \sin A \cos B)$$

$$= \sin^2 A - \sin^2(A - B) \sin(A + B)$$

$$= \sin^2 A - (\sin^2 A - \sin^2 B)$$

$$= \sin^2 B$$

77. We have

$$\cos 2x \cos 2y + \cos^2(x + y) - \cos^2(x - y)$$

$$= \frac{1}{2} [2 \cos 2x \cos 2y + 2 \cos^2(x + y) - 2 \cos^2(x - y)]$$

$$= \frac{1}{2} [\cos(2x + 2y) + \cos(2x - 2y) + 1 + \cos(2x + 2y) - 1 - \cos(2x - 2y)]$$

$$= \frac{1}{2} [\cos(2x + 2y) + \cos(2x + 2y)]$$

$$= \frac{1}{2} [2 \cos(2x + 2y)]$$

$$= \cos(2x + 2y)$$

78. Given, $\sin \theta = \frac{x - y}{x + y}$

Applying componendo and dividendo, we get,

$$\frac{\sin \theta + 1}{\sin \theta - 1} = \frac{x - y + x + y}{x - y - x - y}$$

$$\Rightarrow \frac{\sin \theta + 1}{\sin \theta - 1} = -\frac{x}{y}$$

$$\Rightarrow \frac{1 + \sin \theta}{1 - \sin \theta} = \frac{x}{y}$$

$$\Rightarrow \frac{\left(\cos\left(\frac{\theta}{2}\right) + \sin\left(\frac{\theta}{2}\right)\right)^2}{\left(\cos\left(\frac{\theta}{2}\right) - \sin\left(\frac{\theta}{2}\right)\right)^2} = \frac{x}{y}$$

$$\Rightarrow \frac{\left(1 + \tan\left(\frac{\theta}{2}\right)\right)^2}{\left(1 - \tan\left(\frac{\theta}{2}\right)\right)^2} = \frac{x}{y}$$

$$\Rightarrow \frac{1 + \tan\left(\frac{\theta}{2}\right)}{1 - \tan\left(\frac{\theta}{2}\right)} = \pm \sqrt{\frac{x}{y}}$$

$$\Rightarrow \tan\left(\frac{\pi}{4} + \frac{\theta}{2}\right) = \pm \sqrt{\frac{x}{y}}$$

79. If $\tan \alpha = \frac{P \sin \beta}{Q \cos \beta}$,

$$\text{prove that } \tan(\beta - \alpha) = \frac{P \sin \alpha}{Q + P \cos \alpha}$$

80. We have

$$\begin{aligned} \cos(\alpha - \beta) + \cos(\beta - \gamma) + \cos(\gamma - \alpha) &= -\frac{3}{2} \\ \Rightarrow 2(\cos(\alpha - \beta) + \cos(\beta - \gamma) + \cos(\gamma - \alpha)) + 3 &= 0 \\ \Rightarrow 2(\cos \alpha \cos \beta + \cos \beta \cos \gamma + \cos \gamma \cos \alpha) \\ &+ 2(\sin \alpha \sin \beta + \sin \beta \sin \gamma + \sin \gamma \sin \alpha) + 3 = 0 \\ \Rightarrow (\cos^2 \alpha + \sin^2 \alpha) + (\cos^2 \beta + \sin^2 \beta) \\ &+ (\cos^2 \gamma + \sin^2 \gamma) + \\ &2(\cos \alpha \cos \beta + \cos \beta \cos \gamma + \cos \gamma \cos \alpha) \\ &+ 2(\sin \alpha \sin \beta + \sin \beta \sin \gamma + \sin \gamma \sin \alpha) = 0 \\ \Rightarrow \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma \\ &+ 2 \cos \alpha \cos \beta + 2 \cos \beta \cos \gamma \\ &+ 2 \cos \gamma \cos \alpha + \sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma \\ &+ 2 \sin \alpha \sin \beta + 2 \sin \beta \sin \gamma \\ &+ 2 \sin \gamma \sin \alpha = 0 \\ \Rightarrow (\cos \alpha + \cos \beta + \cos \gamma)^2 \\ &+ (\sin \alpha + \sin \beta + \sin \gamma)^2 = 0 \\ \Rightarrow (\cos \alpha + \cos \beta + \cos \gamma) &= 0 \\ \text{and } (\sin \alpha + \sin \beta + \sin \gamma) &= 0 \end{aligned}$$

81. We have $\tan(\alpha + \theta) = n \tan(\alpha - \theta)$

$$\begin{aligned} \Rightarrow \frac{\tan(\alpha + \theta)}{\tan(\alpha - \theta)} &= \frac{n}{1} \\ \Rightarrow \frac{\tan(\alpha + \theta) + \tan(\alpha - \theta)}{\tan(\alpha + \theta) - \tan(\alpha - \theta)} &= \frac{n+1}{n-1} \\ \Rightarrow \frac{\sin(\alpha + \theta + \alpha - \theta)}{\sin(\alpha + \theta - \alpha + \theta)} &= \frac{n+1}{n-1} \\ \Rightarrow \frac{\sin(2\alpha)}{\sin(2\theta)} &= \frac{n+1}{n-1} \\ \Rightarrow \frac{\sin(2\theta)}{\sin(2\alpha)} &= \frac{n-1}{n+1} \end{aligned}$$

Hence, the result.

82. Given $\sin \alpha + \sin \beta = a$... (i)

and $\cos \alpha + \cos \beta = b$... (ii)

(i) divides by (ii), we get,

$$\begin{aligned} \frac{\sin \alpha + \sin \beta}{\cos \alpha + \cos \beta} &= \frac{a}{b} \\ \Rightarrow \frac{2 \sin\left(\frac{\alpha + \beta}{2}\right) \cos\left(\frac{\alpha - \beta}{2}\right)}{2 \cos\left(\frac{\alpha + \beta}{2}\right) \cos\left(\frac{\alpha - \beta}{2}\right)} &= \frac{a}{b} \\ \Rightarrow \tan\left(\frac{\alpha + \beta}{2}\right) &= \frac{a}{b} \end{aligned}$$

(i) Now, $\cos(\alpha + \beta)$

$$\begin{aligned} &= \frac{1 - \tan^2\left(\frac{\alpha + \beta}{2}\right)}{1 + \tan^2\left(\frac{\alpha + \beta}{2}\right)} \\ &= \frac{1 - \frac{a^2}{b^2}}{1 + \frac{a^2}{b^2}} = \frac{b^2 - a^2}{a^2 + b^2} \end{aligned}$$

(ii) $\sin(\alpha + \beta)$

$$\begin{aligned} &= \frac{2 \tan\left(\frac{\alpha + \beta}{2}\right)}{1 + \tan^2\left(\frac{\alpha + \beta}{2}\right)} \\ &= \frac{2(b/a)}{1 + (b/a)^2} = \frac{2ab}{a^2 + b^2} \end{aligned}$$

83. Given equation is $a \cos \theta + b \sin \theta = c$

$$a \left(\frac{1 - \tan^2(\theta/2)}{1 + \tan^2(\theta/2)} \right) + b \left(\frac{2 \tan(\theta/2)}{1 + \tan^2(\theta/2)} \right) = c$$

$$a(1 - \tan^2(\theta/2)) + 2b \tan(\theta/2) = c(1 + \tan^2(\theta/2))$$

$$(a + c) \tan^2(\theta/2) - 2b \tan(\theta/2) + (c - a) = 0$$

Let its roots be $\tan\left(\frac{\alpha}{2}\right), \tan\left(\frac{\beta}{2}\right)$

$$\text{Thus, } \tan\left(\frac{\alpha}{2}\right) + \tan\left(\frac{\beta}{2}\right) = \frac{2b}{a + c}$$

$$\tan\left(\frac{\alpha}{2}\right) \tan\left(\frac{\beta}{2}\right) = \frac{c - a}{a + c}$$

$$\begin{aligned} \tan\left(\frac{\alpha + \beta}{2}\right) &= \frac{\tan(\alpha/2) + \tan(\beta/2)}{1 - \tan(\alpha/2) \tan(\beta/2)} \\ &= \frac{\frac{2b}{a + c}}{1 - \frac{c - a}{a + c}} = \frac{2b}{2a} = \frac{b}{a} \end{aligned}$$

(i) Now, $\cos(\alpha + \beta)$

$$\begin{aligned} &= \frac{1 - \tan^2\left(\frac{\alpha + \beta}{2}\right)}{1 + \tan^2\left(\frac{\alpha + \beta}{2}\right)} \\ &= \frac{1 - \frac{b^2}{a^2}}{1 + \frac{b^2}{a^2}} = \frac{a^2 - b^2}{a^2 + b^2} \end{aligned}$$

(ii) Now, $\tan\left(\frac{\alpha - \beta}{2}\right)$

$$\begin{aligned}
 &= \frac{\tan\left(\frac{\alpha}{2}\right) - \tan\left(\frac{\beta}{2}\right)}{1 + \tan\left(\frac{\alpha}{2}\right) \cdot \tan\left(\frac{\beta}{2}\right)} \\
 &= \frac{\sqrt{\left(\tan\left(\frac{\alpha}{2}\right) + \tan\left(\frac{\beta}{2}\right)\right)^2 - 4 \tan\left(\frac{\alpha}{2}\right) \tan\left(\frac{\beta}{2}\right)}}{1 + \tan\left(\frac{\alpha}{2}\right) \cdot \tan\left(\frac{\beta}{2}\right)} \\
 &= \frac{\sqrt{\left(\frac{2b}{a+c}\right)^2 - 4\left(\frac{c-a}{c+a}\right)}}{1 + \frac{c-a}{c+a}} \\
 &= \frac{\sqrt{4b^2 - 4(c^2 - a^2)}}{2c} \\
 &= \frac{\sqrt{a^2 + b^2 - c^2}}{c}
 \end{aligned}$$

Now,

$$\begin{aligned}
 \cos(\alpha - \beta) &= \frac{1 - \tan^2\left(\frac{\alpha - \beta}{2}\right)}{1 + \tan^2\left(\frac{\alpha - \beta}{2}\right)} \\
 &= \frac{1 - \left(\frac{a^2 + b^2 - c^2}{c^2}\right)}{1 + \left(\frac{a^2 + b^2 - c^2}{c^2}\right)} \\
 &= \frac{2c^2 - (a^2 + b^2)}{a^2 + b^2}
 \end{aligned}$$

84. Given, $a \tan \theta + b \sec \theta = c$

$$\begin{aligned}
 \Rightarrow (a \tan \theta - c)^2 &= (-b \sec \theta)^2 \\
 \Rightarrow a^2 \tan^2 \theta - 2ac \tan \theta + c^2 &= b^2 \sec^2 \theta \\
 \Rightarrow a^2 \tan^2 \theta - 2ac \tan \theta + c^2 &= b^2 + b^2 \tan^2 \theta \\
 \Rightarrow (a^2 - b^2) \tan^2 \theta - 2ac \tan \theta + (c^2 - b^2) &= 0
 \end{aligned}$$

Let its roots be $\tan \alpha, \tan \beta$

$$\text{So, } \tan \alpha + \tan \beta = \frac{2ac}{a^2 - b^2}$$

$$\text{and } \tan \alpha \cdot \tan \beta = \frac{c^2 - b^2}{a^2 - b^2}$$

Now, $\tan(\alpha + \beta)$

$$\begin{aligned}
 &= \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} \\
 &= \frac{\frac{2ac}{a^2 - b^2}}{1 - \frac{c^2 - b^2}{a^2 - b^2}} \\
 &= \frac{2ac}{a^2 - b^2 - c^2 + b^2} = \frac{2ac}{a^2 - c^2}
 \end{aligned}$$

85. Given, $\tan(\pi \cos \theta) = \cot(\pi \sin \theta)$

$$\begin{aligned}
 \Rightarrow \tan(\pi \cos \theta) &= \tan\left(\pm \frac{\pi}{2} - \pi \sin \theta\right) \\
 \Rightarrow (\pi \cos \theta) &= \left(\pm \frac{\pi}{2} - \pi \sin \theta\right) \\
 \Rightarrow \cos \theta + \sin \theta &= \pm \frac{1}{2} \\
 \Rightarrow \frac{1}{\sqrt{2}} \cos \theta + \frac{1}{\sqrt{2}} \sin \theta &= \pm \frac{1}{2\sqrt{2}} \\
 \Rightarrow \cos\left(\frac{\pi}{4} - \theta\right) &= \pm \frac{1}{2\sqrt{2}}
 \end{aligned}$$

86. We have $\tan \theta = \frac{x \sin \phi}{1 - x \cos \phi}$

$$\begin{aligned}
 \Rightarrow \frac{\sin \theta}{\cos \theta} &= \frac{x \sin \phi}{1 - x \cos \phi} \\
 \Rightarrow \sin \theta - x \sin \theta \cos \phi &= x \cos \theta \sin \phi \\
 \Rightarrow x \sin(\theta + \phi) &= \sin \theta \\
 \Rightarrow x &= \frac{\sin \theta}{\sin(\theta + \phi)}
 \end{aligned}$$

$$\text{Similarly, } y = \frac{\sin \phi}{\sin(\theta + \phi)}$$

Dividing the above relations, we get,

$$\begin{aligned}
 \frac{x}{y} &= \frac{\sin \theta}{\sin \phi} \\
 \Rightarrow x \sin \phi &= y \sin \theta
 \end{aligned}$$

Hence, the result.

87. We have $\tan(\alpha + \beta) = c$

$$\begin{aligned}
 \Rightarrow \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} &= c \\
 \Rightarrow \frac{a}{1 - \tan \alpha \tan \beta} &= c \\
 \Rightarrow \tan \alpha \tan \beta &= \frac{a}{c} - 1 = \frac{a - c}{c} \quad \dots(i)
 \end{aligned}$$

Now, $\cot \alpha + \cot \beta = b$

$$\begin{aligned}
 \Rightarrow \frac{1}{\tan \alpha} + \frac{1}{\tan \beta} &= b \\
 \Rightarrow \frac{\tan \alpha + \tan \beta}{\tan \alpha \cdot \tan \beta} &= b \\
 \Rightarrow \tan \alpha \cdot \tan \beta &= \frac{a}{b} \quad \dots(ii)
 \end{aligned}$$

$$\text{From (i) and (ii), we get, } \frac{a - c}{c} = \frac{a}{b}$$

$$\Rightarrow ac + bc = ab$$

Which is the required relation.

88. We have,

$$\begin{aligned}\tan(\alpha - \beta) &= \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \cdot \tan \beta} \\&= \frac{\frac{\sin \alpha}{\cos \alpha} - \left(\frac{n \sin \alpha \cos \alpha}{1 - n \sin^2 \alpha} \right)}{1 + \frac{\sin \alpha}{\cos \alpha} \times \left(\frac{n \sin \alpha \cos \alpha}{1 - n \sin^2 \alpha} \right)} \\&= \frac{\sin \alpha (1 - n \sin^2 \alpha) - n \sin \alpha \cos^2 \alpha}{\cos \alpha (1 - n \sin^2 \alpha) + n \sin^2 \alpha \cdot \cos \alpha} \\&= \frac{\sin \alpha - n \sin \alpha (\sin^2 \alpha + \cos^2 \alpha)}{\cos \alpha - n \sin^2 \alpha \cdot \cos \alpha + n \sin^2 \alpha \cdot \cos \alpha} \\&= \frac{\sin \alpha - n \sin \alpha}{\cos \alpha} \\&= \frac{(1 - n) \sin \alpha}{\cos \alpha} \\&= (1 - n) \tan \alpha\end{aligned}$$

89. Let $A = x + y - z$, $B = y + z - x$, $C = x + y - z$

$$\begin{aligned}\text{Then, } A + B + C &= (x + y - z) + (y + z - x) + (x + y - z) \\&= (x + y + z) = 0\end{aligned}$$

$$\Rightarrow A + B = -C$$

$$\Rightarrow \cot(A + B) = \cot(-C)$$

$$\Rightarrow \frac{\cot A \cot B - 1}{\cot A + \cot B} = \cot(-C)$$

$$\Rightarrow \frac{\cot A \cot B - 1}{\cot A + \cot B} = -\cot C$$

$$\Rightarrow \cot A \cot B - 1 = -\cot A \cot C - \cot B \cot C$$

$$\Rightarrow \cot A \cot B + \cot A \cot C + \cot B \cot C = 1$$

$$\Rightarrow \cot(x + y - z) \cot(y + z - x)$$

$$+ \cot(y + z - x) \cot(zx + x - y)$$

$$+ \cot(zx + x - y) \cot(x + y - z) = 1$$

90. We have $\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \cdot \tan \beta}$

$$\begin{aligned}&= \frac{\frac{3}{2} \tan \beta - \tan \beta}{1 + \frac{3}{2} \tan \beta \tan \beta} \\&= \frac{\tan \beta}{2 + 3 \tan^2 \beta} \\&= \frac{\frac{\sin \beta}{\cos \beta}}{2 + 3 \frac{\sin^2 \beta}{\cos^2 \beta}} \\&= \frac{\sin \beta \cos \beta}{2 \cos^2 \beta + 3 \sin^2 \beta} \\&= \frac{\sin \beta \cos \beta}{2 + \sin^2 \beta}\end{aligned}$$

$$\begin{aligned}&= \frac{2 \sin \beta \cos \beta}{4 + 2 \sin^2 \beta} \\&= \frac{\sin 2\beta}{4 + 1 - \cos 2\beta} \\&= \frac{\sin 2\beta}{5 - \cos 2\beta}\end{aligned}$$

91. We have

$$\begin{aligned}\frac{\sin 5A - \sin 3A}{\cos 5A + \cos 3A} &= \frac{2 \cos 4A \sin A}{2 \cos 4A \cos A} \\&= \tan A\end{aligned}$$

92. We have

$$\begin{aligned}\frac{\sin A + \sin 3A}{\cos A + \cos 3A} &= \frac{2 \sin 2A \cos A}{2 \cos 2A \cos A} \\&= \tan 2A\end{aligned}$$

93. We have

$$\begin{aligned}\frac{\sin A + \sin B}{\cos A + \cos B} &= \frac{2 \sin \left(\frac{A+B}{2} \right) \cos \left(\frac{A-B}{2} \right)}{2 \cos \left(\frac{A+B}{2} \right) \cos \left(\frac{A-B}{2} \right)} \\&= \tan \left(\frac{A+B}{2} \right)\end{aligned}$$

94. We have $\sin 38^\circ + \sin 22^\circ$

$$= 2 \sin(30^\circ) \cos(8^\circ)$$

$$= 2 \times \frac{1}{2} \times \cos(90^\circ - 82^\circ)$$

$$= \sin(82^\circ)$$

95. We have $\sin 105^\circ + \cos 105^\circ$

$$= \sin(105^\circ) - \sin(15^\circ)$$

$$= 2 \cos(60^\circ) \sin(45^\circ)$$

$$= 2 \times \frac{1}{2} \times \sin(45^\circ)$$

$$= \sin(45^\circ)$$

$$= \cos(45^\circ)$$

96. We have $\cos 55^\circ + \cos 65^\circ + \cos 175^\circ$

$$= \cos(65^\circ) + \cos(55^\circ) - \cos(5^\circ)$$

$$= 2 \cos(60^\circ) \cos(5^\circ) - \cos(5^\circ)$$

$$= 2 \times \frac{1}{2} \times \cos(5^\circ) - \cos(5^\circ)$$

$$= \cos(5^\circ) - \cos(5^\circ)$$

$$= 0$$

97. We have $\cos 20^\circ + \cos 100^\circ + \cos 140^\circ$

$$= \cos(100^\circ) + \cos(20^\circ) - \cos(40^\circ)$$

$$= 2 \cos(60^\circ) \cos(40^\circ) - \cos(40^\circ)$$

$$= 2 \times \frac{1}{2} \times \cos(40^\circ) - \cos(40^\circ)$$

$$= \cos(40^\circ) - \cos(40^\circ)$$

$$= 0$$

98. We have

$$\begin{aligned} & \sin 50^\circ - \sin 70^\circ + \sin 10^\circ \\ &= \sin (50^\circ) + \sin (10^\circ) - \sin (70^\circ) \\ &= 2 \sin (30^\circ) \cos (20^\circ) - \sin (70^\circ) \\ &= 2 \times \frac{1}{2} \times \cos (20^\circ) - \sin (90^\circ - 20^\circ) \\ &= \cos (20^\circ) - \cos (20^\circ) \\ &= 0. \end{aligned}$$

99. We have

$$\begin{aligned} & \sin (47^\circ) + \cos (77^\circ) = \cos (17^\circ) \\ &= \sin (47^\circ) + \sin (13^\circ) \\ &= 2 \sin \left(\frac{47^\circ + 13^\circ}{2} \right) \cos \left(\frac{47^\circ - 13^\circ}{2} \right) \\ &= 2 \sin (30^\circ) \cos (17^\circ) \\ &= 2 \times \frac{1}{2} \times \cos (17^\circ) \\ &= \cos (17^\circ) \end{aligned}$$

100. We have

$$\begin{aligned} & \cos (80^\circ) + \cos (40^\circ) - \cos (20^\circ) \\ &= 2 \cos \left(\frac{80^\circ + 40^\circ}{2} \right) \cos \left(\frac{80^\circ - 40^\circ}{2} \right) - \cos (20^\circ) \\ &= 2 \cos (60^\circ) \cos (20^\circ) - \cos (20^\circ) \\ &= 2 \times \frac{1}{2} \times \cos (20^\circ) - \cos (20^\circ) \\ &= \cos (20^\circ) - \cos (20^\circ) \\ &= 0. \end{aligned}$$

101. We have

$$\begin{aligned} & \sin (10^\circ) + \sin (20^\circ) + \sin (40^\circ) \\ &+ \sin (50^\circ) - \sin (70^\circ) - \sin (80^\circ) \\ &= \{ \sin (50^\circ) + \sin (10^\circ) \} + \{ \sin (40^\circ) + \sin (20^\circ) \} \\ &- \sin (70^\circ) - \sin (80^\circ) \\ &= 2 \sin \left(\frac{50^\circ + 10^\circ}{2} \right) \cos \left(\frac{50^\circ - 10^\circ}{2} \right) \\ &+ 2 \sin \left(\frac{40^\circ + 20^\circ}{2} \right) \cos \left(\frac{40^\circ - 20^\circ}{2} \right) \\ &- \sin (70^\circ) - \sin (80^\circ) \\ &= 2 \sin (30^\circ) \cos (20^\circ) + 2 \sin (30^\circ) \cos (10^\circ) \\ &- \sin (70^\circ) - \sin (80^\circ) \\ &= \cos (20^\circ) + \cos (10^\circ) - \sin (70^\circ) - \sin (80^\circ) \\ &= \cos (20^\circ) + \cos (10^\circ) - \cos (20^\circ) - \cos (10^\circ) \\ &= 0 \end{aligned}$$

102. We have

$$\begin{aligned} & (\cos \alpha + \cos \beta)^2 + (\sin \alpha + \sin \beta)^2 \\ &= 2 + 2 \cos (\alpha - \beta) \end{aligned}$$

$$\begin{aligned} &= 2 + (1 + \cos (\alpha - \beta)) \\ &= 2 + 2 \cos^2 \left(\frac{\alpha - \beta}{2} \right) \\ &= 4 \cos^2 \left(\frac{\alpha - \beta}{2} \right) \end{aligned}$$

103. We have $(\cos \alpha - \cos \beta)^2 + (\sin \alpha - \sin \beta)^2$

$$\begin{aligned} &= 2 - 2 \cos (\alpha - \beta) \\ &= 2(1 - \cos (\alpha - \beta)) \\ &= 2 \times 2 \sin^2 \left(\frac{\alpha - \beta}{2} \right) \\ &= 4 \sin^2 \left(\frac{\alpha - \beta}{2} \right) \end{aligned}$$

104. Do yourself.

105. We have $\cos (20^\circ) \cos (40^\circ) \cos (80^\circ)$

$$\begin{aligned} &= \cos (40^\circ) \cos (20^\circ) \cos (80^\circ) \\ &= \frac{1}{4} [4 \cos (60^\circ - 20^\circ) \cos (20^\circ) \cos (60^\circ + 20^\circ)] \\ &= \frac{1}{4} \times \cos (20^\circ \times 3) \\ &= \frac{1}{4} \times \cos (60^\circ) \\ &= \frac{1}{4} \times \frac{1}{2} = \frac{1}{8} \end{aligned}$$

106. We have $\cos 25^\circ \cos 35^\circ \cos 65^\circ$

$$\begin{aligned} &= \cos 25^\circ \cos 35^\circ \cos 65^\circ \\ &= \frac{1}{4} [4 \cos (60^\circ - 25^\circ) \cos 25^\circ \cos (60^\circ + 25^\circ)] \\ &= \frac{1}{4} \times \cos (25^\circ \times 3) \\ &= \frac{1}{4} \times \cos (75^\circ) \\ &= \frac{1}{4} \times \sin (15^\circ) \\ &= \frac{\sqrt{3} - 1}{8\sqrt{2}} \end{aligned}$$

107. We have $\sin (20^\circ) \sin (40^\circ) \sin (80^\circ)$

$$\begin{aligned} &= \sin (20^\circ) \sin (40^\circ) \sin (80^\circ) \\ &= \frac{1}{4} [4 \sin (60^\circ - 20^\circ) \sin (20^\circ) \sin (60^\circ + 20^\circ)] \\ &= \frac{1}{4} \times \sin (60^\circ) \\ &= \frac{\sqrt{3}}{8} \end{aligned}$$

108. We have $\sin(10^\circ) \sin(50^\circ) \sin(60^\circ) \sin(70^\circ)$

$$\begin{aligned} &= \frac{\sqrt{3}}{2} [\sin(50^\circ) \sin(10^\circ) \sin(70^\circ)] \\ &= \frac{\sqrt{3}}{2} [\sin(60^\circ - 10^\circ) \sin(10^\circ) \sin(60^\circ + 10^\circ)] \\ &= \frac{\sqrt{3}}{8} [4 \sin(60^\circ - 10^\circ) \sin(10^\circ) \sin(60^\circ + 10^\circ)] \\ &= \frac{\sqrt{3}}{8} \times \sin(30^\circ) \\ &= \frac{\sqrt{3}}{8} \times \frac{1}{2} = \frac{\sqrt{3}}{16} \end{aligned}$$

109. We have $\cos(10^\circ) \cos(30^\circ) \cos(50^\circ) \cos(70^\circ)$

$$\begin{aligned} &= \frac{\sqrt{3}}{2} [\cos(50^\circ) \cos(10^\circ) \cos(70^\circ)] \\ &= \frac{\sqrt{3}}{2} [\cos(60^\circ - 10^\circ) \cos(10^\circ) \cos(60^\circ + 10^\circ)] \\ &= \frac{\sqrt{3}}{8} [4 \cos(60^\circ - 10^\circ) \cos(10^\circ) \cos(60^\circ + 10^\circ)] \\ &= \frac{\sqrt{3}}{8} \times \cos(30^\circ) \\ &= \frac{\sqrt{3}}{8} \times \frac{\sqrt{3}}{2} \\ &= \frac{3}{16} \end{aligned}$$

110. We have $\frac{\sin A + \sin 3A + \sin 5A}{\cos A + \cos 3A + \cos 5A}$

$$\begin{aligned} &= \frac{\sin 5A + \sin 3A + \sin A}{\cos 5A + \cos 3A + \cos A} \\ &= \frac{(\sin 5A + \sin A) + \sin 3A}{(\cos 5A + \cos A) + \cos 3A} \\ &= \frac{2 \sin 3A \cos 2A + \sin 3A}{2 \cos 3A \cos 2A + \cos 3A} \\ &= \frac{\sin 3A (2 \cos 2A + 1)}{\cos 3A (2 \cos 2A + 1)} \\ &= \tan 3A \end{aligned}$$

111. We have $\frac{\cos 4x + \cos 3x + \cos 2x}{\sin 4x + \sin 3x + \sin 2x}$

$$\begin{aligned} &= \frac{(\cos 4x + \cos 2x) + \cos 3x}{(\sin 4x + \sin 2x) + \sin 3x} \\ &= \frac{2 \cos 3x \cos x + \cos 3x}{2 \sin 3x \cos x + \sin 3x} \\ &= \frac{\cos 3x (2 \cos x + 1)}{\sin 3x (2 \cos x + 1)} \\ &= \cot 3x \end{aligned}$$

112. We have $\frac{\sin A + \sin 3A + \sin 5A + \sin 7A}{\cos A + \cos 3A + \cos 5A + \cos 7A}$

$$\begin{aligned} &= \frac{(\sin 7A + \sin A) + (\sin 5A + \sin 3A)}{(\cos 7A + \cos A) + (\cos 5A + \cos 3A)} \\ &= \frac{2 \sin 4A \cos 3A + 2 \sin 4A \cos A}{2 \cos 4A \cos 3A + 2 \cos 4A \cos A} \\ &= \frac{2 \sin 4A (\cos 3A + \cos A)}{2 \cos 4A (\cos 3A + \cos A)} \\ &= \tan 4A \end{aligned}$$

113. We have $\frac{\sin A + \sin 2A + \sin 4A + \sin 5A}{\cos A + \cos 2A + \cos 4A + \cos 5A}$

$$\begin{aligned} &= \frac{(\sin 5A + \sin A) + (\sin 4A + \sin 2A)}{(\cos 5A + \cos A) + (\cos 4A + \cos 2A)} \\ &= \frac{2 \sin 3A \cos 2A + 2 \sin 3A \cos A}{2 \cos 3A \cos 2A + 2 \cos 3A \cos A} \\ &= \frac{\sin 3A (\cos 2A + \cos A)}{\cos 3A (\cos 2A + \cos A)} \\ &= \frac{\sin 3A}{\cos 3A} \\ &= \tan 3A \end{aligned}$$

Hence, the result.

114. Given $\sin A - \sin B = \frac{1}{2}$... (i)

and $\cos A - \cos B = \frac{1}{3}$... (ii)

Dividing (i) by (ii) we get,

$$\begin{aligned} &\frac{\sin A - \sin B}{\cos A - \cos B} = \frac{1/2}{1/3} = \frac{3}{2} \\ \Rightarrow &\frac{2 \cos\left(\frac{A+B}{2}\right) \sin\left(\frac{A-B}{2}\right)}{-2 \sin\left(\frac{A+B}{2}\right) \sin\left(\frac{A-B}{2}\right)} = \frac{3}{2} \\ \Rightarrow &\frac{\cos\left(\frac{A+B}{2}\right)}{\sin\left(\frac{A+B}{2}\right)} = -\frac{3}{2} \\ \Rightarrow &\cot\left(\frac{A+B}{2}\right) = -\frac{3}{2} \\ \Rightarrow &\tan\left(\frac{A+B}{2}\right) = -\frac{2}{3} \end{aligned}$$

115. Given, $\sin A + \sin B = \frac{1}{4}$,

and $\cos A + \cos B = \frac{1}{2}$

Dividing, the above relations, we get,

$$\begin{aligned} \frac{\sin A + \sin B}{\cos A + \cos B} &= \frac{1}{2} \\ \Rightarrow \frac{2 \sin \left(\frac{A+B}{2} \right) \cos \left(\frac{A-B}{2} \right)}{2 \cos \left(\frac{A+B}{2} \right) \cos \left(\frac{A-B}{2} \right)} &= \frac{1}{2} \\ \Rightarrow \tan \left(\frac{A+B}{2} \right) &= \frac{1}{2} \end{aligned}$$

Hence, the result.

116. Given,

$$\begin{aligned} \operatorname{cosec} A + \sec A &= \operatorname{cosec} B + \sec B \\ \Rightarrow \operatorname{cosec} A + \operatorname{cosec} B &= \sec B + \sec A \\ \Rightarrow \frac{1}{\sin A} + \frac{1}{\sin B} &= \frac{1}{\cos B} + \frac{1}{\cos A} \\ \Rightarrow \frac{\sin B - \sin A}{\sin A \sin B} &= \frac{\cos A - \cos B}{\cos A \cos B} \\ \Rightarrow \frac{\sin A \sin B}{\cos A \cos B} &= \frac{\sin B - \sin A}{\cos A - \cos B} \\ \Rightarrow \tan A \tan B &= \frac{2 \cos \left(\frac{A+B}{2} \right) \sin \left(\frac{A-B}{2} \right)}{2 \sin \left(\frac{A+B}{2} \right) \sin \left(\frac{A-B}{2} \right)} \\ \Rightarrow \tan A \tan B &= \cot \left(\frac{A+B}{2} \right) \end{aligned}$$

Hence, the result.

117. Given,

$$\begin{aligned} \sin 2A &= \lambda \sin 2B \\ \Rightarrow \frac{\sin 2A}{\sin 2B} &= \frac{\lambda}{1} \\ \Rightarrow \frac{\sin 2A + \sin 2B}{\sin 2A - \sin 2B} &= \frac{\lambda + 1}{\lambda - 1} \\ \Rightarrow \frac{2 \sin (A+B) \cos (A-B)}{2 \cos (A+B) \sin (A-B)} &= \frac{\lambda + 1}{\lambda - 1} \\ \Rightarrow \tan (A+B) \cot (A-B) &= \frac{\lambda + 1}{\lambda - 1} \\ \Rightarrow \frac{\tan (A+B)}{\tan (A-B)} &= \frac{\lambda + 1}{\lambda - 1} \end{aligned}$$

118. We have

$$\sqrt{3} \cot (20^\circ) - 4 \cos (20^\circ)$$

$$\begin{aligned} &= \frac{\sqrt{3} \cos (20^\circ) - 4 \sin (20^\circ) \cos (20^\circ)}{\sin (20^\circ)} \\ &= \frac{2 \left(\frac{\sqrt{3}}{2} \cos (20^\circ) - 2 \sin (20^\circ) \cos (20^\circ) \right)}{\sin (20^\circ)} \\ &= \frac{2 (\sin (60^\circ) \cos (20^\circ) - \sin (40^\circ))}{\sin (20^\circ)} \\ &= \frac{(2 \sin (60^\circ) \cos (20^\circ) - 2 \sin (40^\circ))}{\sin (20^\circ)} \\ &= \frac{(\sin (80^\circ) + \sin (40^\circ) - 2 \sin (40^\circ))}{\sin (20^\circ)} \\ &= \frac{(\sin (80^\circ) - \sin (40^\circ))}{\sin (20^\circ)} \\ &= \frac{2 \cos (60^\circ) \sin (20^\circ)}{\sin (20^\circ)} \\ &= 1 \end{aligned}$$

119. Given,

$$\begin{aligned} \sin \alpha + \sin \beta &= a \quad \dots(i) \\ \text{and } \cos \alpha + \cos \beta &= b \quad \dots(ii) \end{aligned}$$

(i) divides by (ii), we get,

$$\begin{aligned} \frac{\sin \alpha + \sin \beta}{\cos \alpha + \cos \beta} &= \frac{a}{b} \\ \Rightarrow \frac{2 \sin \left(\frac{\alpha + \beta}{2} \right) \cos \left(\frac{\alpha - \beta}{2} \right)}{2 \cos \left(\frac{\alpha + \beta}{2} \right) \cos \left(\frac{\alpha - \beta}{2} \right)} &= \frac{a}{b} \\ \Rightarrow \tan \left(\frac{\alpha + \beta}{2} \right) &= \frac{a}{b} \end{aligned}$$

Now,

$$\begin{aligned} \cos (\alpha + \beta) &= \frac{1 - \tan^2 \left(\frac{\alpha + \beta}{2} \right)}{1 + \tan^2 \left(\frac{\alpha + \beta}{2} \right)} \\ &= \frac{1 - \frac{a^2}{b^2}}{1 + \frac{a^2}{b^2}} = \frac{b^2 - a^2}{a^2 + b^2} \end{aligned}$$

120. We have

$$\begin{aligned} \sin A &= \sqrt{1 - \frac{1}{4} \left(x + \frac{1}{x} \right)^2} \\ &= i \sqrt{\frac{1}{4} \left(x + \frac{1}{x} \right)^2 - 1} \\ &= i \sqrt{\frac{1}{4} \left(x - \frac{1}{x} \right)^2} \\ &= \frac{i}{2} \left(x - \frac{1}{x} \right) \end{aligned}$$

$$\text{Similarly, } \sin B = \frac{i}{2} \left(y - \frac{1}{y} \right)$$

Now, $\cos(A - B)$

$$= \cos A \cos B + \sin A \sin B$$

$$= \frac{1}{4} \left(x + \frac{1}{x} \right) \left(y + \frac{1}{y} \right) + \frac{1}{4} \left(x - \frac{1}{x} \right) \left(y - \frac{1}{y} \right)$$

$$= \frac{1}{4} \left(2xy + \frac{2}{xy} \right)$$

$$= \frac{1}{2} \left(xy + \frac{1}{xy} \right)$$

121. We have

$$\begin{aligned} \sin(47^\circ) + \sin(61^\circ) - \sin(11^\circ) - \sin(25^\circ) \\ &= (\sin(61^\circ) + \sin(47^\circ)) - (\sin(25^\circ) + \sin(11^\circ)) \\ &= 2 \sin(54^\circ) \cos(7^\circ) - 2 \sin(18^\circ) \cos(7^\circ) \\ &= 2 \cos(7^\circ) [\sin(54^\circ) - \sin(18^\circ)] \\ &= 2 \cos(7^\circ) [\cos(36^\circ) - \sin(18^\circ)] \\ &= 2 \cos(7^\circ) \left[\frac{\sqrt{5}+1}{4} - \frac{\sqrt{5}-1}{4} \right] \\ &= 2 \cos(7^\circ) \times \frac{1}{2} \\ &= \cos(7^\circ) \end{aligned}$$

122. Do yourself.

123. We have

$$\begin{aligned} -\sqrt{7^2+5^2} \leq 2k+1 \leq \sqrt{7^2+5^2} \\ -\sqrt{74} \leq 2k+1 \leq \sqrt{74} \\ \frac{-\sqrt{74}-1}{2} \leq k \leq \frac{\sqrt{74}-1}{2} \\ k = -4, -3, -2, -1, 0, 1, 2, 3 \end{aligned}$$

Hence, the number of integral values of

124. We have

$$\begin{aligned} \cos \alpha + \cos \beta + \cos \gamma + \cos(\alpha + \beta + \gamma) \\ &= (\cos \alpha + \cos \beta) + (\cos(\alpha + \beta + \gamma) + \cos \gamma) \\ &= 2 \cos \left(\frac{\alpha + \beta}{2} \right) \cos \left(\frac{\alpha - \beta}{2} \right) \\ &\quad + 2 \cos \left(\frac{\alpha + \beta + \gamma + \gamma}{2} \right) \cos \left(\frac{\alpha + \beta + \gamma - \gamma}{2} \right) \\ &= 2 \cos \left(\frac{\alpha + \beta}{2} \right) \cos \left(\frac{\alpha - \beta}{2} \right) \\ &\quad + 2 \cos \left(\frac{\alpha + \beta + 2\gamma}{2} \right) \cos \left(\frac{\alpha + \beta}{2} \right) \\ &= 2 \cos \left(\frac{\alpha + \beta}{2} \right) \times \\ &\quad \left\{ \cos \left(\frac{\alpha - \beta}{2} \right) + 2 \cos \left(\frac{\alpha + \beta + 2\gamma}{2} \right) \right\} \end{aligned}$$

$$\begin{aligned} &= 2 \cos \left(\frac{\alpha + \beta}{2} \right) \times \cos \left(\frac{\frac{\alpha - \beta}{2} + \frac{\alpha + \beta + 2\gamma}{2}}{2} \right) \\ &\quad \cos \left(\frac{\frac{\alpha - \beta}{2} - \frac{\alpha + \beta + 2\gamma}{2}}{2} \right) \\ &= 2 \cos \left(\frac{\alpha + \beta}{2} \right) \times \left\{ \cos \left(\frac{\alpha + \gamma}{2} \right) \cos \left(\frac{\beta + \gamma}{2} \right) \right\} \end{aligned}$$

125. We have

$$\begin{aligned} \frac{1 - \cos 2\theta}{\sin 2\theta} &= \frac{2 \sin^2 \theta}{2 \sin \theta \cos \theta} \\ &= \frac{\sin \theta}{\cos \theta} = \tan \theta \end{aligned}$$

126. We have

$$\begin{aligned} \frac{1 + \cos 2\theta}{\sin 2\theta} &= \frac{2 \cos^2 \theta}{2 \sin \theta \cos \theta} \\ &= \frac{\cos \theta}{\sin \theta} = \cot \theta \end{aligned}$$

127. We have $(\cot \theta - \tan \theta)$

$$\begin{aligned} &= \left(\frac{\cos \theta}{\sin \theta} - \frac{\sin \theta}{\cos \theta} \right) \\ &= \frac{\cos^2 \theta - \sin^2 \theta}{\sin \theta \cos \theta} \\ &= \frac{2 \cos 2\theta}{2 \sin \theta \cos \theta} \\ &= \frac{2 \cos 2\theta}{\sin 2\theta} \\ &= 2 \cot 2\theta \end{aligned}$$

128. We have

$$\begin{aligned} \tan \theta + 2 \tan(2\theta) + 4 \tan(4\theta) + 8 \cot 8\theta \\ &= \cot \theta (\cot \theta - \tan \theta) + 2 \tan 2\theta + 4 \tan 4\theta + 8 \cot 8\theta \\ &= \cot \theta - 2 \cot 2\theta + 2 \tan 2\theta + 4 \tan 4\theta + 8 \cot 8\theta \\ &= \cot \theta - 2(\cot 2\theta - \tan 2\theta) + 4 \tan 4\theta + 8 \cot 8\theta \\ &= \cot \theta - 4 \cot 4\theta + 4 \tan 4\theta + 8 \cot 8\theta \\ &= \cot \theta - 4(\cot 4\theta \tan 4\theta) + 8 \cot 8\theta \\ &= \cot \theta - 8 \cot 8\theta + 8 \cot 8\theta \\ &= \cot \theta \end{aligned}$$

129. We have

$$\begin{aligned} a \cos(2\theta) + b \sin(2\theta) \\ &= a \left(\frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} \right) + b \left(\frac{2 \tan \theta}{1 + \tan^2 \theta} \right) \\ &= a \left(\frac{1 - \frac{b^2}{a^2}}{1 + \frac{b^2}{a^2}} \right) + b \left(\frac{2 \frac{b}{a}}{1 + \frac{b^2}{a^2}} \right) \end{aligned}$$

$$\begin{aligned}
 &= a \left(\frac{a^2 - b^2}{a^2 + b^2} \right) + b \left(\frac{2ab}{a^2 + b^2} \right) \\
 &= \frac{a(a^2 - b^2 + 2b^2)}{a^2 + b^2} \\
 &= \frac{a(a^2 + b^2)}{a^2 + b^2} \\
 &= a
 \end{aligned}$$

130. We have

$$\begin{aligned}
 &\sqrt{3} \operatorname{cosec}(20^\circ) - \sec(20^\circ) \\
 &= \frac{\sqrt{3}}{\sin(20^\circ)} - \frac{1}{\cos(20^\circ)} \\
 &= \frac{\sqrt{3} \cos(20^\circ) - \sin(20^\circ)}{\sin(20^\circ) \cos(20^\circ)} \\
 &= \frac{4 \left(\frac{\sqrt{3}}{2} \cos(20^\circ) - \frac{1}{2} \sin(20^\circ) \right)}{2 \sin(20^\circ) \cos(20^\circ)} \\
 &= \frac{4(\sin(20^\circ) \cos(20^\circ) - \cos(60^\circ) \sin(20^\circ))}{2 \sin(20^\circ) \cos(20^\circ)} \\
 &= \frac{4(\sin(60^\circ - 20^\circ))}{\sin(40^\circ)} \\
 &= 4
 \end{aligned}$$

131. We have

$$\begin{aligned}
 &\tan(9^\circ) + \tan(27^\circ) - \tan(63^\circ) + \tan(81^\circ) \\
 &= \{\tan(9^\circ) + \tan(81^\circ)\} - \{\tan(27^\circ) + \tan(63^\circ)\} \\
 &= \{\tan(9^\circ) + \cot(9^\circ)\} - \{\tan(27^\circ) + \cot(27^\circ)\} \\
 &= \left\{ \frac{\sin(9^\circ)}{\cos(9^\circ)} + \frac{\cos(9^\circ)}{\sin(9^\circ)} \right\} - \left\{ \frac{\sin(27^\circ)}{\cos(27^\circ)} + \frac{\cos(27^\circ)}{\sin(27^\circ)} \right\} \\
 &= \left\{ \frac{\sin^2(9^\circ) + \cos^2(9^\circ)}{\sin(9^\circ) \cos(9^\circ)} \right\} - \left\{ \frac{\sin^2(27^\circ) + \cos^2(27^\circ)}{\sin(27^\circ) \cos(27^\circ)} \right\} \\
 &= \left\{ \frac{2}{2 \sin(9^\circ) \cos(9^\circ)} \right\} - \left\{ \frac{2}{2 \sin(27^\circ) \cos(27^\circ)} \right\} \\
 &= \left\{ \frac{2}{\sin(18^\circ)} \right\} - \left\{ \frac{2}{\sin(54^\circ)} \right\} \\
 &= \left\{ \frac{2}{\left(\frac{\sqrt{5}-1}{4} \right)} \right\} - \left\{ \frac{2}{\left(\frac{\sqrt{5}+1}{4} \right)} \right\} \\
 &= \left\{ \frac{8}{\sqrt{5}-1} \right\} - \left\{ \frac{8}{\sqrt{5}+1} \right\} \\
 &= \left\{ \frac{8(\sqrt{5}+1-\sqrt{5}+1)}{5-1} \right\} \\
 &= 4
 \end{aligned}$$

132. We have $\left(\frac{\sec 8A - 1}{\sec 4A - 1} \right)$

$$\begin{aligned}
 &= \frac{\frac{1}{\cos 8A} - 1}{\frac{1}{\cos 4A} - 1} \\
 &= \frac{1 - \cos 8A}{1 - \cos 4A} \times \frac{\cos 4A}{\cos 8A} \\
 &= \frac{2 \sin^2 4A}{2 \sin^2 2A} \times \frac{\cos 4A}{\cos 8A} \\
 &= \frac{2 \sin 4A \cos 4A}{\cos 8A} \times \frac{\sin 4A}{2 \sin^2 2A} \\
 &= \frac{\sin 8A}{\cos 8A} \times \frac{2 \sin 2A \cos 2A}{2 \sin^2 2A} \\
 &= \frac{\sin 8A}{\cos 8A} \times \frac{\cos 2A}{\sin 2A} \\
 &= \tan 8A \times \cot 2A \\
 &= \frac{\tan 8A}{\tan 2A}
 \end{aligned}$$

133. We have

$$\begin{aligned}
 &\cos^2(\theta) + \cos^2\left(\frac{2\pi}{3} - \theta\right) + \cos^2\left(\frac{2\pi}{3} + \theta\right) \\
 &= \frac{1}{2} \left(2 \cos^2(\theta) + 2 \cos^2\left(\frac{2\pi}{3} - \theta\right) + 2 \cos^2\left(\frac{2\pi}{3} + \theta\right) \right) \\
 &= \frac{1}{2} (1 + \cos(2\theta)) + \frac{1}{2} \left(1 + \cos\left(\frac{4\pi}{3} - 2\theta\right) \right) \\
 &\quad + \frac{1}{2} \left(1 + \cos\left(\frac{4\pi}{3} + 2\theta\right) \right) \\
 &= \frac{1}{2} \left(3 + \left(\cos 2\theta + \cos\left(\frac{4\pi}{3} - 2\theta\right) + \cos\left(\frac{4\pi}{3} + 2\theta\right) \right) \right) \\
 &= \frac{1}{2} \left(3 + \left(\cos 2\theta + 2 \cos\left(\frac{4\pi}{3}\right) \cos(2\theta) \right) \right) \\
 &= \frac{1}{2} \left(3 + \left(\cos 2\theta + 2 \left(-\frac{1}{2} \right) \cos(2\theta) \right) \right) \\
 &= \frac{3}{2}
 \end{aligned}$$

134. We have

$$\begin{aligned}
 &\sin^2 \theta + \sin^2(120^\circ + \theta) + \sin^2(240^\circ + \theta) \\
 &= \frac{1}{2} \left(2 \sin^2 \theta + 2 \sin^2\left(\frac{2\pi}{3} + \theta\right) \right) \\
 &\quad + \frac{1}{2} \left(2 \sin^2\left(\frac{4\pi}{3} + \theta\right) \right)
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{2} \left((1 - \cos 2\theta) + \left(1 - \cos \left(\frac{4\pi}{3} + 2\theta \right) \right) \right) \\
 &\quad + \frac{1}{2} \left(\left(1 - \cos \left(\frac{8\pi}{3} + 2\theta \right) \right) \right) \\
 &= \frac{3}{2} - \frac{1}{2} \cos 2\theta - \frac{1}{2} \left[\cos \left(\frac{4\pi}{3} + 2\theta \right) + \cos \left(\frac{8\pi}{3} + 2\theta \right) \right] \\
 &= \frac{3}{2} - \frac{1}{2} \cos 2\theta - \frac{1}{2} (2 \cos (120^\circ) \cos 2\theta) \\
 &= \frac{3}{2} - \frac{1}{2} \cos 2\theta + \frac{1}{2} \cos (2\theta) \\
 &= \frac{3}{2}
 \end{aligned}$$

135. We have

$$\begin{aligned}
 &4 \sin (\theta) \sin \left(\frac{\pi}{3} - \theta \right) \sin \left(\frac{\pi}{3} + \theta \right) \\
 &= 4 \sin (\theta) \times \left(\sin \left(\frac{\pi}{3} - \theta \right) \sin \left(\frac{\pi}{3} + \theta \right) \right) \\
 &= 4 \sin (\theta) \times \left(\sin^2 \left(\frac{\pi}{3} \right) - \sin^2 \theta \right) \\
 &= 4 \sin (\theta) \times \left(\frac{3}{4} - \sin^2 \theta \right) \\
 &= 4 \times \left(\frac{3}{4} \sin \theta - \sin^3 \theta \right) \\
 &= (3 \sin \theta - 4 \sin^3 \theta) \\
 &= \sin (3\theta)
 \end{aligned}$$

136. We have

$$\begin{aligned}
 &\sin (20^\circ) \sin (40^\circ) \sin (80^\circ) \\
 &= \frac{1}{4} (\sin (3 \cdot 20^\circ)) \\
 &= \frac{1}{4} (\sin (60^\circ)) \\
 &= \frac{\sqrt{3}}{8}
 \end{aligned}$$

137. We have,

$$\begin{aligned}
 &4 \cos (\theta) \cdot \cos (60^\circ - \theta) \cdot \cos (60^\circ + \theta) \\
 &= 4 \cos (\theta) \cdot (\cos (60^\circ - \theta) \cdot \cos (60^\circ + \theta)) \\
 &= 4 \cos (\theta) \cdot (\cos^2 (60^\circ) - \sin^2 \theta) \\
 &= 4 \cos (\theta) \cdot \left(\frac{1}{4} - 1 + \cos^2 \theta \right) \\
 &= 4 \cos (\theta) \cdot \left(-\frac{3}{4} + \cos^2 \theta \right) \\
 &= \cos (\theta) \cdot (-3 + 4 \cos^2 \theta) \\
 &= (4 \cos^3 \theta - 3 \cos (\theta)) \\
 &= \cos (3\theta)
 \end{aligned}$$

138. We have

$$\begin{aligned}
 &\cos (10^\circ) \cdot \cos (50^\circ) \cdot \cos (70^\circ) \\
 &= \cos (10^\circ) \cdot \cos (60^\circ - 10^\circ) \cdot \cos (60^\circ + 10^\circ) \\
 &= \frac{1}{4} (4 \cos (10^\circ) \cdot \cos (60^\circ - 10^\circ) \cdot \cos (60^\circ + 10^\circ)) \\
 &= \frac{1}{4} (\cos (3 \cdot 10^\circ)) \\
 &= \frac{\sqrt{3}}{8}
 \end{aligned}$$

139. We have

$$\begin{aligned}
 &\tan (\theta) - \tan (60^\circ - \theta) + \tan (60^\circ + \theta) \\
 &= \tan (\theta) - \frac{\tan (60^\circ) - \tan (\theta)}{1 + \tan (60^\circ) \cdot \tan (\theta)} \\
 &\quad + \frac{\tan (60^\circ) + \tan (\theta)}{1 - \tan (60^\circ) \cdot \tan (\theta)} \\
 &= \tan (\theta) - \frac{\sqrt{3} - \tan (\theta)}{1 + \sqrt{3} \cdot \tan (\theta)} + \frac{\sqrt{3} + \tan (\theta)}{1 - \sqrt{3} \cdot \tan (\theta)} \\
 &\quad - \sqrt{3} + 3 \tan (\theta) + \tan (\theta) - \sqrt{3} \tan^2 (\theta) \\
 &= \frac{+\sqrt{3} + 3 \tan (\theta) + \tan (\theta) + \sqrt{3} \tan^2 (\theta)}{(1 - 3 \tan^2 \theta)} \\
 &= \tan (\theta) + \frac{8 \tan (\theta)}{1 - 3 \tan^2 (\theta)} \\
 &= \frac{\tan (\theta) - 3 \tan^3 (\theta) + 8 \tan (\theta)}{1 - 3 \tan^2 (\theta)} \\
 &= 3 \times \left(\frac{3 \tan (\theta) - \tan^3 (\theta)}{1 - 3 \tan^2 (\theta)} \right) \\
 &= 3 \tan (3\theta)
 \end{aligned}$$

140. We have

$$\begin{aligned}
 &\cos (\theta) \cos (2\theta) \cdot \cos (2^2\theta) \cdot \cos (2^3\theta) \dots \cos (2^{n-1}\theta) \\
 &= \frac{1}{2 \sin \theta} (2 \sin \theta \cos \theta) (\cos 2\theta \cdot \cos (2^2\theta) \dots \cos (2^{n-1}\theta)) \\
 &= \frac{1}{2^2 \sin \theta} (2 \sin 2\theta \cos 2\theta) (\cos (2^2\theta) \dots \cos (2^{n-1}\theta)) \\
 &= \frac{1}{2^3 \sin \theta} (2 \sin 4\theta \cos 4\theta) (\cos (2^3\theta) \dots \cos (2^{n-1}\theta)) \\
 &= \frac{1}{2^4 \sin \theta} (2 \sin 2^3\theta \cos 2^3\theta) (\cos (2^4\theta) \dots \cos (2^{n-1}\theta)) \\
 &\quad \vdots \\
 &= \frac{1}{2^n \sin \theta} (2 \sin 2^{n-1}\theta \cos 2^{n-1}\theta) \\
 &= \frac{1}{2^n \sin \theta} (\sin 2^n\theta) \\
 &= \frac{\sin (2^n\theta)}{2^n \sin \theta}
 \end{aligned}$$

$$\begin{aligned}
 141. \text{ We have } & \cos\left(\frac{2\pi}{7}\right)\cos\left(\frac{4\pi}{7}\right)\cos\left(\frac{8\pi}{7}\right) \\
 &= \cos\left(\frac{2\pi}{7}\right)\cos\left(\frac{4\pi}{7}\right)\cos\left(\pi + \frac{\pi}{7}\right) \\
 &= -\cos\left(\frac{\pi}{7}\right)\cos\left(\frac{2\pi}{7}\right)\cos\left(\frac{4\pi}{7}\right) \\
 &= -\frac{1}{2\sin\left(\frac{\pi}{7}\right)}\left(2\sin\left(\frac{\pi}{7}\right)\cos\left(\frac{\pi}{7}\right)\right)\cos\left(\frac{2\pi}{7}\right)\cos\left(\frac{4\pi}{7}\right) \\
 &= -\frac{1}{2^2\sin\left(\frac{\pi}{7}\right)}\left(2\sin\left(\frac{2\pi}{7}\right)\cos\left(\frac{2\pi}{7}\right)\right)\cos\left(\frac{4\pi}{7}\right) \\
 &= -\frac{1}{2^3\sin\left(\frac{\pi}{7}\right)}\left(2\sin\left(\frac{4\pi}{7}\right)\cos\left(\frac{4\pi}{7}\right)\right) \\
 &= -\frac{\sin\left(\frac{8\pi}{7}\right)}{2^3\sin\left(\frac{\pi}{7}\right)} \\
 &= -\frac{\sin\left(\pi + \frac{\pi}{7}\right)}{2^3\sin\left(\frac{\pi}{7}\right)} \\
 &= -\frac{\sin\left(\frac{\pi}{7}\right)}{8\sin\left(\frac{\pi}{7}\right)} \\
 &= -\frac{1}{8}
 \end{aligned}$$

$$\begin{aligned}
 142. \text{ Let } z &= \cos\left(\frac{2\pi}{7}\right) + \cos\left(\frac{4\pi}{7}\right) + \cos\left(\frac{6\pi}{7}\right) \\
 \Rightarrow 2z \sin\left(\frac{\pi}{7}\right) &= 2\sin\left(\frac{\pi}{7}\right)\cos\left(\frac{2\pi}{7}\right) + 2\sin\left(\frac{\pi}{7}\right)\cos\left(\frac{4\pi}{7}\right) \\
 &\quad + 2\sin\left(\frac{\pi}{7}\right)\cos\left(\frac{6\pi}{7}\right) \\
 &= \sin\left(\frac{3\pi}{7}\right) - \sin\left(\frac{\pi}{7}\right) + \sin\left(\frac{5\pi}{7}\right) - \sin\left(\frac{3\pi}{7}\right) \\
 &\quad + \sin\left(\frac{7\pi}{7}\right) - \sin\left(\frac{5\pi}{7}\right) \\
 &= -\sin\left(\frac{\pi}{7}\right) \\
 \text{Thus, } 2z \sin\left(\frac{\pi}{7}\right) &= -\sin\left(\frac{\pi}{7}\right)
 \end{aligned}$$

$$\begin{aligned}
 \Rightarrow z &= -\frac{1}{2} \\
 \Rightarrow \cos\left(\frac{2\pi}{7}\right) + \cos\left(\frac{4\pi}{7}\right) + \cos\left(\frac{6\pi}{7}\right) &= -\frac{1}{2}
 \end{aligned}$$

143. Given

$$\begin{aligned}
 M &= \sqrt{a^2 \cos^2 \theta + b^2 \sin^2 \theta} + \sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta} \\
 \Rightarrow M^2 &= a^2 \cos^2 \theta + b^2 \sin^2 \theta + a^2 \sin^2 \theta + b^2 \cos^2 \theta \\
 &\quad + 2\sqrt{(a^2 \cos^2 \theta + b^2 \sin^2 \theta)(a^2 \sin^2 \theta + b^2 \cos^2 \theta)} \\
 \Rightarrow M^2 &= a^2 + b^2 + 2\sqrt{(a^2 \cos^2 \theta + b^2 \sin^2 \theta)(a^2 \sin^2 \theta + b^2 \cos^2 \theta)} \\
 \Rightarrow M^2 &= a^2 + b^2 + 2[(a^4 + b^4) \sin^2 \theta \cos^2 \theta + a^2 b^2 (\sin^4 \theta + \cos^4 \theta)]^{1/2} \\
 \Rightarrow M^2 &= a^2 + b^2 + 2[(a^4 + b^4) \sin^2 \theta \cos^2 \theta + a^2 b^2 (1 - 2 \sin^2 \theta + \cos^2 \theta)]^{1/2} \\
 \Rightarrow M^2 &= a^2 + b^2 + 2((a^4 + b^4 - 2a^2 b^2) \sin^2 \theta \cos^2 \theta + a^2 b^2)^{1/2} \\
 \Rightarrow M^2 &= a^2 + b^2 + \sqrt{(4(a^4 + b^4 - 2a^2 b^2) \sin^2 \theta \cos^2 \theta + 4a^2 b^2)} \\
 \Rightarrow M^2 &= a^2 + b^2 + \sqrt{((a^2 - b^2)^2 + (\sin(2\theta))^2 + 4a^2 b^2)}
 \end{aligned}$$

$$\begin{aligned}
 \text{Thus, maximum } (M^2) &= a^2 + b^2 + (a^2 + b^2) \\
 &= 2(a^2 + b^2)
 \end{aligned}$$

$$\text{and minimum } (M^2) = a^2 + b^2 + 2ab = (a + b)^2$$

$$\begin{aligned}
 \text{Hence, the value of } m_1 - m_2 &= \text{maximum } (M^2) - \text{minimum } (M^2) \\
 &= 2(a^2 + b^2) - (a + b)^2
 \end{aligned}$$

144. We have

$$\begin{aligned}
 \tan(4\theta) &= \tan(3\theta + \theta) \\
 &= \frac{\tan 3\theta + \tan \theta}{1 - \tan 3\theta \tan \theta} \\
 &= \frac{\frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta} + \tan \theta}{1 - \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta} \tan \theta} \\
 &= \frac{\frac{3 \tan \theta - \tan^3 \theta + \tan \theta - 3 \tan^3 \theta}{1 - 3 \tan^2 \theta}}{\frac{1 - 3 \tan^2 \theta - 3 \tan^2 \theta + \tan^4 \theta}{1 - 3 \tan^2 \theta}} \\
 &= \frac{4 \tan \theta - 4 \tan^3 \theta}{1 - 6 \tan^2 \theta + \tan^4 \theta}
 \end{aligned}$$

145. We have

$$\begin{aligned}
 & \left(\frac{\sin x}{\cos 3x} + \frac{\sin 3x}{\cos 9x} + \frac{\sin 9x}{\cos 27x} \right) \\
 &= \frac{1}{2} \left(\frac{2 \sin x \cos x}{\cos 3x \cos x} + \frac{2 \sin 3x \cos 3x}{\cos 3x \cos 9x} + \frac{2 \sin 9x \cos 9x}{\cos 9x \cos 27x} \right) \\
 &= \frac{1}{2} \left(\frac{\sin 2x}{\cos 3x \cos x} + \frac{\sin 6x}{\cos 3x \cos 9x} + \frac{\sin 18x}{\cos 9x \cos 27x} \right) \\
 &= \frac{1}{2} \left(\frac{\sin (3x - x)}{\cos 3x \cos x} + \frac{\sin (9x - 3x)}{\cos 3x \cos 9x} + \frac{\sin (27x - 9x)}{\cos 9x \cos 27x} \right) \\
 &= \frac{1}{2} \left(\frac{\sin 3x \cos x - \cos 3x \sin x}{\cos 3x \cos x} \right. \\
 &\quad \left. + \frac{\sin 9x \cos 3x - \cos 9x \sin 3x}{\cos 3x \cos 9x} \right. \\
 &\quad \left. + \frac{\sin 27x \cos 9x - \cos 27x \sin 9x}{\cos 9x \cos 27x} \right) \\
 &= \frac{1}{2} (\tan 3x - \tan x + \tan 9x - \tan 3x + \tan 27x - \tan 9x) \\
 &= \frac{1}{2} (\tan 27x - \tan x)
 \end{aligned}$$

146. We have

$$\tan \left(\frac{\theta}{2} \right) (1 + \sec \theta) (1 + \sec 2\theta) (1 + \sec 2^2\theta) \dots \sec (2^n\theta)$$

$$\begin{aligned}
 \text{Now, } \tan \left(\frac{\theta}{2} \right) (1 + \sec \theta) &= \frac{\sin (\theta/2)}{\cos (\theta/2)} \times \frac{2 \cos^2 (\theta/2)}{\cos \theta} \\
 &= \frac{2 \sin (\theta/2) \cos (\theta/2)}{\cos \theta} \\
 &= \frac{\sin \theta}{\cos \theta} = \tan \theta
 \end{aligned}$$

 Also, $\tan \theta \times (1 + \sec 2\theta)$

$$\begin{aligned}
 &= \frac{\sin \theta}{\cos \theta} \times \left(\frac{1 + \cos 2\theta}{\cos 2\theta} \right) \\
 &= \frac{\sin \theta}{\cos \theta} \times \left(\frac{2 \cos^2 \theta}{\cos 2\theta} \right) \\
 &= \frac{2 \sin \theta \cos \theta}{\cos 2\theta} \\
 &= \frac{\sin 2\theta}{\cos 2\theta} = \tan 2\theta
 \end{aligned}$$

$$\begin{aligned}
 \text{Thus, } \tan \left(\frac{\theta}{2} \right) (1 + \sec \theta) (1 + \sec 2\theta) &\dots \sec (2^n\theta) \\
 &= \tan (2^n\theta)
 \end{aligned}$$

147.

 (i) Here, $a = 3$, $b = 4$ and $c = 10$

 Thus, the minimum values of $f(x)$

$$= -\sqrt{a^2 + b^2} + c = -5 + 10 = 5$$

and the maximum values of

$$f(x) = \sqrt{a^2 + b^2} + c = 5 + 10 = 15.$$

 (ii) Maximum value = $\sqrt{3^2 + 4^2} + 10 = 15$

$$\text{Minimum value} = -\sqrt{3^2 + 4^2} + 10 = 5$$

 (iii) Maximum value = $3 + 4 = 7$

$$\text{Minimum value} = -3 + 4 = 1$$

 (iv) Maximum value = $2 + 5 = 7$

$$\text{Minimum value} = -2 + 5 = 3$$

 (v) Maximum value = $\sqrt{2}$

$$\text{Minimum value} = -\sqrt{2}$$

 (vi) Maximum value = $\sqrt{2}$

$$\text{Minimum value} = -\sqrt{2}$$

 (vii) Maximum value = $\sin 1$

$$\text{Minimum value} = -\sin 1$$

 (viii) Maximum value = $\cos 1$

$$\text{Minimum value} = 0$$

 (ix) Maximum value = $\sqrt{2}$

$$\text{Minimum value} = -\sqrt{2}$$

 (x) Given $f(x) = \cos (\sin x) + \sin (\cos x)$.

$$= \cos (\sin x) + \sin (\sin x) + \sin (\cos x) - \sin (\sin x)$$

$$\text{Maximum value} = \sqrt{2} + \sin 1$$

$$\text{Minimum value} = \sqrt{2} - \sin 1$$

 148. $R_f = [\text{minimum } f(x), \text{maximum } f(x)]$

$$= [-\sqrt{2} + 3, \sqrt{2} + 3]$$

 149. Given $2 \sin^2 \theta + 3 \cos^2 \theta$

$$= 2(\sin^2 \theta + \cos^2 \theta) + \cos^2 \theta$$

$$= 2 + \cos^2 \theta$$

$$\text{Maximum value} = 2 + 1 = 3$$

$$\text{Minimum value} = 2 + 0 = 2$$

 150. Let $f(\theta) = 5 \cos \theta + 3 \cos \left(\theta + \frac{\pi}{3} \right) + 3$

$$= 5 \cos \theta + \frac{3}{2} \cos \theta - \frac{3\sqrt{3}}{2} \sin \theta + 3$$

$$= \frac{13}{2} \cos \theta - \frac{3\sqrt{3}}{2} \sin \theta + 3$$

$$\text{Maximum value} = \sqrt{\frac{169}{4} + \frac{27}{4}} + 3 = \sqrt{\frac{196}{4}} + 3 = 10$$

$$\text{Minimum value} = -\sqrt{\frac{169}{4} + \frac{27}{4}} + 3 = -7 + 3 = -4$$

$$\begin{aligned} 151. \text{ Let } f(\theta) &= \cos^2 \theta + 3 \sin^2 \theta - 3 \sin 2\theta + 2 \\ &= 1 + 2 \sin^2 \theta - 3 \sin 2\theta + 2 \\ &= 3 + 1 - \cos 2\theta - 3 \sin 2\theta \\ &= 4 - (\cos 2\theta + 3 \sin 2\theta) \end{aligned}$$

$$\text{Maximum value} = 4 + \sqrt{10}$$

$$\text{Minimum value} = 4 - \sqrt{10}$$

$$\begin{aligned} 152. \text{ Let } f(x) &= \operatorname{cosec}^2 x + 25 \sec^2 x \\ &= 1 + \cot^2 x + 25 + 25 \tan^2 x \\ &= 26 + \cot^2 x + 25 \tan^2 x \\ &\geq 26 + 10 = 36 \end{aligned}$$

Hence, the minimum value is 26.

$$\begin{aligned} 153. \text{ Given expression is } &2 - \cos x + \sin^2 x \\ &= 2 - \cos x + 1 - \cos^2 x \\ &= 3 - \cos x - \cos^2 x \\ &= -(\cos^2 x + \cos x - 3) \\ &= -\left(\left(\cos x + \frac{1}{2}\right)^2 - 3 - \frac{1}{4}\right) \\ &= \frac{13}{4} - \left(\cos x + \frac{1}{2}\right)^2 \end{aligned}$$

$$\text{Maximum value} = \frac{13}{4}$$

$$\text{Minimum value} = \frac{13}{4} - \frac{1}{4} = 3$$

$$\begin{aligned} 154. \text{ Given } y &= 4 \sin^2 \theta - \cos 2\theta \\ &= 2(2 \sin^2 \theta) - \cos 2\theta \\ &= 2(1 - \cos 2\theta) - \cos 2\theta \\ &= 2 - 3 \cos 2\theta \\ &= 2 + 3(-\cos 2\theta) \end{aligned}$$

$$\text{Maximum value} = 2 + 3 = 5$$

$$\text{Minimum value} = 2 - 3 = -1$$

Hence y lies in $[-1, 5]$

$$155. \text{ Here, } m = -3 + 5 = 2$$

$$\text{and } n = 3 + 2 = 5$$

Hence, the value of $(m + n + 2) = 9$

$$\begin{aligned} 156. \text{ Given } f(x) &= \sin^2 x + \cos^4 x \\ &= \frac{1}{2}(2 \sin^2 x) + \frac{1}{4}(2 \cos^2 x)^2 \\ &= \frac{1}{2}(1 - \cos 2x) + \frac{1}{4}(1 + \cos 2x)^2 \\ &= \frac{1}{2}(1 - \cos 2x) + \frac{1}{4}(1 + 2 \cos 2x + \cos^2 2x) \\ &= \frac{1}{2} + \frac{1}{4} + \frac{1}{4} \cos^2 2x \\ &= \frac{3}{4} + \frac{1}{4} \cos^2 2x \end{aligned}$$

$$\text{Maximum value} = \frac{3}{4} + \frac{1}{4} \cdot 1 = \frac{3}{4} + \frac{1}{4} = \frac{4}{4} = 1$$

$$\text{Minimum value} = \frac{3}{4} + \frac{1}{4} \cdot 0 = \frac{3}{4}$$

$$157. \text{ Given } f(x) = \cos^2 x + \sin^4 x$$

$$= \frac{1}{2}(2 \cos^2 x) + \frac{1}{4}(2 \sin^2 x)^2$$

$$= \frac{1}{2}(1 + \cos 2x) + \frac{1}{4}(1 - \cos 2x)^2$$

$$= \frac{1}{2}(1 + \cos 2x) + \frac{1}{4}(1 - 2 \cos 2x + \cos^2 2x)$$

$$= \frac{1}{2} + \frac{1}{4} + \frac{1}{4} \cos^2 2x$$

$$= \frac{3}{4} + \frac{1}{4} \cos^2 2x$$

$$\text{Maximum value} = \frac{3}{4} + \frac{1}{4} \cdot 1 = \frac{3}{4} + \frac{1}{4} = \frac{4}{4} = 1$$

$$\text{Minimum value} = \frac{3}{4} + \frac{1}{4} \cdot 0 = \frac{3}{4}$$

$$\begin{aligned} 158. \text{ Given } f(x) &= \sin^4 x + \cos^4 x \\ &= (\sin^2 x + \cos^2 x)^2 - 2 \sin^2 x \cos^2 x \\ &= 1 - \frac{1}{2} \sin^2 2x \end{aligned}$$

$$\text{Maximum value} = 1 + 0 = 1$$

$$\text{Minimum value} = 1 - \frac{1}{2} = \frac{1}{2}$$

$$\begin{aligned} 159. \text{ We have, } f(\theta) &= \sin^6 \theta + \cos^6 \theta \\ &= (\sin^2 \theta)^3 + (\cos^2 \theta)^3 \\ &= (\sin^2 \theta + \cos^2 \theta)^3 - 3 \sin^2 \theta \cos^2 \theta (\sin^2 \theta + \cos^2 \theta) \\ &= 1 - 3 \sin^2 \theta \cos^2 \theta \\ &= 1 - \frac{3}{4} (4 \sin^2 \theta \cos^2 \theta) \\ &= 1 - \frac{3}{4} (\sin^2 2\theta) \\ &= 1 + \frac{3}{4} (-\sin^2 2\theta) \end{aligned}$$

As we know, $-1 \leq (-\sin^2 2\theta) \leq 0$

$$\Rightarrow -\frac{3}{4} \leq \frac{3(-\sin^2 2\theta)}{4} \leq 0$$

$$\Rightarrow 1 - \frac{3}{4} \leq 1 + \frac{3(-\sin^2 2\theta)}{4} \leq 1$$

$$\Rightarrow \frac{1}{4} \leq f(\theta) \leq 1$$

Hence, the maximum value = 1 and the minimum value = $\frac{1}{4}$

160. Now, A

$$\begin{aligned} &= \cos^2 \theta + \sin^4 \theta \\ &= \frac{1}{2}(2 \sin^2 \theta) + \frac{1}{4}(4 \sin^4 \theta) \\ &= \frac{1}{2}(1 - \cos(2\theta)) + \frac{1}{4}(1 + \cos(2\theta))^2 \\ &= \frac{1}{2}(1 + \cos(2\theta)) + \frac{1}{4}(1 - 2 \cos(2\theta) + \cos^2(2\theta)) \\ &= \frac{1}{2} + \frac{1}{2} \cos(2\theta) + \frac{1}{4} - \frac{1}{2} \cos(2\theta) + \frac{1}{4}(\cos^2(2\theta)) \\ &= \frac{3}{4} + \frac{1}{4}(\cos^2(2\theta)) \end{aligned}$$

$$\text{Maximum value of } A = m_1 = \frac{3}{4} + \frac{1}{4} \cdot 1 = 1$$

Also, B

$$\begin{aligned} &= \sin^2 \theta + \cos^4 \theta \\ &= \frac{1}{2}(2 \sin^2 \theta) + \frac{1}{4}(4 \sin^4 \theta) \\ &= \frac{1}{2}(2 \sin^2 \theta) + \frac{1}{4}(2 \cos^2 \theta)^2 \\ &= \frac{1}{2}(1 - \cos(2\theta)) + \frac{1}{4}(1 + \cos(2\theta))^2 \\ &= \frac{1}{2}(1 - \cos(2\theta)) + \frac{1}{4}(1 + 2 \cos(2\theta) + \cos^2(2\theta)) \\ &= \frac{1}{2} + \frac{1}{4} + \frac{1}{4} \cos^2(2\theta) \\ &= \frac{3}{4} + \frac{1}{4} \cos^2(2\theta) \end{aligned}$$

$$\text{Thus, the minimum value of } B = m = \frac{3}{4} + \frac{1}{4} \cdot 0 = \frac{3}{4}$$

Now, the value of $m_1^2 + m_2^2 + m_1 m_2$

$$\begin{aligned} &= 1 + \frac{9}{16} + \frac{3}{4} \\ &= \frac{37}{16} \end{aligned}$$

161. Given $f(x) = (\sin x + \cos x + \operatorname{cosec} 2x)^3$

As we know that,

AM $\geq$ GM

$$\left(\frac{\sin x + \cos x + \operatorname{cosec} 2x}{3} \right) \geq \sqrt[3]{(\sin x \cdot \cos x \cdot \operatorname{cosec} 2x)}$$

$$\left(\frac{\sin x + \cos x + \operatorname{cosec} 2x}{3} \right) \geq \sqrt[3]{\frac{1}{2}}$$

$$\left(\frac{\sin x + \cos x + \operatorname{cosec} 2x}{3} \right)^3 \geq \frac{1}{2}$$

$$(\sin x + \cos x + \operatorname{cosec} 2x)^3 \geq \frac{27}{2}$$

Hence, the minimum value of $f(x)$ is $\frac{27}{2}$

162. Find the maximum and minimum values of

$$f(x) = \frac{5}{\sin^2 \theta - 6 \sin \theta \cos \theta + 3 \cos^2 \theta}$$

$$\begin{aligned} 163. \text{ Given } f(x) &= \frac{a^2}{\cos^2 x} + \frac{b^2}{\sin^2 x} \\ &= a^2 \sec^2 x + b^2 \operatorname{cosec}^2 x \\ &= a^2 + a^2 \tan^2 x + b^2 + b^2 \cot^2 x \\ &= a^2 + b^2 + (a^2 \tan^2 x + b^2 \cot^2 x) \\ &\geq a^2 + b^2 + 2ab = (a + b)^2 \end{aligned}$$

Hence, the minimum value of $f(x)$ is $(a + b)^2$.

$$164. \text{ We have, } f(x) = \frac{x^2 \sin^2 x + 4}{x \sin x}$$

$$= x \sin x + \frac{4}{x \sin x} \geq 4$$

Hence, the minimum values of $f(x)$ is 4

165. Given $f(x) = \log_x y + \log_y x$

As we know,

AM $\geq$ GM

$$\frac{\log_x y + \log_y x}{2} \geq \sqrt{\log_x y \cdot \log_y x} = 1$$

$$\frac{\log_x y + \log_y x}{2} \geq 1$$

$$\log_x y + \log_y x \geq 2$$

Hence, the minimum value of $f(x)$ is 2.

166. Given $f(x) = 2 \log_{10} x - \log_x (0.01)$, $x > 1$

$$\begin{aligned} &= 2 \log_{10} x - \log_x (10)^{-2} \\ &= 2 \log_{10} x + 2 \log_x (10) \\ &= 2(\log_{10} x + \log_x (10)) \\ &\geq 2 \cdot 2 = 4 \end{aligned}$$

Hence, the minimum value of $f(x)$ is 4

$$167. \text{ Given } f(x, y, z) = \frac{(x^2 + 1)(y^2 + 1)(z^2 + 1)}{xyz}$$

$$= \left(\frac{x^2 + 1}{x} \right) \left(\frac{y^2 + 1}{y} \right) \left(\frac{z^2 + 1}{z} \right)$$

$$= \left(x + \frac{1}{x} \right) \left(y + \frac{1}{y} \right) \left(z + \frac{1}{z} \right)$$

$$\geq 2 \cdot 2 \cdot 2 = 8$$

Hence, the minimum value is 2.

$$168. \text{ Given } f(x, y, z) = \frac{(x^3 + 2)(y^3 + 2)(z^3 + 2)}{xyz}$$

$$\begin{aligned}
 &= \left(\frac{x^3+2}{x} \right) \left(\frac{y^3+2}{y} \right) \left(\frac{z^3+2}{z} \right) \\
 &= \left(x^2 + \frac{2}{x} \right) \left(y^2 + \frac{2}{y} \right) \left(z^2 + \frac{2}{z} \right) \\
 &\geq 3.3.3 = 27
 \end{aligned}$$

Hence, the minimum value is 27.

169. We have $f(a, b, c, d)$

$$\begin{aligned}
 &= \frac{(a^2+1)(b^2+1)(c^2+1)(d^2+1)}{abcd} \\
 &= \frac{(a^2+1)}{a} \times \frac{(b^2+1)}{b} \times \frac{(c^2+1)}{c} \times \frac{(d^2+1)}{d} \\
 &= \left(a + \frac{1}{a} \right) \left(b + \frac{1}{b} \right) \left(c + \frac{1}{c} \right) \left(d + \frac{1}{d} \right) \\
 &\geq 2.2.2.2 = 16
 \end{aligned}$$

Hence, the minimum value is 16.

170. Do yourself.

171. We have $\sin^2(24^\circ) - \sin^2(6^\circ)$

$$\begin{aligned}
 &= \sin(24^\circ + 6^\circ) \sin(24^\circ - 6^\circ) \\
 &= \sin(30^\circ) \times \sin(18^\circ) \\
 &= \frac{1}{2} \times \sin(18^\circ) \\
 &= \frac{1}{2} \times \frac{\sqrt{5}-1}{4} \\
 &= \frac{(\sqrt{5}-1)}{8}
 \end{aligned}$$

172. We have $\sin^2(48^\circ) - \cos^2(12^\circ)$

$$\begin{aligned}
 &= \cos(48^\circ + 12^\circ) \times \cos(48^\circ - 12^\circ) \\
 &= \cos(60^\circ) \times \cos(36^\circ) \\
 &= \frac{1}{2} \times \frac{\sqrt{5}+1}{4} \\
 &= \frac{\sqrt{5}+1}{8}
 \end{aligned}$$

173. We have $\sin(12^\circ) \cdot \sin(48^\circ) \cdot \sin(54^\circ)$

$$\begin{aligned}
 &= \frac{1}{\sin(72^\circ)} (\sin(12^\circ) \cdot \sin(48^\circ) \cdot \sin(72^\circ)) (\sin(54^\circ)) \\
 &= \frac{1}{4 \sin(72^\circ)} \\
 &= \frac{1}{4 \sin(72^\circ)} (\sin(36^\circ) \cdot \cos(36^\circ)) \\
 &= \frac{1}{8 \sin(72^\circ)} (2 \sin(36^\circ) \cdot \cos(36^\circ)) \\
 &= \frac{1}{8 \sin(72^\circ)} (\sin(72^\circ)) \\
 &= \frac{1}{8}
 \end{aligned}$$

174. We have

$$\begin{aligned}
 &\sin(6^\circ) \cdot \sin(42^\circ) \cdot \sin(66^\circ) \sin(78^\circ) \\
 &= \frac{1}{4 \sin(54^\circ)} \\
 &= \frac{1}{4 \sin(54^\circ)} (4 \sin(6^\circ) \cdot \sin(60^\circ - 6^\circ) \cdot \sin(60^\circ + 6^\circ)) \times (\sin(78^\circ) \cdot \sin(42^\circ)) \\
 &= \frac{1}{4 \cos(36^\circ)} (\sin(18^\circ) \sin(72^\circ) \cdot \sin(42^\circ)) \\
 &= \frac{1}{16 \cos(36^\circ)} \\
 &= \frac{1}{16 \cos(36^\circ)} (4 \sin(18^\circ) \cdot \sin(60^\circ + 18^\circ) \cdot \sin(60^\circ - 18^\circ)) \\
 &= \frac{1}{16 \cos(36^\circ)} (\sin(54^\circ)) \\
 &= \frac{1}{16 \cos(36^\circ)} (\cos(36^\circ)) \\
 &= \frac{1}{16}
 \end{aligned}$$

175. We have $4(\sin(24^\circ) + \cos(6^\circ))$

$$\begin{aligned}
 &= 4(\sin(24^\circ) + \sin(84^\circ)) \\
 &= 4 \left(2 \sin\left(\frac{24^\circ + 84^\circ}{2}\right) \cos\left(\frac{24^\circ - 84^\circ}{2}\right) \right) \\
 &= 8(\sin(54^\circ) \cos(30^\circ)) \\
 &= 8(\sin(36^\circ) \cos(30^\circ)) \\
 &= 8 \left(\frac{\sqrt{5}+1}{4} \times \frac{1}{2} \right) \\
 &= (\sqrt{5}+1)
 \end{aligned}$$

176. We have

$$\begin{aligned}
 &\tan(6^\circ) \cdot \tan(42^\circ) \cdot \tan(66^\circ) \tan(78^\circ) \\
 &= (\tan(6^\circ) \cdot \tan(66^\circ)) \times (\tan(42^\circ) \cdot \tan(78^\circ)) \\
 &= \frac{1}{\tan(54^\circ)} \times (\tan(6^\circ) \cdot \tan(54^\circ) \cdot \tan(66^\circ)) \\
 &= \frac{1}{\tan(54^\circ)} \times (\tan(6^\circ) \cdot \tan(60^\circ - 6^\circ) \cdot \tan(60^\circ + 6^\circ)) \times (\tan(42^\circ) \cdot \tan(78^\circ)) \\
 &= \frac{1}{\tan(54^\circ)} (\tan(18^\circ)) \times (\tan(42^\circ) \cdot \tan(78^\circ)) \\
 &= \frac{1}{\tan(54^\circ)} (\tan(18^\circ) \cdot \tan(42^\circ) \cdot \tan(78^\circ)) \\
 &= \frac{1}{\tan(54^\circ)} \\
 &= \frac{1}{\tan(54^\circ)} (\tan(60^\circ - 18^\circ) \cdot \tan(18^\circ) \cdot \tan(60^\circ + 18^\circ)) \\
 &= \frac{1}{\tan(54^\circ)} \times (\tan(54^\circ)) \\
 &= 1
 \end{aligned}$$

$$\begin{aligned}
 177. \text{ We have } & \left(1 + \cos\left(\frac{\pi}{8}\right)\right) \cdot \left(1 + \cos\left(\frac{3\pi}{8}\right)\right) \\
 & \cdot \left(1 + \cos\left(\frac{5\pi}{8}\right)\right) \cdot \left(1 + \cos\left(\frac{7\pi}{8}\right)\right) \\
 & = \left(1 + \cos\left(\frac{\pi}{8}\right)\right) \cdot \left(1 + \cos\left(\frac{7\pi}{8}\right)\right) \\
 & \quad \cdot \left(1 + \cos\left(\frac{3\pi}{8}\right)\right) \cdot \left(1 + \cos\left(\frac{5\pi}{8}\right)\right) \\
 & = \left(1 + \cos\left(\frac{\pi}{8}\right)\right) \cdot \left(1 - \cos\left(\frac{\pi}{8}\right)\right) \\
 & \quad \cdot \left(1 + \cos\left(\frac{3\pi}{8}\right)\right) \cdot \left(1 - \cos\left(\frac{3\pi}{8}\right)\right) \\
 & = \left(1 - \cos^2\left(\frac{\pi}{8}\right)\right) \cdot \left(1 - \cos^2\left(\frac{3\pi}{8}\right)\right) \\
 & = \left(\sin^2\left(\frac{\pi}{8}\right)\right) \cdot \left(\sin^2\left(\frac{3\pi}{8}\right)\right) \\
 & = \frac{1}{4} \left(2 \sin^2\left(\frac{\pi}{8}\right)\right) \cdot \left(2 \sin^2\left(\frac{3\pi}{8}\right)\right) \\
 & = \frac{1}{4} \left(1 - \cos\left(\frac{\pi}{4}\right)\right) \cdot \left(1 - \cos\left(\frac{3\pi}{4}\right)\right) \\
 & = \frac{1}{4} \left(1 - \frac{1}{\sqrt{2}}\right) \cdot \left(1 + \frac{1}{\sqrt{2}}\right) \\
 & = \frac{1}{4} \left(1 - \frac{1}{2}\right) \\
 & = \frac{1}{8}.
 \end{aligned}$$

178. We have

$$\begin{aligned}
 & \sin^4\left(\frac{\pi}{8}\right) + \sin^4\left(\frac{3\pi}{8}\right) + \sin^4\left(\frac{5\pi}{8}\right) + \sin^4\left(\frac{7\pi}{8}\right) \\
 & = \sin^4\left(\frac{\pi}{8}\right) + \sin^4\left(\frac{3\pi}{8}\right) + \sin^4\left(\frac{3\pi}{8}\right) + \sin^4\left(\frac{\pi}{8}\right) \\
 & = 2 \left(\sin^4\left(\frac{\pi}{8}\right) + \sin^4\left(\frac{3\pi}{8}\right) \right) \\
 & = 2 \left(\sin^4\left(\frac{\pi}{8}\right) + \cos^4\left(\frac{\pi}{8}\right) \right) \\
 & = 2 \left(1 - 2 \sin^2\left(\frac{\pi}{8}\right) \cdot \cos^2\left(\frac{\pi}{8}\right) \right) \\
 & = \left(2 - 2 \sin^2\left(\frac{\pi}{8}\right) \cdot 2 \cos^2\left(\frac{\pi}{8}\right) \right) \\
 & = \left(2 - \left(2 \sin^2\left(\frac{\pi}{8}\right) \right) \cdot \left(2 \cos^2\left(\frac{\pi}{8}\right) \right) \right)
 \end{aligned}$$

$$\begin{aligned}
 & = \left(2 - \left(1 - \cos\left(\frac{\pi}{4}\right) \right) \cdot \left(1 + \cos\left(\frac{\pi}{4}\right) \right) \right) \\
 & = \left(2 - \left(1 - \frac{1}{\sqrt{2}} \right) \cdot \left(1 + \frac{1}{\sqrt{2}} \right) \right) \\
 & = \left(2 - \left(1 - \frac{1}{2} \right) \right) \\
 & = \frac{3}{2}.
 \end{aligned}$$

179. We have

$$\begin{aligned}
 & \cos^4\left(\frac{\pi}{8}\right) + \cos^4\left(\frac{3\pi}{8}\right) + \cos^4\left(\frac{5\pi}{8}\right) + \cos^4\left(\frac{7\pi}{8}\right) \\
 & = \cos^4\left(\frac{\pi}{8}\right) + \cos^4\left(\frac{3\pi}{8}\right) + \cos^4\left(\frac{3\pi}{8}\right) + \cos^4\left(\frac{\pi}{8}\right) \\
 & \quad + \cos^4\left(\frac{3\pi}{8}\right) + \cos^4\left(\frac{\pi}{8}\right) \\
 & = 2 \left(\cos^4\left(\frac{\pi}{8}\right) + \cos^4\left(\frac{3\pi}{8}\right) \right) \\
 & = 2 \left(\cos^4\left(\frac{\pi}{8}\right) + \sin^4\left(\frac{\pi}{8}\right) \right) \\
 & = 2 \left(1 - 2 \cos^2\left(\frac{\pi}{8}\right) \cdot \sin^2\left(\frac{\pi}{8}\right) \right) \\
 & = \left(2 - \left(2 \cos^2\left(\frac{\pi}{8}\right) \right) \cdot \left(2 \sin^2\left(\frac{\pi}{8}\right) \right) \right) \\
 & = \left(2 - \left(1 + \cos\left(\frac{\pi}{4}\right) \right) \cdot \left(1 + \cos\left(\frac{\pi}{4}\right) \right) \right) \\
 & = \left(2 - \left(1 + \frac{1}{\sqrt{2}} \right) \cdot \left(1 - \frac{1}{\sqrt{2}} \right) \right) \\
 & = \left(2 - \left(1 - \frac{1}{2} \right) \right) \\
 & = \frac{3}{2}.
 \end{aligned}$$

180. We have $\tan(20^\circ) \tan(80^\circ)$

$$\begin{aligned}
 & = \frac{1}{\tan(40^\circ)} [\tan(40^\circ) \tan(20^\circ) \tan(80^\circ)] \\
 & = \frac{1}{\tan(40^\circ)} \times \tan(60^\circ) \\
 & = \frac{\sqrt{3}}{\tan(40^\circ)} \\
 & = \sqrt{3} \cot(40^\circ) \\
 & = \sqrt{3} \tan(60^\circ)
 \end{aligned}$$

181. We have $\tan(10^\circ) \tan(70^\circ)$

$$\begin{aligned} &= \frac{1}{\tan(50^\circ)} [\tan(50^\circ) \tan(10^\circ) \tan(70^\circ)] \\ &= \frac{1}{\tan(50^\circ)} \times \tan(30^\circ) \\ &= \frac{1}{\sqrt{3}} \times \cot(50^\circ) \\ &= \frac{1}{\sqrt{3}} \times \tan(40^\circ) \end{aligned}$$

182. We have $\sin 55^\circ - \sin 19^\circ + \sin 53^\circ - \sin 17^\circ$

$$\begin{aligned} &= (\sin(55^\circ) + \sin(53^\circ)) - (\sin(19^\circ) + \sin(17^\circ)) \\ &= 2 \sin(54^\circ) \cos(1^\circ) - 2 \sin(18^\circ) \cos(1^\circ) \\ &= 2 \cos(1^\circ) [\sin(54^\circ) - \sin(18^\circ)] \\ &= 2 \cos(1^\circ) \left[\frac{\sqrt{5}+1}{4} - \frac{\sqrt{5}-1}{4} \right] \\ &= 2 \cos(1^\circ) \times \frac{1}{2} = \cos(1^\circ) \end{aligned}$$

183. We have

$$\begin{aligned} &\cos\left(\frac{2\pi}{7}\right) \cos\left(\frac{4\pi}{7}\right) \cos\left(\frac{6\pi}{7}\right) \\ &= -\cos\left(\frac{\pi}{7}\right) \cos\left(\frac{2\pi}{7}\right) \cos\left(\frac{4\pi}{7}\right) \\ &= -\frac{1}{2 \sin\left(\frac{\pi}{7}\right)} \times \left[2 \sin\left(\frac{\pi}{7}\right) \cos\left(\frac{\pi}{7}\right) \right] \cos\left(\frac{2\pi}{7}\right) \cos\left(\frac{4\pi}{7}\right) \\ &= -\frac{1}{2^2 \sin\left(\frac{\pi}{7}\right)} \times \left[2 \sin\left(\frac{2\pi}{7}\right) \cos\left(\frac{2\pi}{7}\right) \right] \cos\left(\frac{4\pi}{7}\right) \\ &= -\frac{1}{2^3 \sin\left(\frac{\pi}{7}\right)} \times \left[2 \sin\left(\frac{4\pi}{7}\right) \cos\left(\frac{4\pi}{7}\right) \right] \\ &= -\frac{1}{2^3 \sin\left(\frac{\pi}{7}\right)} \times \sin\left(\frac{8\pi}{7}\right) \\ &= -\frac{1}{2^3 \sin\left(\frac{\pi}{7}\right)} \times \sin\left(\pi + \frac{\pi}{7}\right) \\ &= -\frac{1}{2^3 \sin\left(\frac{\pi}{7}\right)} \times -\sin\left(\frac{\pi}{7}\right) \\ &= \frac{1}{2^3} = \frac{1}{8} \end{aligned}$$

184. Let $S = \cos\left(\frac{2\pi}{7}\right) + \cos\left(\frac{4\pi}{7}\right) + \cos\left(\frac{6\pi}{7}\right)$

$$\begin{aligned} &\Rightarrow 2 \sin\left(\frac{\pi}{7}\right) S = 2 \sin\left(\frac{\pi}{7}\right) \cos\left(\frac{2\pi}{7}\right) \\ &\quad + 2 \sin\left(\frac{\pi}{7}\right) \cos\left(\frac{4\pi}{7}\right) + 2 \sin\left(\frac{\pi}{7}\right) \cos\left(\frac{6\pi}{7}\right) \\ &\Rightarrow 2 \sin\left(\frac{\pi}{7}\right) S = \left(\sin\left(\frac{3\pi}{7}\right) - \sin\left(\frac{\pi}{7}\right) \right) \\ &\quad + \left(\sin\left(\frac{5\pi}{7}\right) - \sin\left(\frac{3\pi}{7}\right) \right) + \left(\sin\left(\frac{7\pi}{7}\right) - \sin\left(\frac{5\pi}{7}\right) \right) \\ &\Rightarrow 2 \sin\left(\frac{\pi}{7}\right) S = -\sin\left(\frac{\pi}{7}\right) \\ &\Rightarrow S = -\frac{1}{2} \end{aligned}$$

Hence, the result.

185. We have $\tan\left(7\frac{1}{2}^\circ\right) + \cot\left(7\frac{1}{2}^\circ\right)$

$$\begin{aligned} &= \frac{\sin\left(7\frac{1}{2}^\circ\right)}{\cos\left(7\frac{1}{2}^\circ\right)} + \frac{\cos\left(7\frac{1}{2}^\circ\right)}{\sin\left(7\frac{1}{2}^\circ\right)} \\ &= \frac{\sin^2\left(7\frac{1}{2}^\circ\right) + \cos^2\left(7\frac{1}{2}^\circ\right)}{\sin\left(7\frac{1}{2}^\circ\right) \cos\left(7\frac{1}{2}^\circ\right)} \\ &= \frac{2}{2 \sin\left(7\frac{1}{2}^\circ\right) \cos\left(7\frac{1}{2}^\circ\right)} \\ &= \frac{2}{\sin(15^\circ)} \\ &= \frac{2}{\frac{\sqrt{3}-1}{2\sqrt{2}}} \\ &= \frac{4\sqrt{2}}{\sqrt{3}-1} \\ &= \frac{4\sqrt{2}(\sqrt{3}+1)}{2} = 2\sqrt{2}(\sqrt{3}+1) \end{aligned}$$

186. Let $y = \cos\left(\frac{x}{2}\right) - \sqrt{3} \sin\left(\frac{x}{2}\right)$

$$\begin{aligned} &= 2 \left(\frac{1}{2} \cos\left(\frac{x}{2}\right) - \frac{\sqrt{3}}{2} \sin\left(\frac{x}{2}\right) \right) \\ &= 2 \cos\left(\frac{x}{2} + \frac{\pi}{3}\right) \end{aligned}$$

$$\text{Now, } \frac{dy}{dx} = -\sin\left(\frac{x}{2} + \frac{\pi}{3}\right)$$

For maximum or minimum, $\frac{dy}{dx} = 0$

$$\Rightarrow \sin\left(\frac{x}{2} + \frac{\pi}{3}\right) = 0$$

$$\Rightarrow \left(\frac{x}{2} + \frac{\pi}{3}\right) = \pi$$

$$\Rightarrow \frac{x}{2} = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$$

$$\Rightarrow x = \frac{4\pi}{3}$$

Hence, the value of x is $\frac{4\pi}{3}$

187. If α and β be two different roots

$a \cos \theta + b \sin \theta = c$, then prove that

$$\sin(\alpha + \beta) = \frac{2ab}{a^2 + b^2}.$$

188. We have, $\sin 2A + \sin 2B + \sin 2C$

$$\begin{aligned} &= (\sin 2A + \sin 2B) + \sin 2C \\ &= 2(\sin(A+B) \cos(A-B)) + \sin 2C \\ &= 2(\sin(\pi - C) \cdot \cos(A-B)) + 2 \sin C \cos C \\ &= 2(\sin C \cdot \cos(A-B)) + 2 \sin C \cos C \\ &= 2(\sin C(\cos(A-B) + \cos C)) \\ &= 2 \sin C(\cos(A-B) + \cos(\pi - (A+B))) \\ &= 2 \sin C(\cos(A-B) - \cos(A+B)) \\ &= 2 \sin C(2 \sin A \sin B) \\ &= 4 \sin A \cdot \sin B \cdot \sin C \end{aligned}$$

189. We have $\cos 2A + \cos 2B + \cos 2C$

$$\begin{aligned} &= (\cos 2A + \cos 2B) + \cos 2C \\ &= 2 \cos(A+B) \cos(A-B) + \cos 2C \\ &= 2 \cos\{\pi - C\} \cos(A-B) + \cos 2C \\ &= -2 \cos C \cos(A+B) + 2 \cos^2 C - 1 \\ &= -1 - 2 \cos C(\cos(A-B) - \cos C) \\ &= -1 - 2 \cos C(\cos(A-B) + \cos(A+B)) \\ &= -1 - 2 \cos C(2 \cos A \cdot \cos B) \\ &= -1 - 4 \cos A \cdot \cos B \cdot \cos C \end{aligned}$$

190. We have $\sin^2 A + \sin^2 B - \sin^2 C$

$$\begin{aligned} &= \sin^2 A + (\sin^2 B - \sin^2 C) \\ &= \sin^2 A + \sin(B+C) \sin(B-C) \\ &= \sin^2 A + \sin(\pi - A) \sin(B-C) \\ &= \sin^2 A + \sin A \sin(B-C) \\ &= \sin A(\sin A + \sin(B-C)) \\ &= \sin A(\sin(B+C) + \sin(B-C)) \\ &= \sin A(2 \sin B \cos C) \\ &= 2 \sin A \cdot \sin B \cdot \cos C \end{aligned}$$

191.(i) We have $\sin^2 A + \sin^2 B + \sin^2 C$

$$\begin{aligned} &= 1 - \cos^2 A + \sin^2 B + \sin^2 C \\ &= 1 - (\cos^2 A - \sin^2 B) + (1 - \cos^2 C) \end{aligned}$$

$$\begin{aligned} &= 2 - (\cos^2 A - \sin^2 B) - \cos^2 C \\ &= 2 - (\cos(A+B) \cdot \cos(A-B)) - \cos^2 C \\ &= 2 - (\cos(\pi - C) \cdot \cos(A-B)) - \cos^2 C \\ &= 2 + \cos C(\cos(A-B) - \cos C) \\ &= 2 + \cos C(\cos(A-B) + \cos(A+B)) \\ &= 2 + \cos C(2 \cos A \cdot \cos B) \\ &= 2 + 2 \cos A \cdot \cos B \cdot \cos C \end{aligned}$$

192. We have $\sin 2A + \sin 2B + \sin 2C$

$$= 4 \sin A \sin B \sin C$$

Also, $\cos A + \cos B + \cos C - 1$

$$\begin{aligned} &= 2 \cos\left(\frac{A+B}{2}\right) \cos\left(\frac{A-B}{2}\right) - 2 \sin^2\left(\frac{C}{2}\right) \\ &= 2 \cos\left(\frac{\pi}{2} - \frac{C}{2}\right) \cos\left(\frac{A-B}{2}\right) - 2 \sin^2\left(\frac{C}{2}\right) \\ &= 2 \sin\left(\frac{C}{2}\right) \cos\left(\frac{A-B}{2}\right) - 2 \sin^2\left(\frac{C}{2}\right) \\ &= 2 \sin\left(\frac{C}{2}\right) \left(\cos\left(\frac{A-B}{2}\right) - \sin\left(\frac{C}{2}\right)\right) \\ &= 2 \sin\left(\frac{C}{2}\right) \left(\cos\left(\frac{A-B}{2}\right) - \cos\left(\frac{A+B}{2}\right)\right) \\ &= 2 \sin\left(\frac{C}{2}\right) 2 \sin\left(\frac{A}{2}\right) \sin\left(\frac{B}{2}\right) \\ &= 4 \sin\left(\frac{A}{2}\right) \sin\left(\frac{B}{2}\right) \sin\left(\frac{C}{2}\right) \end{aligned}$$

Thus, $\frac{\sin 2A + \sin 2B + \sin 2C}{\cos A + \cos B + \cos C - 1}$

$$\begin{aligned} &= \frac{4 \sin A \sin B \sin C}{4 \sin(A/2) \sin(B/2) \sin(C/2)} \\ &= 8 \cos\left(\frac{A}{2}\right) \cos\left(\frac{B}{2}\right) \cos\left(\frac{C}{2}\right) \end{aligned}$$

193.

(i) We have $A + B + C = \pi$

$$\Rightarrow A + B = \pi - C$$

$$\Rightarrow \tan(A+B) = \tan(\pi - C)$$

$$\Rightarrow \frac{\tan A + \tan B}{1 - \tan A \cdot \tan B} = -\tan C$$

$$\Rightarrow \tan A + \tan B = -\tan C(1 - \tan A \cdot \tan B)$$

$$\Rightarrow \tan A + \tan B = -\tan C + \tan A \cdot \tan B \cdot \tan C$$

$$\Rightarrow \tan A + \tan B + \tan C = \tan A \cdot \tan B \cdot \tan C$$

(ii) As we know that,

$$\tan A + \tan B + \tan C = \tan A \cdot \tan B \cdot \tan C$$

Dividing both the sides by

' $\tan A \cdot \tan B \cdot \tan C$ ', we get,

$$\begin{aligned} &= \frac{\tan A}{\tan A \cdot \tan B \cdot \tan C} + \frac{\tan B}{\tan A \cdot \tan B \cdot \tan C} \\ &\quad + \frac{\tan C}{\tan A \cdot \tan B \cdot \tan C} = 1 \end{aligned}$$

$$\Rightarrow \frac{1}{\tan B \cdot \tan C} + \frac{1}{\tan A \cdot \tan C} + \frac{1}{\tan A \cdot \tan B} = 1$$

$$\Rightarrow \cot B \cdot \cot C + \cot A \cdot \cot C + \cot A \cdot \cot B = 1$$

194. We have $A + B + C = 2\pi$

$$\Rightarrow A + B + C = 2\pi$$

$$\Rightarrow A + B = 2\pi - (C + D)$$

$$\Rightarrow \tan(A + B) = \tan\{2\pi - (C + D)\}$$

$$\Rightarrow \tan(A + B) = -\tan(C + D)$$

$$\Rightarrow \frac{\tan A + \tan B}{1 - \tan A \cdot \tan B} = -\frac{\tan C + \tan D}{1 - \tan C \cdot \tan D}$$

$$\Rightarrow (\tan A + \tan B)(1 - \tan C \cdot \tan D) = -(\tan C + \tan D)(1 - \tan A \cdot \tan B)$$

$$\Rightarrow \tan A + \tan B - \tan A \cdot \tan C \cdot \tan D - \tan B \cdot \tan C \cdot \tan D = -\tan C - \tan D + \tan A \cdot \tan B \cdot \tan C + \tan A \cdot \tan B \cdot \tan D$$

$$\Rightarrow \tan A + \tan B + \tan C + \tan D = \tan A \cdot \tan C \cdot \tan D + \tan B \cdot \tan C \cdot \tan D + \tan A \cdot \tan B \cdot \tan C + \tan A \cdot \tan B \cdot \tan D$$

$$\Rightarrow \frac{\tan A + \tan B + \tan C + \tan D}{\tan A \cdot \tan B \cdot \tan C \cdot \tan D}$$

$$= \frac{\tan A \cdot \tan B \cdot \tan C}{\tan A \cdot \tan B \cdot \tan C \cdot \tan D}$$

$$+ \frac{\tan A \cdot \tan C \cdot \tan D}{\tan A \cdot \tan B \cdot \tan C \cdot \tan D}$$

$$+ \frac{\tan A \cdot \tan B \cdot \tan D}{\tan A \cdot \tan B \cdot \tan C \cdot \tan D}$$

$$+ \frac{\tan B \cdot \tan C \cdot \tan D}{\tan A \cdot \tan B \cdot \tan C \cdot \tan D}$$

$$\Rightarrow \frac{\tan A + \tan B + \tan C + \tan D}{\tan A \cdot \tan B \cdot \tan C \cdot \tan D}$$

$$= \frac{1}{\tan A} + \frac{1}{\tan B} + \frac{1}{\tan C} + \frac{1}{\tan D}$$

$$\Rightarrow \frac{\tan A + \tan B + \tan C + \tan D}{\tan A \cdot \tan B \cdot \tan C \cdot \tan D}$$

$$= \cot A + \cot B + \cot C + \cot D$$

$$\Rightarrow \frac{\tan A + \tan B + \tan C + \tan D}{\cot A + \cot B + \cot C + \cot D}$$

$$= \tan A \cdot \tan B \cdot \tan C \cdot \tan D$$

194. We have $(\cot A + \cot B) = \frac{\cos A}{\sin A} + \frac{\cos B}{\sin B}$

$$= \frac{\cos A \sin B + \sin A \cdot \cos B}{\sin A \sin B}$$

$$= \frac{\sin(A + B)}{\sin A \sin B} = \frac{\sin C}{\sin A \sin B}$$

$$\text{Similarly, } (\cot B + \cot C) = \frac{\sin A}{\sin B \sin C}$$

$$\text{and } (\cot C + \cot A) = \frac{\sin B}{\sin A \sin C}$$

Thus, $(\cot A + \cot B)(\cot B + \cot C)$

$(\cot C + \cot A)$

$$= \frac{\sin C}{\sin A \sin B} \times \frac{\sin B}{\sin A \sin C} \times \frac{\sin A}{\sin B \sin C}$$

$$= \frac{1}{\sin A \sin B \sin C}$$

$$= \operatorname{cosec} A \operatorname{cosec} B \operatorname{cosec} C$$

195. Put $x = \tan A$, $y = \tan B$ and $z = \tan C$

Given, $xy + yz + zx = 1$

$$\Rightarrow \tan A \cdot \tan B + \tan B \cdot \tan C + \tan C \cdot \tan A = 1$$

$$\Rightarrow \tan B \cdot \tan C + \tan C \cdot \tan A = 1 - \tan A \cdot \tan B$$

$$\Rightarrow \tan C (\tan B + \tan A) = 1 - \tan A \cdot \tan B$$

$$\Rightarrow \frac{\tan A + \tan B}{1 - \tan A \cdot \tan B} = \frac{1}{\tan C}$$

$$\Rightarrow \tan(A + B) = \cot C = \tan\left(\frac{\pi}{2} - C\right)$$

$$\Rightarrow (A + B) = \left(\frac{\pi}{2} - C\right)$$

$$\Rightarrow (A + B + C) = \frac{\pi}{2}$$

Now,

$$\begin{aligned} \text{LHS} &= \frac{x}{1-x^2} + \frac{y}{1-y^2} + \frac{z}{1-z^2} \\ &= \frac{\tan A}{1-\tan^2 A} + \frac{\tan B}{1-\tan^2 B} + \frac{\tan C}{1-\tan^2 C} \\ &= \frac{1}{2} \left(\frac{2 \tan A}{1-\tan^2 A} + \frac{2 \tan B}{1-\tan^2 B} + \frac{2 \tan C}{1-\tan^2 C} \right) \\ &= \frac{1}{2} (\tan 2A + \tan 2B + \tan 2C) \\ &= \frac{1}{2} (\tan 2A \cdot \tan 2B \cdot \tan 2C) \\ &= \frac{1}{2} \left(\frac{2 \tan A}{1-\tan^2 A} \cdot \frac{2 \tan B}{1-\tan^2 B} \cdot \frac{2 \tan C}{1-\tan^2 C} \right) \\ &= \left(\frac{4 \tan A \cdot \tan B \cdot \tan C}{(1-\tan^2 A)(1-\tan^2 B)(1-\tan^2 C)} \right) \\ &= \frac{4xyz}{(1-x^2)(1-y^2)(1-z^2)} \end{aligned}$$

Hence, the result.

196. Put $x = \tan A$, $y = \tan B$ and $z = \tan C$

Given, $xy + yz + zx = 1$

$$\Rightarrow \tan A \cdot \tan B + \tan B \cdot \tan C + \tan C \cdot \tan A = 1$$

$$\Rightarrow A + B + C = \frac{\pi}{2}$$

Now, LHS

$$\begin{aligned} &= \frac{x}{1+x^2} + \frac{y}{1+y^2} + \frac{z}{1+z^2} \\ &= \frac{1}{2} \left(\frac{2x}{1+x^2} + \frac{2y}{1+y^2} + \frac{2z}{1+z^2} \right) \\ &= \frac{1}{2} \left(\frac{2 \tan A}{1+\tan^2 A} + \frac{2 \tan B}{1+\tan^2 B} + \frac{2 \tan C}{1+\tan^2 C} \right) \\ &= \frac{1}{2} (\sin 2A + \sin 2B + \sin 2C) \\ &= \frac{1}{2} (4 \cos A \cdot \cos B \cdot \cos C) \\ &= 2 \cos A \cdot \cos B \cdot \cos C \\ &= \frac{2}{\sec A \cdot \sec B \cdot \sec C} \\ &= \frac{2}{\sqrt{(1+\tan^2 A)(1+\tan^2 B)(1+\tan^2 C)}} \\ &= \frac{2}{\sqrt{(1+x^2)(1+y^2)(1+z^2)}} \end{aligned}$$

Hence, the result.

197. Let $A = \pi - \beta$, $B = \beta - \gamma$, $C = \gamma - \alpha$

Now, $A + B + C = 0$

$$\Rightarrow A + B = -C$$

$$\Rightarrow \tan(A + B) = \tan(-C) = -\tan C$$

$$\Rightarrow \frac{\tan A + \tan B}{1 - \tan A \tan B} = -\tan C$$

$$\Rightarrow \tan A + \tan B = -\tan C + \tan A \tan B \tan C$$

$$\begin{aligned} \Rightarrow \tan A + \tan B + \tan C \\ = \tan A \tan B \tan C \end{aligned}$$

$$\begin{aligned} \Rightarrow \tan(\alpha - \beta) + \tan(\beta - \gamma) + \tan(\gamma - \alpha) \\ = \tan(\alpha - \beta) \tan(\beta - \gamma) \tan(\gamma - \alpha) \end{aligned}$$

198. We have $\cot A + \cot B + \cot C = \sqrt{3}$

$$\Rightarrow (\cot A + \cot B + \cot C)^2 = 3$$

$$\begin{aligned} \Rightarrow \cot^2 A + \cot^2 B + \cot^2 C \\ + 2(\cot A \cot B + \cot B \cot C + \cot C \cot A) = 3 \end{aligned}$$

$$\Rightarrow \cot^2 A + \cot^2 B + \cot^2 C + 2 = 3$$

$$\Rightarrow \cot^2 A + \cot^2 B + \cot^2 C = 1$$

$$\begin{aligned} \Rightarrow \cot^2 A + \cot^2 B + \cot^2 C \\ = (\cot A \cot B + \cot B \cot C + \cot C \cot A) \end{aligned}$$

$$\Rightarrow \frac{1}{2} [(\cot A - \cot B)^2 + (\cot A - \cot C)^2 + (\cot B - \cot C)^2] = 0$$

$$\Rightarrow (\cot A - \cot B)^2 = 0, (\cot B - \cot C)^2 = 0, (\cot C - \cot A)^2 = 0$$

$$\begin{aligned} \Rightarrow \cot A = \cot B, \cot B = \cot C, \\ \cot C = \cot A \end{aligned}$$

$$\Rightarrow \cot A = \cot B = \cot C$$

So, Δ is an equilateral.

199. Given expression is

$$x + y + z = xyz$$

Put $x = \tan A$, $y = \tan B$ and $z = \tan C$

So, $\tan A + \tan B + \tan C = \tan A \tan B \tan C$

$$\tan A + \tan B = -\tan C(1 - \tan A \tan B)$$

$$\frac{\tan A + \tan B}{(1 - \tan A \tan B)} = \tan(\pi - C)$$

$$\tan(A + B) = \tan(\pi - C)$$

$$(A + B) = (\pi - C)$$

$$(A + B + C) = \pi$$

$$(3A + 3B + 3C) = 3\pi$$

$$(3A + 3B) = 3\pi - 3C$$

$$\tan(3A + 3B) = \tan(3\pi - 3C)$$

$$\frac{\tan 3A + \tan 3B}{1 - \tan 3A \tan 3B} = -\tan 3C$$

$$\tan 3A + \tan 3B + \tan 3C$$

$$= \tan 3A \cdot \tan 3B \cdot \tan 3C$$

$$\frac{3 \tan A - \tan^3 A}{1 - 3 \tan^2 A} + \frac{3 \tan B - \tan^3 B}{1 - 3 \tan^2 B} + \frac{3 \tan C - \tan^3 C}{1 - 3 \tan^2 C}$$

$$= \frac{3 \tan A - \tan^3 A}{1 - 3 \tan^2 A} \cdot \frac{3 \tan B - \tan^3 B}{1 - 3 \tan^2 B} \cdot \frac{3 \tan C - \tan^3 C}{1 - 3 \tan^2 C}$$

$$\frac{3x - x^3}{1 - 3x^2} \cdot \frac{3y - y^3}{1 - 3y^2} \cdot \frac{3z - z^3}{1 - 3z^2}$$

$$= \frac{3x - x^3}{1 - 3x^2} \cdot \frac{3y - y^3}{1 - 3y^2} \cdot \frac{3z - z^3}{1 - 3z^2}$$

Hence, the result.

200. We have $1 + \cos 56^\circ + \cos 58^\circ - \cos 66^\circ$

$$= (1 - \cos(66^\circ)) + (\cos(58^\circ)) + \cos(56^\circ)$$

$$= 2 \sin^2(33^\circ) + 2 \cos(57^\circ) \cos(1^\circ)$$

$$= 2 \sin^2(33^\circ) + 2 \sin(33^\circ) \cos(1^\circ)$$

$$= 2 \sin(33^\circ)(\sin(33^\circ) + \cos(1^\circ))$$

$$= 2 \sin(33^\circ)(\cos(57^\circ) + \cos(1^\circ))$$

$$= 2 \sin(33^\circ)(2 \cos(29^\circ) \cos(28^\circ))$$

$$= 4 \cos(29^\circ) \sin(28^\circ) \sin(33^\circ)$$

Hence, the result.

201. Let $S = \sin \alpha + \sin(\alpha + \beta) + \sin(\alpha + 2\beta)$

$$+ \sin(\alpha + 3\beta) + \dots + \sin(\alpha + (n-1)\beta)$$

Now,

$$2 \sin \alpha \sin\left(\frac{\beta}{2}\right) = \cos\left(\alpha - \frac{\beta}{2}\right) - \cos\left(\alpha + \frac{\beta}{2}\right)$$

$$2 \sin(\alpha + \beta) \cdot \sin\left(\frac{\beta}{2}\right) = \cos\left(\alpha + \frac{\beta}{2}\right) - \cos\left(\alpha + \frac{3\beta}{2}\right)$$

$$2 \sin(\alpha + 2\beta) \cdot \sin\left(\frac{\beta}{2}\right) = \cos\left(\alpha + \frac{3\beta}{2}\right) - \cos\left(\alpha + \frac{5\beta}{2}\right)$$

⋮